

A prospect of Cuba: Past and Future

Alexei Vazquez

For those who stayed, and those who left.

Preface

Cuba is one of the few countries about which almost nobody writes carefully. The literature divides into two libraries that do not read each other. One records a small poor island that abolished illiteracy, built a health system that outperformed its income by a wide margin, sent doctors to forty countries, and did it under six decades of economic siege by the largest economy on earth. The other records a one-party state that jailed its poets, criminalised its markets, drove a fifth of its people into exile, and ended the period with a grid that could not keep the lights on. Both libraries are accurate. Neither is honest, because each is complete only by omission.

This book tries to hold both at once, and it does so for a practical reason rather than a scholarly one. The last part of the book is a design: a proposed socio-political and economic system for Cuba from 2026 forward. A design is only as good as its account of the initial conditions. If you believe the first library, you will design as though the only thing wrong is the embargo, and you will produce a plan that fails the moment it meets a Cuban who wants to open a shop. If you believe the second, you will design as though the only thing wrong is socialism, and you will produce a plan that dismantles a public health system it cannot replace, hands the assets to whoever is standing closest, and reproduces in ten years the inequality that made 1959 possible. Both failures have been run as experiments elsewhere. Neither needs a second trial.

So Parts I and II are not throat-clearing. They are the constraint set. Part I covers 1959 to 1990, the years in which Cuba built its social state and paid for it with a Soviet subsidy and a closed

politics. Part II covers 1990 to 2026, the long adjustment to the loss of that subsidy: two full cycles of liberalise-under-duress and recentralise-when-the-pressure-lifts, ending in the worst material crisis the revolution has faced, worse in some dimensions than the Special Period, and the largest emigration in Cuban history. Part III asks what can be built on what is left.

What is left is more than the ruin makes it look. Cuba has a population that can read, calculate and be trained quickly; a medical and biotechnological establishment with real products and a reputation that is still worth money; some of the best beaches, reefs and colonial cities in the hemisphere, underexploited; eleven hundred kilowatt-hours of sunlight per square metre per year falling on a country that currently burns imported fuel oil; a diaspora of roughly two million people, many of them wealthy, most of them one flight from home; and, unusually for a poor country, low violent crime and a functioning public order. Those are the assets. The book's argument is that they compose: that education plus connectivity plus a residence regime for mobile knowledge workers is a second export track alongside tourism, that solar plus storage converts the country's largest recurring hard-currency cost into a domestic one, and that biotechnology is the one sector where Cuba already holds something the world will pay for and has never been organised to sell.

The book's argument is also that none of it works without the boring part. Every asset listed above has been available for twenty years and has not been converted, and the reason is not the embargo alone. It is that Cuba lacks the institutional machinery that turns an asset into an investment: enforceable contracts, a currency that means something, a customs service that does not take a cut, a court a foreigner can lose in fairly, a procurement system that publishes what it bought. A country can have sunshine and doctors and beaches and still be poor, and Cuba is the proof. So the longest chapters in Part III are not the ones about solar panels. They are the ones about money, about property, and about corruption, which is the failure mode every transition of this type has actually died of.

One commitment in Part III is not up for argument in this book, and it should be stated at the front rather than arrived at in

the middle. The design keeps a universal welfare state at Western European standard — health, education, pensions, disability, income support — and everything else in Part III is arranged around being able to afford it. This is not sentiment about the revolution. It is the reading of the record in Parts I and II: the social state is the one asset those sixty-seven years produced that is unambiguously worth keeping, it is the thing that protects the Cubans most exposed if a transition goes badly, and it is also an economic input rather than only a transfer, since the education system is precisely what the second track and the biotechnology sector are built out of.

What changes is not the benefit but who pays for it. Cuba's welfare state has never been financed by Cubans. It was financed by Soviet terms of trade until 1990, by Venezuelan oil in the 2000s, and in the intervals by printing money and by not maintaining anything, which is why it has come close to collapse twice in living memory. An entitlement funded by an external rent lasts exactly as long as the rent. The European systems this book takes as its standard are financed by taxing a domestic base, which is why they survive changes of government, of commodity price and of patron. So the proposal is a European benefit structure on a European tax structure — a broad-based, simply administered tax system of a kind Cuba has never had — and the cost of that commitment, a tax burden well above the Latin American norm, is stated in the chapters rather than hidden in them. A country that wants Nordic outcomes has to collect Nordic revenue, and there is no version of this in which the arithmetic is pleasant.

I write as a Cuban of the diaspora and a physicist by training, which shapes the book in two ways worth declaring. It is why Part III is written as a set of numbered institutional rules rather than as an essay: a design that cannot be stated as rules is not a design, and rules can be argued with one at a time. It is also why the book is not neutral about outcomes even where it tries to be neutral about facts. It wants Cuba to be prosperous, open and its own. It does not assume those three are automatically compatible, and where they conflict, the text says so instead of choosing quietly.

Nothing here is a prediction. The prospect of the title is the older sense of the word: a view of what lies ahead from where one is

standing, drawn to scale.

A note on sources and numbers

Cuban statistics are a problem with three layers, and a book that quotes them without saying so is misleading its reader by arithmetic rather than by argument.

The first layer is ordinary. Cuba's national statistics office, the Oficina Nacional de Estadística e Información (ONEI), publishes an annual *Anuario Estadístico* that is detailed, internally consistent, and by the standards of the region reasonably competent. Health and education series in particular are collected by an administrative apparatus that reaches every municipality, and the vital-registration coverage is close to complete, which is more than can be said for most countries at Cuba's income level. Where this book quotes life expectancy, infant mortality, school enrolment or physician counts, ONEI is usually the source and the figure is usually trustworthy in the narrow sense that the state measured what it says it measured.

The second layer is the exchange rate, and it corrupts everything monetary. For most of the period covered here the Cuban state carried an official rate of one peso to one dollar for enterprise accounting while households changed money at twenty-four or twenty-five, and after 2021 at rates that in the informal market passed three hundred. Any figure denominated in Cuban pesos and converted at an official rate — GDP, GDP per capita, wages, the value of the social wage, the cost of a subsidy — is therefore not a measurement but an accounting convention, and cross-country comparisons built on it

are close to meaningless. Cuba's reported GDP per capita has for decades placed it in the upper-middle band of Latin America while its consumption basket has looked like the lower. Both facts are real; the contradiction is in the conversion. This book gives peso figures with the rate attached, prefers physical quantities to monetary ones wherever a physical quantity exists — tons of sugar, barrels of oil, megawatts, doctors per thousand, calories per day — and treats any single figure for Cuban GDP as an argument rather than a datum.

The third layer is political, and it runs in both directions. The state does not publish, or publishes late and partially, series that embarrass it: the scale of the 1993–94 contraction was reconstructed years afterwards, the prison population is not disclosed, the count of political detentions is produced by opposition monitors and by no one else, and after 2020 several routine series in the *Anuario* simply stopped appearing. Against this, the exile literature has its own inflations, some of them very large and very durable — the twenty thousand dead attributed to Batista's last years, a figure that originated in a newspaper and has never been supported by anything resembling a count, is the best-known example, and it is repeated to this day by people who would never accept a Cuban government figure on the same evidence. Where a contested number appears in this book it appears as a range with the disagreement named, and where the honest answer is that nobody knows, the text says nobody knows.

Three practical conventions follow.

Figures asserted from a secondary source that has not been traced to a primary one are marked in the manuscript and listed in `verify.md` in the book's source directory. The mark prints nothing, so it does not clutter the page; it exists so that the list of unverified claims is mechanically extractable rather than a matter of memory.

Where a range is given, the endpoints are the two most defensible published estimates and not the extremes of what has been claimed. A figure that has been claimed only by a party with an interest in it, and never independently, is described that way rather than averaged into a range as though it were evidence.

Where the book states a causal claim — that the embargo cost Cuba this much, that the 2021 monetary reform caused that much

inflation — it states what would have to be true for the claim to hold, because on this subject almost every causal claim is contested by someone whose objection is not frivolous. The two standing questions, how much of Cuba's poverty is the embargo and how much is the model, and how much of Cuba's social achievement is socialism and how much is what any determined state can buy with a large enough external subsidy, are not settled in this book. They are, however, kept visible, because Part III's design has to work under either answer.

Contents

Preface	iii
A note on sources and numbers	vii
I 1959–1990	1
1 The inheritance	3
2 Three years, 1959–1962	11
3 The social state and its price	19
4 The Soviet economy	27
5 Abroad	35
II 1990–2026	41
6 The Special Period	43
7 Reform, and its reversal	51
8 The Venezuelan decade	59
9 Raúl’s decade	65

10 The crisis	71
11 The balance sheet	83
 III The prospect	 91
12 Premises	93
13 The polity	101
14 The social contract	113
15 Money	127
16 Sun	137
17 Visitors	147
18 The second track	155
19 Life sciences	165
20 Land and food	173
21 Capital and property	181
22 Integrity	191
23 Nature	199
24 The sequence	205
Bibliography	220

Part I

1959–1990

Chapter 1

The inheritance

1.1 A country that was not poor, and was not developed

On the last day of 1958 Cuba had about 6.6 million people, a fifth of the population of Spain on an island slightly larger than Portugal, and it was by most aggregate measures one of the four or five richest countries in Latin America. It had more television sets per head than any country in the region and among the highest counts in the world; it had one of the densest railway networks in the hemisphere, inherited from sugar; it had roughly one physician for every thousand inhabitants, a ratio comparable with much of Western Europe at the time; it had a literate majority, 76 per cent of the population over ten by the 1953 census; and it had a life expectancy at birth somewhere around 62 years (Pérez, 2015), higher than every mainland Latin American country except Argentina and Uruguay.

Every one of those numbers is a national average, and every one of them is concealing the thing that actually mattered, which is that Cuba in 1958 was not one country but two economies bolted together at the port of Havana.

The urban economy, and above all the capital, was genuinely modern. Havana had department stores, an air-conditioned bus

system, a stock exchange, medical specialists, a professional press of a dozen daily papers, a university with a real research tradition, and a middle class of professionals, shopkeepers and unionised industrial workers whose standard of living would not have been out of place in southern Europe. Cuban organised labour was strong — the sugar and transport unions in particular — and had extracted job protections and wage floors that were unusual in the region and that would, a decade later, become one of the constraints the revolution had to negotiate with rather than around.

The rural economy was something else. The most-quoted single source on it is a survey of a thousand rural wage-worker families conducted in 1956–57 by the Agrupación Católica Universitaria, a lay Catholic organisation with no sympathy for what came after, which is why the figures survived scrutiny. It found that among rural workers 4 per cent ate meat with any regularity, 1 per cent ate fish, 11 per cent drank milk; that 43 per cent could not read; that about 60 per cent lived in dwellings with earth floors and thatch roofs; that under 10 per cent had electricity and about 2 per cent running water (Pérez-Stable, 2012); and that the median family's annual income was a fraction of the urban wage. The survey was published in a Catholic magazine in 1957 and its findings were not seriously contested at the time. Approximately 43 per cent of Cubans lived in the countryside.

So the honest description of pre-revolutionary Cuba is neither the plundered colony of the official histories nor the prosperous republic of the exile memoirs. It is a middle-income country with a first-world capital, a third-world interior, and a distribution so wide that the national average describes almost nobody.

1.2 Sugar, and the shape it gave everything

The reason the two economies existed is sugar, and the reason sugar dominated is a policy instrument in Washington called the quota.

Sugar was roughly four-fifths of Cuban export earnings and had been for half a century. Under the reciprocal trade arrangements that

governed Cuban–American commerce, the United States bought a guaranteed tonnage at a price above the world market and in return Cuban tariffs favoured American manufactures. The arrangement was, in the short run, extremely good for Cuba: it converted a volatile commodity into a stable annuity and it underwrote the consumption of the urban middle class. In the long run it did three things that compounded.

It suppressed industry. Why build a Cuban shoe factory when the tariff schedule made American shoes cheap and the quota made sugar dollars certain? Cuban manufacturing outside sugar processing, tobacco and a handful of consumer goods was thin, and what existed was largely assembly.

It froze the land. Sugar wants large flat holdings and cheap seasonal labour, and it got both. By the 1946 agricultural census something like 8 per cent of farms held about 70 per cent of the farmland, and a substantial share of the cane land — the standard estimates run from a quarter to two-fifths of milling capacity — was owned by American companies. Much of the best land was held out of production entirely as a reserve against a good year, in a country that imported a third of its food.

And it made unemployment structural. The cane harvest, the *zafra*, ran roughly January to May. The rest of the year was the *tiempo muerto*, the dead season, in which several hundred thousand cane workers had no work at all. Annual average unemployment in the mid-1950s ran around 16 per cent and the seasonal swing was far wider than that. A country cannot build a stable politics on an economy that lays off a fifth of its workforce every June.

To this was added the ownership question, which mattered less economically than it did politically. American capital owned most of the electric and telephone utilities, most of the nickel, a large share of the banking and the refineries, and the milling capacity noted above. Whether this was extraction or investment is a genuinely arguable question — the utilities were real utilities and the mills were real mills. What is not arguable is that it made the commanding heights of the Cuban economy answerable to decisions taken in another country, and that this was intolerable to a generation of Cubans who had grown up on the nationalism of the 1933 revolution and

the constitution of 1940.

1.3 The republic that could not hold

Cuba's 1940 constitution was, on paper, among the most advanced documents of its era anywhere: it guaranteed the eight-hour day, paid holidays, the right to strike, equal pay for women, free compulsory education, and it prohibited latifundia and set limits on foreign landholding. Very little of the social programme was ever implemented by enabling legislation, and the latifundia clause was ignored outright. This gap between a superb constitution and a government that did not execute it is the central fact of Cuban republican politics, and it is why the constitution of 1940 became the banner of every subsequent opposition, including Fidel Castro's.

The governments of Ramón Grau San Martín (1944–48) and Carlos Prío Socarrás (1948–52) were elected, were not dictatorships, and were spectacularly corrupt. Public money disappeared on a scale that was openly discussed in the press and never prosecuted. Political violence between armed factions, the *grupos de acción*, was routine in Havana. The Ortodoxo party of Eduardo Chibás built a mass following on a single-issue platform of public honesty — his radio slogan was *vergüenza contra dinero*, shame against money — and when Chibás shot himself on air in August 1951 the country lost the one figure who might have channelled the anti-corruption current into an electoral outcome. Castro was an Ortodoxo candidate for congress in the election that was never held.

On 10 March 1952, three months before that election, Fulgencio Batista, who had run Cuba from behind and then in front between 1933 and 1944 and who was polling third, took power in a bloodless dawn coup. This is the hinge. The coup destroyed the constitutional route, and it destroyed it against a president who was corrupt but elected and a constitution the country was proud of. Every insurrectionary argument made in Cuba between 1952 and 1959 begins there, and the argument was correct: after March 1952 there was no legal way to remove the government.

Batista's second period was a dictatorship, and it got worse as it

went on. Congress was suspended, then restored as a rubber stamp through a rigged 1954 election that the opposition boycotted; the press was censored in waves; and after the guerrilla insurgency began the police and army responded with torture, disappearance and public killings, most notoriously by the Havana police under Esteban Ventura and by the army in Oriente. The number of people killed by the regime between 1952 and 1958 is genuinely unknown. The figure of 20,000, which entered circulation via a Havana newspaper in January 1959 and was then repeated for sixty years in official Cuban texts, has never been supported by any count and is not believed by serious historians in any camp; careful estimates run from about two thousand to about five thousand, including combatants (Thomas, 1971). That is a smaller number and an entirely sufficient one.

Meanwhile Havana became the offshore pleasure district of the United States. It had had casinos before; under Batista, and under an explicit 1955 law offering gaming licences and tax holidays to hotel builders, American organised crime — Meyer Lansky and Santo Trafficante most prominently — became a structural part of the tourist economy, with the regime taking its share. This is where the tourism chapter of Part III has to start, because the memory of that Havana is still doing political work in Cuba seventy years later, and any proposal to make tourism the country's anchor industry has to answer it.

1.4 Race, which the republic did not solve

The 1953 census recorded roughly 27 per cent of Cubans as black or mixed-race by self-identification, a figure most demographers regard as an undercount. Slavery had ended in 1886, later than anywhere in the Americas except Brazil, and within living memory of people alive in 1958. The republic had abolished legal racial distinction and had produced a genuinely multiracial army, police and political class — Batista himself was of mixed race and was blackballed from Havana's white social clubs while running the country.

What it had not abolished was the private colour bar: the beaches,

clubs, hotels and neighbourhoods that were white by practice; the hiring in banking, commerce and the professions that was white by preference; the shape of the job ladder in which Afro-Cubans were concentrated in cane cutting, construction, domestic service and the informal economy. An attempt to legislate against discrimination in employment in the late 1940s failed (de la Fuente, 2001). When the revolution desegregated the beaches and the clubs in 1959 it was addressing something real, it did so quickly, and it earned a loyalty among Afro-Cubans that has outlasted almost every other source of the government's support — a fact that Part III has to take seriously, because the constituency most exposed to a badly designed transition is the same one.

1.5 The insurrection

The armed opposition to Batista was not one organisation and Castro's was not initially the largest.

It began on 26 July 1953 with an attack by about 130 young Ortodoxos on the Moncada barracks in Santiago, which failed completely; roughly half the attackers were killed, most after capture. Castro's trial defence, later published as *La historia me absolverá*, laid out a programme — restore the 1940 constitution, agrarian reform, profit-sharing, education, nationalise the utilities — that was radical-nationalist and not communist, and that Cubans could and did read as a continuation of Chibás. He was sentenced, amnestied in 1955 in a gesture of confidence by a regime that did not expect what followed, and went to Mexico.

The *Granma* landed 82 men in Oriente on 2 December 1956; a dozen or so reached the Sierra Maestra. From there the July 26th Movement fought a small guerrilla war — never more than a few hundred fighters in the mountains — while a much larger urban underground, the *llano*, ran sabotage, finance, propaganda and supply in the cities under Frank País until his killing by police in July 1957 (Sweig, 2002). A separate organisation, the Directorio Revolucionario of the university students, nearly killed Batista in an assault on the Presidential Palace in March 1957 and lost most

of its leadership doing it. The old Communist party, the Partido Socialista Popular, opposed the armed struggle until late and joined it only in 1958, a fact that would matter enormously in the 1960s.

Two things decided the war, and neither was military in the ordinary sense. The first was that the general strike the movement called in April 1958 failed, which pushed the centre of gravity decisively from the cities to the Sierra and from the *llano*'s leadership to Castro's. The second was that in March 1958 the United States, under congressional pressure over the regime's atrocities, embargoed arms sales to Batista. The army was not defeated in the field so much as it stopped being an army: the summer 1958 offensive into the Sierra collapsed, whole units surrendered or changed sides, and when Che Guevara's column took Santa Clara at the end of December the road to Havana was open. Batista flew to the Dominican Republic in the small hours of 1 January 1959.

1.6 What was actually inherited

It is worth stating plainly, because both the later histories distort it.

The ledger: the inheritance, 1 January 1959

Assets. A middle-income economy with functioning infrastructure, a literate urban majority, a real professional class, strong unions, a dense rail and road network, a substantial sugar industry with modern mills, mineral resources, and a tourism plant that already worked. A constitution with mass legitimacy that everyone claimed to be restoring. Low external debt.

Liabilities. A monoculture export economy dependent on a single purchaser's political goodwill; structural seasonal unemployment near a fifth of the workforce; a rural half of the country at Haitian levels of nutrition, literacy and housing; concentrated landownership with much of the best land idle; foreign ownership of the utilities, mines and much of the milling; an unsolved colour bar; a political culture in which corruption was assumed and the constitutional route had just been closed by force; and no institution whatever with the legitimacy to arbitrate what came next — the army was Batista's, the congress

was a fiction, the courts were compromised, and the parties had been irrelevant since 1952.

That last liability is the one that decided the shape of everything after. In January 1959 the only body in Cuba with unambiguous legitimacy was the guerrilla movement, and within it, one man. There was no counterweight, and there had been no counterweight since 10 March 1952. A great deal of what follows in this book — the speed of the radicalisation, the absence of any internal brake on it, the ease with which a plural revolutionary coalition became a single party — is downstream of the fact that Batista's coup had already destroyed the institutions that might have slowed it down. The revolution did not seize a constitutional order. It inherited a vacuum where one had been.

This is the first lesson the book will carry into Part III, and it is not an abstract one: *the sequence in which institutions are rebuilt matters more than the speed*, and a transition that produces one legitimate actor and no others will end where 1959 ended, whatever its intentions. Cuba has now run that experiment twice, in 1952 and in 1959, from opposite directions.

Chapter 2

Three years, 1959–1962

2.1 The question of the turn

The single most argued question in modern Cuban history is whether Fidel Castro was a communist in January 1959 who concealed it, or a radical nationalist whom American reaction drove into the Soviet camp. Sixty years of scholarship has not settled it and this book will not either, but the shape of the evidence is worth stating, because Part III's treatment of the United States relationship depends on which lesson one draws.

Against the concealment thesis: the July 26th Movement's published programme was nationalist and constitutionalist; the Communist party opposed the insurrection almost to the end and was distrusted by the guerrillas for it; the first cabinet was liberal, headed by the judge Manuel Urrutia as president and the lawyer José Miró Cardona as prime minister, with figures like Felipe Pazos — an orthodox economist — at the central bank; and Castro spent April 1959 in the United States assuring audiences he was not a communist, at a moment when saying so had no obvious tactical value he could not have obtained more cheaply.

For it: Raúl Castro and Che Guevara were Marxists well before 1959 and neither was a marginal figure; the parallel structures that displaced the formal cabinet were in place within months; the pace

at which the liberal ministers were removed — Urrutia gone by July 1959, Pazos by November, Huber Matos arrested in October for objecting to communist influence in the army and sentenced to twenty years — was too fast to be improvised; and Castro himself said in 1961 that he had been a Marxist-Leninist all along, though he said many things in 1961.

The most defensible reading is that neither pure version survives the record (Farber, 2006). The revolution's leadership contained committed Marxists and committed liberals, the leader was neither and was above all a nationalist with an extraordinary instinct for where power lay, and the sequence of 1959–60 was a genuine interaction in which each side's move made the other's next move cheaper. The agrarian reform threatened American holdings; the American response threatened the reform; the Soviet alternative made the response survivable; taking it required the internal alliance that produced the party. By the end of 1960 there was no path back for anybody, and the question of what was in Castro's head in January 1959 had stopped mattering.

What matters for Part III is the second-order lesson: Cuba's sovereignty problem was never solved, only re-addressed. The country moved from dependence on one great power to dependence on another, and in 1990 discovered what that had cost. A design for 2026 that does not think hard about the structure of external dependence — who buys, who lends, who supplies the fuel — will reproduce the same vulnerability with different flags.

2.2 1959: the year of the two governments

Formally, Cuba in 1959 had a cabinet, a president and a restored 1940 constitution as amended by a Fundamental Law. Actually it had the cabinet and it had the Rebel Army, and where they disagreed the Rebel Army won. Castro took the prime ministership in February. Urrutia was forced out in July after criticising communist influence, on television, by Castro, in an act of political theatre that established the method for the next decade.

Two things happened in 1959 that fixed everything after.

2.2.1 The tribunals

Batista's police and army had tortured and killed, largely with impunity, and in January 1959 there was overwhelming popular demand for retribution and no functioning court system with the legitimacy to deliver it. The revolutionary government tried several hundred people before summary military tribunals and executed them, most in the first four months. The number is disputed: figures around 550 for 1959 are commonly cited and the Cuba Archive project's compilations run considerably higher for the decade as a whole, while the government has never published a count. The trials were public, were attended by crowds, and were in many individual cases against people who had demonstrably committed the acts charged. They were also summary, appealed to no independent authority, and in the case of the acquitted Batista airmen retried to conviction on Castro's public insistence — which destroyed, in the first weeks, the principle that a verdict was final.

Both facts are true and they should be held together. A revolutionary government facing genuine mass demand for justice against a state that had tortured people is in a real dilemma, and doing nothing was not available. What the tribunals established, however, was that in the new Cuba the law was an instrument of the executive and not a check on it, and that precedent was never subsequently reversed. The Huber Matos trial in October used the same machinery against a revolutionary commander whose offence was political disagreement, ten months later.

2.2.2 The agrarian reform

The First Agrarian Reform Law of 17 May 1959 capped individual holdings at 30 *caballerías*, about 402 hectares, with exceptions to about 1,340 hectares for high-productivity land; expropriated the excess with compensation in twenty-year bonds at 4.5 per cent, valued at the owners' own declared tax assessments; banned foreign landownership; and distributed land to tenants and squatters in

inalienable parcels of about 27 hectares. It created INRA, the National Institute of Agrarian Reform, which within two years was a parallel government running credit, distribution, housing and much of the economy.

The reform was, by Latin American standards of the period, not extreme — the cap was generous, compensation was offered, and comparable reforms were being urged on the region by the United States itself two years later under the Alliance for Progress. What made it a rupture was the valuation and the currency. American owners had for decades declared their land at a fraction of its market value to reduce tax; the law took them at their word, and offered payment in Cuban bonds no one believed would be honoured. Washington demanded prompt, adequate and effective compensation in cash. Havana declined. That disagreement, more than any speech, is where the two countries separated.

2.3 1960: the mechanics of the break

The year runs as a sequence of moves in which each side's action made the other's next one nearly automatic.

February: Anastas Mikoyan brings a Soviet trade delegation to Havana and signs an agreement to buy Cuban sugar and supply crude oil on credit. May: diplomatic relations with Moscow restored. June: the American- and British-owned refineries in Cuba — Esso, Texaco, Shell — refuse, on instruction, to refine Soviet crude; Cuba seizes them. July: the United States cancels the balance of the Cuban sugar quota. Also July: the Soviet Union announces it will buy the sugar. August: Cuba nationalises the American utilities, telephone company, sugar mills and refineries. October: the United States imposes an embargo on all exports to Cuba except food and medicine. Also October: Cuba nationalises essentially the remainder of large enterprise — banks, some 380 large domestic firms, and, under the Urban Reform Law, all rental housing, which was converted into ownership for the sitting tenants at their existing rent paid over five to twenty years.

That last measure deserves more attention than it usually gets,

because it is still shaping Cuba. The Urban Reform Law made a majority of Cuban households owners of their dwellings overnight and abolished the residential rental market for six decades. It was massively popular; it is the single most durable redistribution the revolution carried out; and it produced a housing stock in which nobody had any incentive or legal mechanism to invest in maintenance, and which has been falling down ever since. Havana loses buildings to collapse every year. Chapter 21 has to deal with the consequence, which is that Cuba has near-universal home ownership and a catastrophically decayed housing stock, and that the two facts are the same fact.

January 1961: Washington breaks diplomatic relations. February 1962: the embargo is extended to virtually all trade, under the Trading with the Enemy Act, where it remains.

2.4 The exodus begins

Between 1959 and 1962 roughly 250,000 Cubans left, out of 6.6 million (Eckstein, 2003). They were disproportionately the professional and business class: the expropriated, and then their doctors, lawyers, engineers and managers. Around half of Cuba's physicians left in the first years — roughly three thousand of six thousand — which is the origin of the revolution's medical policy, since the state's response was to build medical schools at a rate no comparable country attempted (Feinsilver, 1993).

Two features of this first wave matter later. It was legal and largely orderly, by air to Miami, and it was understood by those leaving as temporary — most expected to return within months, which is why so much property was left in the care of relatives, a fact that will make restitution claims in Chapter 21 unusually complicated. And it included Operation Peter Pan, in which some 14,000 unaccompanied children were flown to the United States between 1960 and 1962 by a Catholic-organised programme, on the strength of a rumour that the state intended to take custody of children. The rumour was false. The programme was real, many of the families were separated for years, and the episode poisoned the

emotional register of the Cuban–American question in a way that has never fully cleared.

2.5 Playa Girón and October 1962

On 17 April 1961 about 1,400 Cuban exiles, trained, armed and transported by the CIA, landed at the Bay of Pigs. The operation had been planned under Eisenhower and inherited by Kennedy, who cancelled the second air strike; the air cover failed, the expected internal rising did not occur — the security services had rounded up tens of thousands of suspected sympathisers in the preceding days — and the brigade was defeated in seventy-two hours. Around 1,200 were captured and later ransomed for food and medicine.

The invasion was the most consequential single American decision of the period, and its consequence was the opposite of its intent. It converted the Cuban government's security paranoia into a demonstrated fact, it justified permanent mobilisation, it discredited the internal opposition by association with a foreign landing, and it gave Castro the occasion — on 16 April, at the funeral of the victims of the preliminary air raid — to declare the socialist character of the revolution for the first time. It also made Cuba's request for Soviet military guarantees reasonable rather than paranoid.

Eighteen months later, in October 1962, American reconnaissance found Soviet medium-range ballistic missiles under construction in Cuba. The thirteen days that followed were the closest the world has come to nuclear war. The resolution — Soviet withdrawal in exchange for an American undertaking not to invade, and a quiet withdrawal of American missiles from Turkey — was negotiated between Moscow and Washington over Havana's head and against Castro's furious objection; he had wanted the missiles to stay and had, at the peak of the crisis, urged Khrushchev to consider a first strike if the United States invaded.

Two legacies. The non-invasion understanding is, in practice, what has kept Cuba's government alive for sixty years, and it was obtained by the Soviet Union rather than by Cuba. And the experience of being traded over taught the Cuban leadership a lesson about

great-power patronage that it then declined to act on for another twenty-eight years.

2.6 Consolidation at home

The same three years built the domestic machinery.

The literacy campaign of 1961 sent about 100,000 volunteers, most of them teenagers, into the countryside for eight months. It reported teaching some 707,000 adults to read and reducing illiteracy from around 23 per cent to under 4. The standard applied was elementary and the figure has been challenged at the margins, but the campaign's reality is not seriously in doubt — it is visible in later cohort data — and it remains the most impressive thing the revolution did in its first decade. It was also, and simultaneously, an instrument of political integration: the brigadistas carried a manual whose reading exercises were about the revolution, and the campaign put a hundred thousand urban adolescents under state discipline in the countryside for a year.

In June 1961 private schools were nationalised, ending Catholic education. The same year saw the founding of the Committees for the Defence of the Revolution, a block-by-block network that combined vaccination drives, blood donation and civil defence with neighbourhood surveillance — one organisation doing both, which is precisely why it worked and precisely why it was feared. Independent press ceased to exist over 1960–61 through a mixture of expropriation, printers' union takeovers and emigration. In March 1962 rationing was introduced, the *libreta*, intended as a temporary emergency measure; it is still in force in 2026.

By 1962 the revolutionary organisations had been merged into the ORI, and its first crisis — the removal of the old communist Aníbal Escalante in March 1962 for staffing the apparatus with his own people, denounced by Castro as “sectarianism” — established who would be running the party. It was not the communists.

The ledger: 1959–1962

Achieved. Land redistributed to over 100,000 tenant families; rents abolished and urban dwellings transferred to occupants; illiteracy effectively eliminated in a single year; racial segregation of public and semi-public space ended by decree and enforced; the rural half of the country brought inside the state's reach for the first time; foreign ownership of the commanding heights ended; and an invasion defeated.

Cost. Summary executions in the hundreds, with no appeal and no published count; the courts subordinated to the executive within weeks and never restored; the independent press extinguished within two years; every political organisation other than the governing one dissolved; a quarter of a million people, including half the physicians, gone; private education abolished; and the beginning of a security apparatus whose block-level reach was without precedent in the hemisphere. Rationing introduced as a temporary measure sixty-four years ago.

Chapter 3

The social state and its price

3.1 What was built

Between 1961 and 1975 the Cuban state constructed the thing it is still known for, and it did so fast enough that the international development literature has been arguing about it ever since. The argument is worth having, but it should not be allowed to obscure the outcomes, which are not in dispute and which were achieved on an income per head that never rose above the middle of the Latin American range.

3.1.1 Health

The starting point was bad and got worse before it got better: roughly half the country's physicians emigrated between 1959 and 1963, leaving about three thousand doctors for six and a half million people, most of them in Havana. The response was structural rather than remedial. Medical education was expanded past the point of surplus and kept there; a rural medical service was made compulsory for new graduates; and the delivery model was rebuilt around the

policlínico, a neighbourhood polyclinic responsible for a defined population, with prevention, vaccination, maternal care and chronic-disease follow-up as its main business rather than as an afterthought to curative medicine. The family-doctor programme that put a physician and nurse on a city block came later, in 1984, but it was the same logic pushed one level down.

The measurable results were large. Infant mortality, around 37 to 40 deaths per thousand live births in the late 1950s by the incomplete registration of the time, was under 20 by the mid-1970s and reached single digits in the 1980s. Cuba eliminated polio in 1962, the first country in the Americas to do so, through a mass campaign run by the block committees; measles, diphtheria, tetanus and rubella followed. Life expectancy rose from around 62 years to over 73 by the mid-1980s, converging on the United States figure and passing every country in the region except Costa Rica (Oficina Nacional de Estadística e Información, 2024a; Feinsilver, 1993).

There are three honest qualifications, and none of them cancels the achievement. Cuba's health indicators were already the best or nearly the best in Latin America in 1958, so the revolution began from a high base and the gains are partly a continuation of a pre-existing trend — though the convergence of rural with urban outcomes, which is the part the pre-existing trend was not delivering, is entirely post-1959. Infant mortality is a statistic sensitive to definitional practice and to a health system with strong incentives to make it low; Cuba's aggressive antenatal management, including very high rates of intervention in pregnancies flagged as risky, contributes to the number in ways that a purely epidemiological reading misses. And the achievement was bought with a share of national output that a country of Cuba's income could only sustain because someone else was paying for the oil.

3.1.2 Education

Universal primary schooling was effectively achieved in the 1960s and universal lower-secondary in the 1970s. University enrolment multiplied by roughly an order of magnitude over three decades. Boarding secondary schools in the countryside, the *escuelas en el*

campo, combined study with agricultural work on a scale no other country attempted, moving hundreds of thousands of adolescents into a state-run residential system for part of every year. Teacher training was industrialised. Textbooks and materials were free.

The quality was real and independently confirmed later. When UNESCO's regional laboratory tested third and fourth graders across thirteen Latin American countries in 1997, Cuban children scored around 350 on a scale whose regional mean was 250 and whose standard deviation was 50 — roughly two standard deviations above the region, in both language and mathematics, with the nearest comparators, Brazil and Chile, near 260. The finding that matters more than the average is distributional: pupils in Cuba's lowest-income schools outperformed most upper-middle-class pupils in the rest of the region (Carnoy et al., 2007). That is the single most important number in this book's argument for Part III, because it is the one that cannot be explained away by subsidy: it says the country built a mass education system that compressed rather than reproduced advantage, and that the system still worked after seven years of the Special Period.

The cost side of education is that it was also, explicitly and by design, a political formation system. The curriculum carried the ideology, access to higher education was mediated by political record as well as academic performance, and entire fields — sociology, economics as practised elsewhere, much of philosophy — were closed or reduced to catechism for long periods. Cuba produced excellent engineers, physicians and mathematicians, and almost no economists, lawyers or social scientists trained to think about how a market or a court works. Part III has to notice this: the country's human capital is superb in exactly the areas a technical economy needs and thin in exactly the areas an institutional transition needs.

3.1.3 Women, and the Family Code

Female labour-force participation rose from around 13 per cent in 1958 to over 30 per cent by the late 1970s and kept rising, driven by free childcare, free education, and a Federation of Cuban Women that was simultaneously a mass organisation, a service provider and

a transmission belt for the party. The Family Code of 1975 required, in law, that husbands share housework and childcare equally — a provision that was unenforceable and that everyone knew was unenforceable, and that was nonetheless a declaration no other country in the hemisphere had made. Abortion was legal, free and safe from 1965, decades before most of Latin America. Cuban women's educational attainment overtook men's and has stayed ahead.

What did not change was the composition of power. The Political Bureau remained overwhelmingly male for decades, and the domestic division of labour changed far less than the statute said it should.

3.1.4 Race, declared solved

The desegregation of 1959–61 was fast, real and enforced, and it is one of the revolution's genuine moral achievements. What followed was a decision with long consequences: in 1962 the leadership declared racial discrimination abolished and the racial question closed, and independent Afro-Cuban organisations — including the societies and clubs that had been the black civil-society tradition since the nineteenth century — were dissolved along with everything else. Race disappeared from the census categories of analysis and from permissible public discussion for thirty years.

The effect was that a real gain was locked in and further progress stopped being measurable. When race returned to Cuban public discussion in the 1990s it did so because the dollar economy had made the disparity impossible to ignore: remittances flowed overwhelmingly to white Cubans, because the 1960s emigration had been overwhelmingly white, and the tourist sector hired by *buena presencia*. Cuba entered its market opening with a racial distribution of foreign-currency access that it had spent thirty years declining to measure (de la Fuente, 2001). Chapter 12 treats this as a hard constraint rather than a footnote.

3.2 What it cost

The chapters of this book that describe Cuban social policy and the chapters that describe Cuban repression are usually written by different people. They describe the same state, in the same years, doing both things with the same apparatus, and often through the same organisation — the block committee that ran the vaccination drive also kept the file on the neighbours.

3.2.1 The camps

Between 1965 and 1968 the Military Units to Aid Production, the UMAP, held young men in agricultural labour camps in Camagüey under military discipline. The categories swept up were religious believers — Jehovah’s Witnesses, Seventh-Day Adventists, practising Catholics, including seminarians — homosexuals, and young men classed as antisocial, which in practice meant long-haired, unemployed, or attached to the wrong music. Something like twenty-five to thirty-five thousand people passed through them. There were deaths, beatings and suicides. The camps were closed in 1968 after protests from Cuban intellectuals and international pressure, and the closure was an admission, though the state has never made a full one; Castro said in 2010 that the persecution of homosexuals in those years was a “great injustice” and that he bore responsibility for it.

3.2.2 The five grey years

The cultural opening of the early 1960s — which was real, and which produced the film institute, the Casa de las Américas, a serious literary magazine culture and a poster art still worth looking at — ended in 1971. The trigger was the case of the poet Heberto Padilla, arrested in March 1971 after winning a national prize for a book the writers’ union had criticised, held for a month, and released after a public self-criticism in which he denounced himself and named his friends. The performance was so evidently coerced that it broke the revolution’s standing with the European and Latin American

left; Sartre, Vargas Llosa, Calvino, Paz, Sontag and dozens of others signed against it, and several of them never came back.

The First National Congress on Education and Culture, a month later, made the line official: art was a weapon of the revolution, homosexuals were unfit for work with young people or for representing Cuba abroad, and religious belief was incompatible with certain professions. What followed, 1971 to about 1976, is known in Cuba as the *quinquenio gris*, the five grey years, and its mechanism was not mass arrest but *parametración* — the setting of parameters. Writers, actors, academics and teachers were assessed against ideological and personal criteria and, if they failed, removed from their professions and reassigned to manual work. Careers were ended quietly, by administrative letter, and many were never restored.

3.2.3 Prison, exit, and the closing of the door

The 1960s saw a rural insurgency in the Escambray mountains, fought from 1960 to 1965 and known officially as the struggle against bandits. It was defeated, and the defeat included the forced relocation of thousands of peasant families from the Escambray to new settlements in Pinar del Río, at the far end of the island. The political prison population in the 1960s ran into the tens of thousands at its peak; the *plantados*, prisoners who refused the re-education programme and its uniform, served terms of twenty and thirty years. No count has ever been published.

Emigration continued and its terms hardened. When Castro opened the port of Camarioca in 1965, Cuban Americans came by boat; the resulting negotiation produced the Freedom Flights, twice-daily charter flights from Varadero to Miami that carried roughly 300,000 people between 1965 and 1973. The regime attached a price to using them. Applying to leave made a person a *gusano*, a worm; the applicant lost their job, was often assigned to agricultural labour for the years-long wait, had their household inventoried and confiscated on departure, and men of military age could not go at all. Families were split by these rules for decades. The emotional economy of the Cuban diaspora — its intensity, its unwillingness to compromise, its transmission across three generations — is not

fully explicable without this policy, and Part III's proposals about the diaspora have to reckon with a grievance that is specific and documented rather than diffuse.

3.2.4 The Revolutionary Offensive of 1968

In March 1968 the state nationalised what remained of small private business: roughly 55,000 to 58,000 establishments, essentially all of them — corner shops, bars, food stalls, repair shops, laundries, street vendors, carpenters. The stated reason was that these were parasitic and a breeding ground for the black market. The effect was to abolish the Cuban service economy overnight, to hand its functions to a state distribution system that had no capacity to perform them, and to eliminate in one law the entire class of person who knew how to run a small enterprise.

This is the most consequential purely economic decision of the period, and it is under-discussed relative to the agrarian reform because it did not involve foreigners. Cuba spent the following fifty-eight years periodically re-legalising, in narrow and reversible increments, categories of activity that were legal and functioning in 1967: self-employment in 1978 and again in 1993, private restaurants in 1993, private taxis, private rooms for rent, agricultural markets in 1980 and again in 1994, and finally, in 2021, small and medium private enterprises with legal personality. Every one of those steps was presented as a bold reform. Every one of them was a partial restoration of something the state had abolished on a single day in March 1968.

The ledger: the social state, 1961–1975

Achieved. Universal free health care with the rural–urban gap closed, infant mortality more than halved, polio and measles eliminated, life expectancy up more than ten years. Universal free education to secondary level with measured attainment two standard deviations above the regional mean and the strongest results at the bottom of the social distribution. Female labour-force participation more than doubled, free childcare, legal abortion, an equality statute ahead of

the hemisphere. Public space desegregated by enforcement rather than declaration. Social security extended to the whole population. *Cost.* Forced-labour camps for believers, homosexuals and nonconformists, 1965–68. A cultural purge, 1971–76, conducted by administrative reassignment rather than arrest and therefore leaving no record and no redress. Political imprisonment on a scale never disclosed. Exit made punishable, with confiscation and years of forced labour as the price of an application, splitting families across three generations. All private enterprise below the corner-shop level abolished in 1968, destroying the country's service economy and its stock of entrepreneurial skill. Race declared solved and then made unmeasurable for thirty years.

Chapter 4

The Soviet economy

4.1 The argument about incentives, and how it was settled

Between 1963 and 1965 the Cuban leadership held, in public and in print, one of the more serious debates in the history of socialist economics. On one side Che Guevara, then Minister of Industries, argued for a budgetary finance system in which enterprises were departments of a single national firm with no accounts of their own, transfers between them were not sales, and workers were motivated by moral incentives — emulation, voluntary labour, the formation of a new kind of person — rather than by pay. On the other, Carlos Rafael Rodríguez and the economists of the old communist party argued for self-finance on the Soviet model of the period, in which enterprises kept books, faced prices, and paid bonuses.

The debate was conducted with more intellectual honesty than the subject usually receives, and Guevara's side of it is worth reading; his objection to using capitalist categories to build socialism is coherent, and his warning that material incentives would reproduce the mentality they were meant to serve turned out to be, in the Soviet case, prescient. He was also wrong about everything operational. A system without prices cannot tell which of two processes is cheaper,

a system without enterprise accounts cannot locate a loss, and a system in which effort is rewarded by a certificate produces, after the first enthusiasm, exactly the effort a certificate is worth.

The debate was not settled by argument. Guevara left the government in 1965; his approach was applied maximally through the late 1960s, culminating in the 1968 abolition of small business and in an economy where money had almost stopped functioning — rents, utilities, local telephone calls, burials, sporting events and school meals were free, wages were paid but there was progressively less to buy with them, and absenteeism reached levels the state could not control. Then, after 1970, the whole apparatus was replaced wholesale with the Soviet system Rodríguez had argued for.

4.2 Ten million tons

The pivot was a sugar harvest.

Having failed at rapid industrialisation in the early 1960s — an attempt to substitute imports across the board, which foundered on the absence of inputs, spares and anyone who had run a factory — Cuba returned to sugar as the source of accumulation, and in 1968 the leadership committed publicly to a harvest of ten million tons in 1970, a figure roughly a third above anything the country had ever produced. The commitment was made in Castro's characteristic register: as a matter of national honour, with a date, in a speech.

The mobilisation to meet it was total. Factories closed and their workers went to the cane. Students, soldiers, office staff, foreign volunteers went to the cane. Transport, fuel, spares and management attention were pulled out of every other sector. The harvest ran into July, wrecking the following season's planting. The country produced 8.5 million tons, the largest harvest in Cuban history before or since, and the effort broke the rest of the economy: output fell across manufacturing, transport and construction, the cane fields were damaged for years, and the mills had been run past maintenance.

On 26 July 1970 Castro made a speech admitting the failure in unusually direct terms, taking personal responsibility, and offering to resign. The crowd refused, as everyone in the square and on the

platform knew it would. What followed the speech, however, was a real change of system, and it went in the opposite direction to the one the speech implied. Cuba stopped trying to invent its own socialism and adopted the Soviet one.

4.3 Institutionalisation, 1972–1976

Cuba joined the Council for Mutual Economic Assistance, Comecon, in July 1972. It adopted five-year plans, a central planning board, and in 1976 the SDPE, the System of Direction and Planning of the Economy, a close copy of the Soviet enterprise model with cost accounting, profit indicators and bonuses. The First Congress of the Communist Party met in 1975, sixteen years after the revolution. A new constitution was approved by referendum in 1976, replacing the 1940 constitution as amended; it defined the party as the leading force of society and state, and created *Poder Popular*, a system of municipal, provincial and national assemblies elected from the base, with direct secret competitive election at municipal level and indirect selection above it. The provinces were redrawn from six into fourteen. A civil and criminal code followed, and a family code.

The period 1971–1985 was, in ordinary economic terms, the best Cuba has had. Growth was solid, consumption improved, the housing programme delivered through the *microbrigadas* — work teams seconded from their enterprises to build apartment blocks, with the flats allocated back to the enterprise — and the social indicators of Chapter 3 reached their peaks. Cubans who are nostalgic for the revolution are, almost always, nostalgic for these fourteen years specifically.

4.4 What was actually paying for it

The arithmetic underneath is the single most important thing to understand about Cuba, because everything in Part II is the consequence of its disappearance.

Cuba's trade with the Soviet Union was not trade at world prices. Under the long-term sugar protocols the USSR bought Cuban sugar

at a price fixed well above the world market — in some years of the early 1980s at four or five times it — and supplied crude oil at a price fixed well below it. Nickel and citrus had similar arrangements. Payment ran through a bilateral clearing account in transferable rubles rather than in convertible currency, so that Cuba's persistent deficit accumulated as a debt that never had to be settled in anything the Soviet Union could spend elsewhere.

On top of the price terms there was a second mechanism, and it is the one that is usually left out. Cuba was allocated more Soviet oil than it consumed, and was permitted to re-export the surplus for hard currency. By the mid-1980s this re-export was one of the largest single sources of Cuba's convertible-currency earnings — in some years the largest, ahead of sugar. Cuba was, in effect, being handed a petroleum rent by a country that had oil and wanted an ally in the Caribbean.

The magnitudes are not small and they are not in serious dispute. In 1987 the Soviet Union paid Cuba the equivalent of about 42 US cents a pound for sugar against a world price of under 7 cents — more than six times the market — and a CIA assessment earlier in the decade found a comparable ratio on a volume of 3.2 million tonnes. Converting the whole relationship into dollars is harder, because it depends on what rate one applies to transferable rubles, which is the problem the note on sources described. But the standard estimates put the combined inflow at around 23 per cent of Cuban output over 1985–88, reaching something over four billion dollars in a single year (Mesa-Lago, 2000). Roughly a quarter of national income, from one patron, sustained for over a decade. By 1990 the accumulated debt to the USSR stood at something over twenty billion transferable rubles, a figure Russia would later claim in dollars and eventually, in 2014, write off ninety per cent of.

Alongside it Cuba had borrowed in hard currency from Western banks and governments in the 1970s, when sugar prices were briefly high and Cuban credit was good. When they fell, it could not pay: Cuba suspended payments on its convertible-currency debt in 1986 and has been in default to the Paris Club essentially ever since. That default is thirty-nine years old in 2026 and it is the reason Cuba cannot borrow, which is the reason it cannot invest, which is

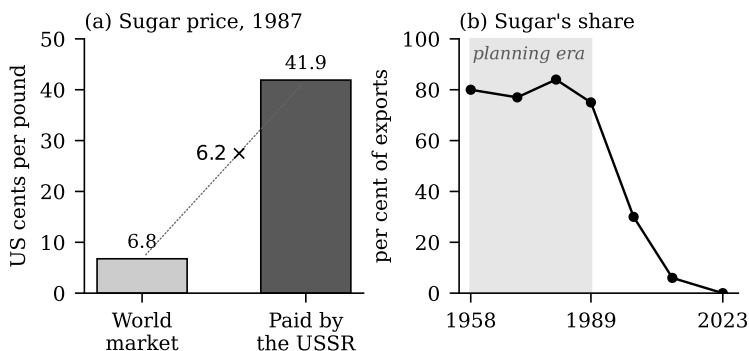


Figure 4.1: The subsidy, and what it bought. (a) The mechanism: in 1987 the Soviet Union paid more than six times the world price for Cuban sugar, and supplied oil below it. (b) The result: sugar's share of Cuban merchandise exports barely moved across thirty years of planning, the highest investment rate in Latin America and a Comecon membership organised around international specialisation — and then collapsed when the buyer went. A guaranteed price removes the only signal that tells a producer to become good at something else. Panel (a) is two measured values rather than a series, because the year-by-year price data is not reliably public and drawing a line through two points would invent the shape; the post-1989 values in (b) are estimates, flagged in `verify.md`. Generated by `figs/comecon.py`.

a constraint Chapter 15 has to solve before anything else in Part III is financeable.

4.5 The failure that mattered: no diversification

Here is the fact that indicts the whole period, and Fig. 4.1(b) is its picture. In 1958, sugar was about four fifths of Cuban exports.

In 1989, after thirty years of planning, three decades of the highest investment rates in Latin America, a Comecon membership explicitly organised around international specialisation, and the best-educated workforce in the region, sugar was still about three quarters of Cuban exports (Oficina Nacional de Estadística e Información, 2024a; Mesa-Lago, 2000).

Everything else was small. Nickel, from the Moa and Nicaro deposits, was second. Tobacco, citrus, fish and rum followed. The industrial base built in the 1970s was assembly and import substitution for the domestic market, ran on Soviet inputs, and could not export outside Comecon because its costs and quality were set by a market that did not test either. A pharmaceutical and biotechnology capability was started in the early 1980s — of which much more in Chapter 19, because it is the one exception and the one asset — but it was a decade from products in 1989.

The reason for the failure is not mysterious and it is not the embargo. A protected market at guaranteed prices removes the only signal that tells a producer to become better at something else. Cuba had, for eighteen years, the sweetest terms of trade any small country has ever been offered, and it used them to remain a sugar economy with a very good school system. That is the central economic lesson of Part I, and Chapter 12 states it as a constraint on the design: *a guaranteed buyer at a political price is not a market, and an economy built on one does not develop*. It applied to the American sugar quota before 1959 and it applied to the Soviet protocols after, and Chapter 8 will show it applying a third time to Venezuelan oil.

4.6 Rectification, 1986

In 1980, under pressure from food shortages, Cuba opened free farmers' markets where cooperatives and private farmers could sell surplus above their state quota at whatever price the market bore. They worked: food appeared. They also produced visible inequality — intermediaries, some farmers doing very well, prices the poorest could not pay — and complaints about profiteering.

In 1986 Castro launched the Rectification of Errors and Negative

Tendencies. It closed the farmers' markets. It banned private house construction for sale and much private artisanal work. It curtailed the enterprise bonus system of the SDPE, denounced material incentives, revived voluntary labour and the contingent brigades, and rehabilitated Guevara's economic thinking explicitly by name.

The timing is the point. In 1986 the Soviet Union was beginning perestroika. Cuba responded to its patron's turn toward markets by turning away from them, and it did so eighteen months before the terms of trade that made the choice affordable began to disappear. Rectification is the clearest single instance of the pattern this book will name in Chapter 11 and that recurs through Part II: *Cuba liberalises under duress and re-centralises as soon as the pressure lifts, and the re-centralisation always arrives just before it is least affordable.*

The ledger: the Soviet economy, 1963–1989

Achieved. Fourteen years of solid growth and the highest living standards the revolution has produced. A universal social state financed to completion. Mass housing through the microbrigades. Full employment, in the sense that everyone had a job. Institutional forms — a constitution, a party congress, elected municipal assemblies, codified law — where before there had been improvisation by speech. An education system whose output was, and remains, the country's best asset. The founding of a biotechnology capability that would outlive everything else here.

Cost. Thirty years of accumulation that left the export basket where it was in 1958, three quarters sugar, because guaranteed prices removed the reason to change it. An economy structurally dependent on transfers running between a fifth and a third of national income, from a single patron, with no plan for their ending. A hard-currency default in 1986 that has locked the country out of credit markets ever since. An enterprise system that could not calculate, and a workforce whose pay had been decoupled from what it did. And a decision in 1986 to reverse the one reform that had visibly worked, taken at the last moment when reversing it was still survivable.

Chapter 5

Abroad

5.1 Why a small poor island had a foreign policy

For thirty years Cuba conducted a foreign policy several orders of magnitude larger than its size, and any account of the period that treats this as peripheral has missed how much of the country's resources, self-understanding and reputation went into it. At its peak Cuba had tens of thousands of soldiers in Africa, doctors and teachers in dozens of countries, and a claim on the attention of the non-aligned world that no state of six million people has had before or since.

Three motives ran together and it is not always possible to separate them. There was conviction: the leadership genuinely believed in a global revolution and acted on that belief at real cost. There was security: a country under permanent threat from a superpower ninety miles away buys allies and buys standing, and standing is a form of deterrence. And there was the patron: after 1972 Cuban military deployments in Africa were logistically impossible without Soviet transport and finance, and they served Soviet purposes, which is not the same as saying Moscow ordered them — the Angolan intervention in particular appears to have been

decided in Havana and presented to Moscow afterwards.

5.2 The guerrilla decade

Through the 1960s Cuba supported armed insurgency across Latin America, in Venezuela, Guatemala, Colombia, Peru, Bolivia and Nicaragua, and in Africa. Almost all of it failed. The doctrine, elaborated by Régis Debray from Guevara's writings, held that a small armed nucleus in the countryside could create the conditions for revolution rather than waiting for them, which was a misreading of Cuba's own case: the Sierra Maestra had succeeded alongside an urban movement, a collapsing army and an isolated dictator, none of which the doctrine required.

Guevara left Cuba in 1965 to apply it, first in the Congo, where the campaign was a failure he documented himself with striking honesty, and then in Bolivia, where he was captured and killed by the Bolivian army with CIA assistance in October 1967. His death converted him into the most reproduced image of the twentieth century and removed from Cuban politics the one figure whose prestige was independent of Castro's.

The policy also cost Cuba with Moscow, which was pursuing peaceful coexistence and had working relations with several of the governments Cuba was trying to overthrow, and with the Latin American communist parties, which Havana was denouncing as passive. The 1966 Tricontinental Conference in Havana was the high point of Cuban third-worldism and close to the low point of Cuban-Soviet relations.

The alignment was purchased back in 1968. In August, when Warsaw Pact tanks entered Czechoslovakia, Castro went on television and endorsed the invasion. He did so in a speech that conceded almost everything — that the invasion violated legality, that the socialist camp had no right to do it under the rules it claimed — and endorsed it anyway on the grounds that the alternative was Czechoslovakia's drift to the West. The speech is the most revealing document of Cuban foreign policy in the period, and the trade it embodied was explicit within months: Soviet oil deliveries were

increased, the economic integration of the 1970s followed, and Cuban support for guerrilla movements in Latin America was wound down.

5.3 Africa

Cuba's military engagement in Africa began in Algeria in 1963 and became large in Angola in 1975.

When Portugal's empire collapsed, Angola's independence was contested by three movements. The Soviet-aligned MPLA held Luanda; the FNLA advanced from Zaire; and UNITA was backed from the south by South Africa, which invaded in October 1975 with an armoured column heading for the capital. Cuba's response, Operation Carlota, began in November (Gleijeses, 2002): troops flown in on Cuban and later Soviet aircraft, stopping the South African advance short of Luanda. Within months there were 36,000 Cuban soldiers in Angola. They stayed for sixteen years. Over the whole period something like 300,000 Cuban military personnel and 50,000 civilians served there (Gleijeses, 2013).

The war's decisive phase came in 1987–88 around Cuito Cuanavale in the south-east, where a Cuban and Angolan force stopped a South African offensive, and then in a Cuban armoured push toward the Namibian border that changed the military balance in the south-west. The battle's outcome is disputed in detail — both sides claim it — but its consequence is not: the quadripartite negotiations that followed produced South African withdrawal from Angola, Namibian independence in 1990, and Cuban withdrawal, and they took place because Pretoria concluded it could no longer hold the line militarily. Nelson Mandela, released in 1990, went to Havana in 1991 and said so directly. Whatever else is in the ledger, Cuba's contribution to ending South African control of Namibia and to the pressure that ended apartheid is real, was paid for in Cuban lives, and was not done for money.

Ethiopia is the other half of the record and it is much worse. In 1977–78 Cuba sent some 16,000 troops to defend Ethiopia against a Somali invasion of the Ogaden — defensible on its own terms, since Somalia was the aggressor and Cuba refused to join the subsequent

offensive into Somalia proper. But the government being defended was the Derg of Mengistu Haile Mariam, which was in the middle of the Red Terror, and Cuba's presence continued through it and through the famine and the forced resettlements of the 1980s. Cuba had also been the ally of Somalia and of the Eritrean independence movement before switching sides, and the Eritreans, who had trained in Cuba, were now on the other end of the guns. Whatever internationalism meant in Angola, in Ethiopia it meant supporting a mass-murdering client of the same patron.

The official Cuban count of dead across all internationalist missions, published in 1991, was 2,289. The families were told what was necessary and the coffins came home together, in a single ceremony, sixteen years of them at once, in December 1989.

5.3.1 The doctors

Running in parallel, and outliving all of it, was medical internationalism. The first Cuban medical brigade went to Algeria in 1963, fifty-six people, sent by a country that had just lost half its own physicians (Kirk, 2015). By the 1980s Cuban doctors were working across Africa, Central America and the Caribbean, mostly without charge, in places where no other country would send anyone. The Latin American School of Medicine, founded later, trained foreign students free.

This is the origin of the reputation that Chapter 19 will argue is Cuba's most valuable intangible asset — and, from the 2000s, of the country's largest export earner, which is a different thing and carries its own problems of pay, consent and conditions that Part II will take up honestly.

5.4 Mariel

In April 1980 a bus crashed through the fence of the Peruvian embassy in Havana. The guards were withdrawn after a dispute over asylum, and within seventy-two hours around 10,800 people were inside the embassy grounds. The scale of it — ten thousand people

in a garden, in a country whose government had said for twenty years that only a handful of *gusanos* wanted to leave — was a shock from which the official narrative never quite recovered.

Castro's response was to open the port of Mariel and invite Cuban Americans to come and collect their relatives by boat. Between April and October 1980 about 125,000 people crossed to Florida. The Cuban state used the opening to empty some of its prisons and psychiatric hospitals into the boats — the number is disputed but the practice is not, and it was deliberate, both to rid itself of a burden and to discredit the emigration as criminal. It also organised *actos de repudio*, mob demonstrations outside the homes of those leaving, in which neighbours and co-workers were assembled to shout, throw eggs and sometimes beat people who had announced their departure. Cubans were made to brutalise their neighbours as a condition of their own standing.

The consequences ran both ways. In the United States, Mariel arrived in an election year, overwhelmed Florida's capacity, was fixed in the public mind by the criminal contingent, and did lasting damage to the standing of Cuban immigrants generally. In Cuba, the memory of the repudiation mobs is one of the sharpest wounds the revolution inflicted on its own society — not because of what the state did, but because of what it made ordinary people do to each other, which is harder to forgive and harder to forget. Any reconciliation process proposed in Part III has to have an answer for the *acto de repudio*, and the answer is not that it was long ago.

5.5 The end of the decade

Two events closed the period.

In June and July 1989 General Arnaldo Ochoa, a hero of the Angolan campaign and one of the three or four most prominent officers in the country, was arrested, tried and executed along with three others, and a group of senior Interior Ministry officials including the de la Guardia brothers was convicted, on charges of drug trafficking and corruption. The Interior Ministry was purged and placed under military control. That the trafficking occurred is not

seriously disputed. Whether the trial was about trafficking is: Ochoa was popular, was associated with the officers who had seen how the Soviet Union was changing, and had reportedly been critical of the leadership. The proceedings were televised, the confessions had the cadence of 1971, and no independent inquiry has ever been possible. What is certain is that the affair removed the only figure in Cuba with a military reputation comparable to the Castros', at the exact moment the Soviet bloc was dissolving.

In April 1989 Mikhail Gorbachev came to Havana. The visit was correct and cold. Cuba had already banned two Soviet magazines, *Moscow News* and *Sputnik*, for spreading perestroika; Castro had already made the speech that gave the period its slogan, *socialismo o muerte*, socialism or death. The Cuban position was that reform in the Soviet Union was a Soviet matter and that Cuba would not follow. It did not follow. Nineteen months after Gorbachev's visit the trade that had underwritten the country for eighteen years began to stop arriving.

The ledger: abroad, 1963–1990

Achieved. A decisive military contribution to the defeat of South African expansion in southern Africa, to Namibian independence, and to the pressure that ended apartheid — acknowledged by Mandela, paid for in Cuban lives, and not paid for by anyone. Medical brigades in dozens of countries from 1963, the seed of Cuba's most durable asset and its eventual largest export. A standing in the non-aligned world out of all proportion to the country's size, which functioned as a real component of its security.

Cost. A decade of exported insurgency that failed everywhere it was tried and killed the people sent to try it. The endorsement of the invasion of Czechoslovakia, made in full knowledge of what it was, as the price of oil. A long deployment in support of the Derg through the Red Terror and the famine. Sixteen years of war fought by conscripts whose families were told little and whose dead came home in one ceremony. And at home, in 1980, the deliberate mobilisation of ordinary Cubans to assault their departing neighbours — the one thing in this ledger for which no external necessity can be pleaded.

Part II

1990–2026

Chapter 6

The Special Period

6.1 The stop

What happened to Cuba between 1990 and 1993 has no close parallel among countries not at war.

The Soviet bloc had taken roughly 85 per cent of Cuban trade. Comecon dissolved in June 1991; the Soviet Union followed in December. The preferential sugar protocols ended, the concessional oil ended, the spare parts ended, the credit lines ended, and the machinery of an entire economy — built over eighteen years to Soviet standards, with Soviet inputs, for a Soviet market — discovered simultaneously that none of its three sides existed any more.

Cuban imports fell from about \$8.1 billion in 1989 to about \$2.2 billion in 1993, a fall of roughly three quarters in four years. Oil imports fell from around 13 million tonnes to around 6. The official Cuban figure for the fall in output is 34.8 per cent between 1989 and 1993, probably the largest four-year decline in twentieth-century Cuba. It is also the *lowest* figure in circulation: Cuban media reported 48 per cent at the time, and independent reconstructions have gone as high as 50 per cent (Mesa-Lago and Pérez-López, 2005). The spread turns on the exchange-rate question of the note on sources rather than on any dispute about what physically happened, and this book uses the official figure because it is the most conservative

one available and the argument does not need a larger number. For comparison, the United States lost about 27 per cent of output in the Great Depression, over four years, with its trading partners still in place.

Fidel Castro named it in 1990: the Special Period in Time of Peace, a phrase lifted from the civil-defence contingency plans drawn up in the 1980s for the event of a war and a total blockade. The name was accurate. The plans had been written for a siege, and a siege is what arrived, except that nobody was besieging.

6.2 What that was like

The abstraction is useless without the texture, and the texture is what Cubans who lived through it still measure everything against.

Electricity went first, because the thermal plants ran on imported fuel. Blackouts of twelve to sixteen hours a day were normal by 1993, scheduled by neighbourhood and then not scheduled at all. Without electricity there is no water, because Cuban water is pumped; no refrigeration, in a climate where food spoils in hours; no lifts in the apartment blocks the microbrigades had built; and no light to read by in a country whose principal remaining asset was that everyone could read.

Transport went next. Buses stopped for want of fuel and tyres. The state imported over a million Chinese bicycles and built factories to assemble more, and Havana — a city with no bicycle tradition at all — became a city of bicycles, with the injuries and the exhaustion that implies in that heat. Later came the *camellos*, articulated trailers hauled by lorry tractors, carrying several hundred standing passengers each. Oxen returned to fields that had been mechanised for twenty years, and the state trained them and the men to drive them, because there was no diesel.

Then food. The ration book had always been a floor and the state economy a ceiling; now the ceiling collapsed onto the floor and the floor did not hold. Average daily energy intake fell from roughly 2,900 kilocalories in 1989 to somewhere around 1,860 in 1993, and protein intake fell by close to half. Cuban adults lost, on

average, between five and ten kilograms. There was no famine in the technical sense — no mass starvation deaths, no localised collapse — but there was a country-wide, sustained, involuntary reduction in food intake to below what an adult needs.

The clinical proof arrived in 1992. An epidemic of optic and peripheral neuropathy spread across the island, blurring and in some cases destroying vision. It ran from 1991 to 1994 and peaked over 1992–93, and the Cuban health ministry's own count was 50,862 cases in a population of 10.8 million (Centers for Disease Control and Prevention, 1994). The cause was nutritional: B-vitamin deficiency, aggravated by tobacco and rough alcohol, in a population that had been eating too little for two years. It was halted by handing out vitamin tablets to the entire population, which is simultaneously an indictment of the situation and a demonstration of what a health system that reaches every block can do in a month. There were no deaths from it, and fewer than one in a thousand of those affected were left with lasting damage — which is the same story as the infant mortality figures below, and should be read the same way.

Everything else followed from those four. Soap and detergent disappeared, and with them a measurable rise in diarrhoeal and skin disease. Tuberculosis incidence roughly tripled from a very low base. Prostitution reappeared, aimed at the tourists who were the only source of hard currency, and with it a word, *jineterismo*, that had not been needed for thirty years. The sugar industry — the thing the whole economy had been built around — could not get fertiliser, fuel or spares, and the harvest fell from 8.1 million tonnes in 1989–90 to 4.3 million by 1993 and 3.3 million by 1994–95. It never recovered, in any year, ever again.

6.3 What did not collapse

Here the book has to be careful, because this is the single strongest piece of evidence in favour of the Cuban model and it is routinely omitted by people who find it inconvenient.

Infant mortality did not rise. It fell. In 1989 it was about 11.1 per thousand live births; through the worst years of the Special Period it

continued to decline, and by 1994 it was under 10. Life expectancy did not fall. Immunisation coverage did not fall. No school closed. The universities stayed open, on bicycles and candlelight. There was no epidemic of the classic mortality signatures of economic collapse — no surge in infant deaths, no collapse of male life expectancy of the kind Russia recorded in exactly the same years from exactly the same shock.

This is not a small thing and it should not be explained away. Russia in 1990–94 lost a comparable share of output and lost roughly six years of male life expectancy with it. Cuba lost more output and lost none. The difference is that Cuba’s health and education systems were not financed out of the collapsing sector; they were a prior claim on whatever remained, defended politically, and staffed by people who kept coming to work in a country with no transport. When this book argues in Chapter 14 that the welfare state is the one thing worth keeping, this is the evidence: a universal system with deep territorial reach is a shock absorber of a kind that nothing else in a poor country’s toolkit can substitute for.

Two qualifications, both real. The systems degraded even where the outcome indicators held — hospitals without anaesthetics, without X-ray film, without functioning sterilisers, and doctors who improvised or did without. And there was one genuinely perverse benefit, documented afterwards and worth stating because it shows how strange the episode was: the forced calorie restriction and the forced physical activity produced a measurable fall in diabetes and cardiovascular mortality across the 1990s, which reversed when weight was regained after 1995 (Franco et al., 2013). A country lost weight because it was hungry and its heart-disease deaths fell. That is not a policy recommendation. It is a reminder that this period does not fit any simple story.

6.4 Option Zero, and what came out of it

The contingency planning went further than what was implemented. Option Zero was the plan for the total loss of imported fuel: no

electricity generation to speak of, transport by animal, food produced and distributed at neighbourhood level through communal kitchens, the country reorganised into self-supplying cells. It was prepared and it was not needed, and that it was prepared says something about how the leadership assessed 1992.

What was implemented, and what outlasted the crisis, was the agricultural improvisation. Without fertiliser, pesticide or fuel, Cuban agriculture was forced into low-input methods: biological pest control, composting, animal traction, and above all the *organopónicos*, raised-bed vegetable gardens on vacant urban lots, which by the 2000s supplied a substantial share of the fresh vegetables consumed in Havana. This became internationally famous and is frequently over-claimed. What it demonstrably did was put vegetables into cities. What it did not do was feed the country: Cuba's food import dependence went up across the whole period, not down, and the case for low-input agriculture has to be made on its merits in Chapter 20 rather than on a 1990s legend.

6.5 August 1994

By the summer of 1994 the currency had gone. The peso, official at one to the dollar, traded on the street at 120 to 150. A state wage bought almost nothing. People were leaving on anything that floated, and the state's practice of intercepting boats had produced a series of violent incidents.

On 13 July 1994 a group of about seventy people hijacked an old tugboat, the *13 de Marzo*, and headed for Florida. It was pursued by state-operated vessels which rammed it and directed high-pressure water hoses at the people on deck, including at children being held up. The tug sank seven miles offshore. Thirty-seven people died, ten of them children. The government's account was that the sinking was accidental and the fault of the hijackers. No independent investigation was permitted. The Inter-American Commission on Human Rights found the state responsible (Inter-American Commission on Human Rights, 1996). Whatever one concludes about the Special Period generally, this happened, and it happened to people trying

to leave a country that would not let them.

On 5 August 1994, after rumours that a boat would be allowed to depart, thousands of people gathered on the Havana Malecón, threw stones, broke windows, and shouted against the government. It was the first mass anti-government demonstration in thirty-five years. It was put down within hours, partly by police and partly by mobilised construction workers with sticks, and Castro himself came to the seafront during it. The *maleconazo* lasted an afternoon and changed the government's calculation permanently.

Within days Castro announced that anyone who wanted to leave by sea could go unhindered. Over the following five weeks something like 30,000 to 35,000 people put to sea on rafts and inner tubes. An unknown number drowned. The United States, unable to absorb them, changed a thirty-year-old policy: rafters intercepted at sea were taken to Guantánamo rather than to Florida, where some 30,000 were held in camps for months. The negotiations that followed produced the migration accords of September 1994 and May 1995, under which the United States undertook to admit a minimum of 20,000 Cubans a year through legal channels and to return those interdicted at sea, while those who reached land could stay — the policy that became known as wet-foot, dry-foot, and that would shape Cuban emigration for the next twenty-two years.

The migration valve is the mechanism this book will keep returning to. Twice before — Camarioca in 1965, Mariel in 1980 — and now a third time, internal pressure was released by opening the door, and each time it worked. It is one of the two instruments that have kept the Cuban political system stable through crises that would otherwise have broken it, the other being the welfare floor. Chapter 10 will show the same valve opening a fourth time, on a scale that finally threatened the thing it was protecting.

The ledger: the Special Period, 1990–1994

Achieved. A loss of over a third of national output absorbed without famine, without epidemic mortality, without the collapse of the state, and without the fall in life expectancy that the identical shock produced in Russia. Infant mortality continued to *fall* throughout. Every

school stayed open. The health system degraded severely and held its outcomes, which is the strongest evidence in this book that a universal system with territorial reach is worth the money it costs. Urban agriculture, born of desperation, became a durable and genuinely useful institution.

Cost. A country of readers reduced to roughly 1,860 kilocalories a day, losing five to ten kilograms a head, with fifty thousand cases of nutritional blindness as the clinical receipt. The sugar industry destroyed and never recovered. A decade of consumption gone. Prostitution and a dollar economy restored as facts of Cuban life. Thirty-seven people, ten of them children, drowned off Havana in an incident the state has never independently investigated. And the moment, on an August afternoon on the Malecón, when the social contract of 1959 stopped being believed by the people it had been written for.

Chapter 7

Reform, and its reversal

7.1 Reform under duress

Between July 1993 and the end of 1995 the Cuban government did more to liberalise its economy than in the previous thirty-four years combined. It did so reluctantly, described the measures throughout as temporary concessions forced by circumstance, and withdrew as many of them as it could as soon as it was able. That sequence is the subject of this chapter, and it is the clearest single instance of the pattern this book has been assembling.

The measures, in order:

July 1993. Possession of hard currency was decriminalised. Until that month a Cuban caught with a dollar could be prosecuted; from that month Cubans could hold, spend and receive dollars legally. This was the pivotal decision, because it legalised remittances — and remittances from the diaspora that thirty years of policy had denounced as worms became, within a decade, one of the largest sources of foreign exchange in the country.

September 1993. Self-employment was re-legalised in 117 categories, later extended to around 157. The list was a list: activity was permitted by enumeration, not by right, and university-trained professions were excluded — a doctor could not practise privately, an engineer could not consult, but either could rent a room or drive

a taxi. This design decision, more than any other, determined what the Cuban private sector became.

September 1993. The state farms, which held most of the agricultural land, were broken into Basic Units of Cooperative Production, the UBPCs, holding land in usufruct with a nominal right to organise their own work. Roughly 40 per cent of agricultural land changed hands on paper. In practice the state retained the procurement monopoly, the input supply and the price, so what was devolved was the loss and not the decision. The UBPCs have underperformed the private and older-cooperative sector continuously ever since.

October 1994. Agricultural markets were reopened at free prices for produce above the compulsory state quota — the same measure that had been introduced in 1980 and abolished in 1986. Food appeared in Havana within weeks, at prices most people could not afford, which was also what had happened in 1980.

1994–95. The fiscal and monetary stabilisation, which is the part of this period that gets least attention and deserves the most. The budget deficit, which had reached something like 30 per cent of output in 1993, was cut to around 7 per cent in 1994 and to 2 per cent by 1996, through price rises, subsidy cuts, new taxes under a 1994 tax law, and the removal of currency by confiscatory means. Exchange houses were opened in 1995 at a market rate. The informal exchange rate, which had reached 120 to 150 pesos to the dollar in mid-1994, came back to around 20 to 25 by 1996 and stayed in that band for the next twenty-five years (Mesa-Lago and Pérez-López, 2005). A currency that had been destroyed was stabilised inside two years without external assistance. Chapter 15 takes this seriously as a precedent, because the same state did the opposite in 2021.

September 1995. Law 77 opened foreign investment, permitting up to full foreign ownership in principle, and creating the joint-venture form that most foreign capital in Cuba has used since. It kept the feature that has done most to deter investors: the foreign partner does not hire its own workers, but contracts them from a state employment agency, pays the agency in hard currency, and the agency pays the worker in pesos, keeping the difference. At an official rate of one to one, this meant a worker whose employer paid the state several hundred dollars a month received the peso equivalent

of perhaps twenty. Chapter 21 argues that this single arrangement has cost Cuba more foreign investment than any American statute.

The measures worked. 1994 was the trough; growth resumed in 1995 and ran through the second half of the decade. The economy that emerged, however, was not the one that had gone in.

7.2 The dual economy

Cuba came out of the Special Period with two currencies, two price levels, two labour markets and two populations.

The convertible peso, the CUC, was introduced in 1994 and pegged at one to the dollar. Alongside it the ordinary peso continued, at twenty-four or twenty-five to the CUC for households and — this is the part that mattered — at one to one for state enterprise accounting. Cuba thus ran a twenty-five-fold discrepancy between the rate its households faced and the rate its own firms used to value imports, exports, costs and profits, for twenty-six years. Every enterprise account in the country was fiction. No Cuban manager could tell whether their operation created or destroyed value, and neither could the ministry above them. This is not a technicality; it is the reason Cuban industry could not be reformed even by people who wanted to reform it, because nobody could identify which parts were worth keeping.

The social consequence was the inversion of the pyramid. A surgeon earned a state salary in pesos. A bellhop earned tips in convertible currency. A professor of literature drove a taxi in the afternoon; a research chemist rented her spare room. The valuation the society had spent thirty years teaching — that education was the route to standing — was contradicted daily and publicly by the exchange rate. Cuba's Gini coefficient, which had been around 0.24 in the mid-1980s and was among the lowest measured anywhere, was around 0.40 by the end of the 1990s (Mesa-Lago and Pérez-López, 2005).

That inequality had a colour. The emigration of 1959–62 had been overwhelmingly white; the remittances that flowed back thirty years later therefore flowed overwhelmingly to white households, by

margins that the available surveys put at four fifths and above. The tourist sector, the other route to hard currency, hired front-of-house staff by *buena presencia* — good appearance — which in practice meant what it has always meant. So the state that had abolished the colour bar in 1959 and then declared the racial question closed in 1962, and had thereby made it unmeasurable for thirty years, entered its market opening with foreign-currency access distributed along the exact line it had stopped looking at. Chapter 12 treats this as a hard constraint on any transition design, and not as a historical note.

Tourism was the visible face of it. Arrivals went from about 340,000 in 1990 to around 745,000 in 1995 and to about 1.8 million by 2000. The plant was built by joint ventures with Spanish, Canadian and Jamaican chains, mostly as all-inclusive beach resorts at Varadero and the northern cays, physically separated from the country around them — and, until 2008, legally separated too, since Cubans were not permitted in tourist hotels. A state that had made its founding claim on the abolition of a foreigners' Cuba had rebuilt one, and this time it was the government enforcing the door.

7.3 The reversal

By 1996 the emergency had passed, and the leadership's assessment of the reforms changed with it.

The signal came in March 1996, when Raúl Castro delivered a report to the Central Committee attacking the Centre for the Study of the Americas, a party research institute whose economists had been analysing the reforms and their possible extension. The institute was dismantled and its researchers dispersed. The message to Cuban social science was received and has never needed repeating.

What followed was a decade of quiet reversal. Self-employment licences, which had peaked at around 209,000 in early 1996, were squeezed by taxation, by inspection, and by the withdrawal of categories, falling to roughly 150,000 by 2003 and lower after (Ritter and Henken, 2015). New licences in most categories were suspended outright in 2003–04. In 2003 enterprises were required to hold and

transact their foreign currency through a single central account, ending the decentralised access that had made the joint-venture sector function. In November 2004 the United States dollar was withdrawn from circulation altogether and a 10 per cent surcharge imposed on converting dollars into CUC, which was presented as a response to American financial pressure and which also happened to tax every remittance.

None of this was a return to 1968; the markets stayed open, the tourism sector grew, the remittances kept coming. But the direction was unmistakable, and its logic was consistent: the reforms of 1993–95 had been concessions extracted by an emergency, and with the emergency over the concessions were withdrawn.

7.4 Politics in the same years

The reversal was not only economic, and the external environment did not help.

On 24 February 1996 Cuban air-force MiGs shot down two civilian aircraft flown by Brothers to the Rescue, an exile organisation that had been searching for rafters and dropping leaflets over Havana, killing four men. The location of the shootdown — inside or outside Cuban airspace — was disputed; that the aircraft were unarmed civilian light planes was not. The political consequence in Washington was immediate: the Cuban Liberty and Democratic Solidarity Act, Helms–Burton, which had been stalled, passed within weeks. It codified the embargo into statute, so that it can now be lifted only by Congress and not by a president; it extended sanctions to foreign companies trafficking in confiscated American property; and its Title III created a right for American nationals to sue such companies in American courts. Every president from Clinton onward suspended Title III at six-month intervals until 2019.

Helms–Burton matters to Part III for a reason beyond its direct effect. It converted American policy toward Cuba from an executive instrument, which can be traded, into a legislative one, which must be repealed. Any Cuban reform design that assumes the embargo will end when relations improve is assuming something the American

constitutional structure does not readily deliver, and Chapter 12 therefore treats the embargo as a variable that may not move.

Inside Cuba the same years produced Law 88 of 1999, which made collaboration with United States policy punishable by up to twenty years and which is sufficiently broadly drafted to cover most forms of independent journalism; and then, in 2002, the Varela Project. Oswaldo Payá and a network of Catholic lay activists used Article 88 of the 1976 constitution, which permitted citizens to propose legislation with 10,000 signatures, to submit 11,020 signatures asking for a referendum on free expression, free association, an amnesty and free elections. It was the most serious attempt to use the system's own rules against it in forty years. The response was a counter-mobilisation that collected some eight million signatures for a constitutional amendment declaring the socialist character of the system irrevocable, which the National Assembly duly passed in June 2002. The route Payá had found was closed by constitutional amendment.

In March 2003, in the week the United States invaded Iraq, Cuban state security arrested seventy-five independent journalists, librarians, trade unionists and opposition organisers, tried them within weeks, and sentenced them to terms of six to twenty-eight years. It is known as the Black Spring. In the same month three young men hijacked a Havana Bay passenger ferry in an attempt to reach Florida, took hostages, harmed nobody, surrendered when the boat ran out of fuel, and were tried and executed by firing squad within nine days. Cuba's standing with the European left, which had survived Padilla, largely did not survive this. The wives and mothers of the seventy-five began walking to mass in white every Sunday, and were still doing so, and being detained for it, twenty years later.

The ledger: reform and reversal, 1993–2004

Achieved. An economy pulled out of free fall in eighteen months. A destroyed currency stabilised without external assistance — deficit from about 30 per cent of output to 2 per cent, informal exchange rate from 150 to 25 — which is a genuine technocratic accomplishment

and the counter-example to Chapter 10. Dollars, self-employment, farmers' markets and foreign investment legalised. Tourism rebuilt from nothing to 1.8 million visitors. A private sector of a few hundred thousand people re-created from zero, and with it the first Cubans in a generation who did not depend on the state for their income.

Cost. A dual currency that made every enterprise account in the country fictitious for twenty-six years, and thereby made the economy unreformable by anyone, including those who wanted to reform it. Inequality up from among the lowest measured anywhere to Latin American levels in a decade, distributed along a racial line the state had spent thirty years declining to measure. Tourist enclaves Cubans were legally barred from entering. A reform programme withdrawn as soon as it was survivable to withdraw it, with the research institute that had analysed it dismantled as a warning. Seventy-five people imprisoned for terms up to twenty-eight years, and three shot, in a single month.

Chapter 8

The Venezuelan decade

8.1 The third patron

In October 2000 Fidel Castro and Hugo Chávez signed an Integral Cooperation Agreement in Caracas. Venezuela would supply Cuba with crude oil and refined products on concessional terms — initially some 53,000 barrels a day, rising over the decade to around 90,000 or 100,000 — and Cuba would pay, in substantial part, not in money but in the services of its people: doctors, nurses, dentists, teachers, sports trainers, agricultural technicians and military and intelligence advisers.

It is worth pausing on the structure of this deal, because it is the third time the same structure appears in this book. In 1934 the United States guaranteed a price for Cuban sugar above the world market and Cuba organised itself around it. From 1972 the Soviet Union guaranteed a price for Cuban sugar and supplied oil below the world market, and Cuba organised itself around that. From 2000 Venezuela supplied oil below the world market and bought Cuban services above their market value, and Cuba organised itself around that. Each arrangement was generous, each was political rather than commercial, each rescued the Cuban economy at the moment it was struck, and each removed the pressure that would otherwise have forced diversification. Chapter 4 stated the rule; this chapter is its

third confirmation.

8.2 Doctors as an export industry

The exchange grew far beyond oil. From 2003, under the Barrio Adentro programme, Cuba placed medical personnel in Venezuelan barrios and rural areas where Venezuelan doctors had never practised — at the peak something in the range of twenty to thirty thousand health workers, with total Cuban civilian personnel in Venezuela reaching perhaps forty thousand. Similar arrangements, smaller and mostly paid in cash, were signed with Brazil, Angola, Algeria, South Africa, Qatar and dozens of other countries, and after 2013 the *Mais Médicos* programme placed some 11,000 Cuban doctors in underserved regions of Brazil.

The economics of this transformed the country. By the early 2010s Cuba's exports of professional services were worth on the order of eight to ten billion dollars a year, of which health services were the large majority — against tourism at roughly two and a half billion and nickel at about one and a half (Oficina Nacional de Estadística e Información, 2024a; Mesa-Lago and Pérez-López, 2013). Cuba's largest export was no longer a commodity. It was its doctors. That is a genuinely remarkable outcome for a country of eleven million, it is the direct dividend of the investment described in Chapter 3, and it is the strongest existing evidence for the proposition in Chapter 18 that Cuba's educated population is an exportable asset.

The arrangement's other half has to be reported as plainly. The host government paid Cuba; Cuba paid the doctor. The doctor's share was typically a small fraction of the contract value — in the Brazilian programme, the most transparently documented, Cuba received roughly four times what it passed to the physician (Kirk, 2015). Doctors on mission were subject to rules that would not be accepted in most labour markets: in several destinations their passports were held by the mission chief; movement, association and relationships with locals were restricted; and a doctor who abandoned the mission was, until the migration reform of 2013, barred from returning to Cuba for eight years, which in practice

meant separation from children. The United States operated a Cuban Medical Professional Parole programme from 2006 to 2017 specifically to induce defections, and several thousand took it.

Both descriptions are true at once, and the honest summary is that Cuba built a real and valuable export industry and financed it partly by underpaying and over-controlling the people who produced it. Chapter 19 argues that the programme should be kept and the labour arrangement changed, and that this is not only the ethical answer but the revenue-maximising one, since the industry's binding constraint is now the supply of willing doctors.

8.3 What the money bought

The Venezuelan relationship coincided with, and paid for, the most politically confident period the Cuban government has had since the 1980s.

Abroad, it bought a bloc. ALBA was founded by Cuba and Venezuela in 2004 and grew to include Bolivia, Nicaragua, Ecuador and several Caribbean states. Petrocaribe, from 2005, extended Venezuelan concessional oil across the Caribbean with Cuban medical missions attached. The regional isolation that had been American policy since 1962 simply dissolved: the Organization of American States revoked Cuba's 1962 suspension in 2009, the Community of Latin American and Caribbean States was founded without the United States in 2011 and Cuba chaired it in 2013, and the European Union's 1996 Common Position, which had conditioned relations on democratisation, was steadily bypassed by member states and formally replaced in 2016.

At home it bought the Battle of Ideas. This began in late 1999 with the case of Elián González, a five-year-old boy who survived the sinking of his mother's boat and was held in Miami by relatives against his father's wishes. Cuba mobilised around his return for seven months, in daily marches and a permanent tribune built opposite the American mission, and won: the boy went home in June 2000. It was the last unambiguous political victory the Cuban government won on its own terms, and the mobilisation machinery it

built was then turned to domestic programmes — an army of young social workers, university courses delivered in every municipality, television as a teaching medium, and a great deal of capital spending decided by mass campaign rather than by plan.

One of those campaigns deserves its own paragraph, because Part III depends on it. By 2005 the Cuban grid was failing: the thermal plants were beyond their design lives, unplanned outages were routine, and a hurricane season could take the country dark. The Energy Revolution of 2006 attacked the problem from both ends at once. On the demand side the state replaced essentially every incandescent bulb in the country — some nine million — with compact fluorescents, distributed millions of efficient rice cookers and pressure cookers, and replaced the oldest refrigerators and air conditioners. On the supply side it installed around four thousand small distributed generator sets, roughly 1,300 megawatts in total, sited across the country rather than concentrated in a few large plants. Blackout hours fell sharply. The programme was expensive, the generators burned expensive fuel, and the strategy was later allowed to decay — but the diagnosis was right, and it was right in a way that matters now: Cuba's grid problem is best attacked by distributed capacity and demand reduction, not by rebuilding large central plants. Chapter 16 builds directly on this precedent, with the generators replaced by photovoltaics and storage.

8.4 Two shocks, and a credit history

In August, September and November of 2008 three hurricanes — Gustav, Ike and Paloma — crossed Cuba within ten weeks. Together they damaged or destroyed something over half a million homes and caused losses conventionally estimated at around ten billion dollars, on the order of a fifth of national output. It was the worst hurricane season in Cuban recorded history, and it arrived in the same quarter as the global financial crisis.

Cuba's response to the resulting liquidity squeeze did lasting damage. Beginning in late 2008 the state froze the bank accounts of foreign companies operating in Cuba — money those companies

had earned and were entitled to repatriate, amounting to something over a billion dollars — and simply stopped paying suppliers. Some balances were released over the following years; some were not. The measure solved a cash problem and created a reputation problem, and the reputation problem has proved much more durable. Foreign companies now price Cuban counterparty risk on the basis of what happened in 2009, and no statement of intent from Havana has changed that. Chapter 21 argues that this is a credit history, that credit histories are repaired by behaviour over years rather than by legislation, and that any investment strategy which does not begin by settling these arrears is proposing to build on a default.

8.5 The end of it

Hugo Chávez was diagnosed with cancer in 2011 and was treated in Havana. He died on 5 March 2013. Venezuela's own economy then entered the collapse that has defined it since, and the terms Cuba had enjoyed began to erode — shipments falling year by year through the mid-2010s, and falling faster after 2016.

By then Cuba had had thirteen years of subsidised oil and a services export boom, and had used them, as it had used the Soviet years, to postpone rather than to restructure. The sugar industry was not rebuilt. Domestic food production did not rise. The dual currency was not unified, although unification was announced as policy in 2011 and prepared for a decade. The energy system was not converted. What the country had instead was one more external rent, larger than it looked, and a decade in which the pressure to change had been removed.

The ledger: the Venezuelan decade, 2000–2013

Achieved. An export industry built out of the education system that displaced sugar, nickel and tourism combined, and that placed Cuban doctors in places no other country would staff — a real service really delivered, and the best evidence available for the argument of Part III. The end of Cuba's regional isolation, with the OAS suspension revoked

and CELAC founded and chaired. The Energy Revolution of 2006, a correct diagnosis and a partially correct cure, whose logic Chapter 16 adopts. Recovery from the worst hurricane season on record without mass casualties, which is what a civil-defence system with block-level reach is for.

Cost. The third guaranteed buyer at a political price in seventy years, taken for the third time, with the third identical result: no diversification, no restructuring, and a cliff at the end of it. An export industry financed by paying its workers a fraction of their contract value and, in several destinations, holding their passports. The freezing of foreign companies' accounts in 2009, which bought a year of liquidity at the price of a credit history that is still being paid for. And thirteen years in which every structural problem this book has named was identified, announced as policy, and not fixed.

Chapter 9

Raúl's decade

9.1 A different kind of governing

Fidel Castro fell ill on 31 July 2006 and never resumed office; Raúl Castro took over provisionally that day and formally in February 2008. The change in style was immediate and larger than the change in policy.

Fidel governed by speech: problems were named in public, solutions were announced as campaigns, and the campaign was the instrument. Raúl governed by meeting. He delegated, set deadlines, demanded that ministries answer for targets, and — most unusually — said in public that the system did not work. The Camagüey speech of 26 July 2007 is the text: he told an audience that Cuban wages were insufficient to live on, that structural and conceptual changes were needed, and, in the example everyone remembers, that a country that could not put a glass of milk on a child's table every day had a problem it should stop explaining and start solving. Coming from the second man of the revolution, in the anniversary speech, this was licence to discuss things that had not been discussable.

He also opened a consultation. In late 2010 the draft economic guidelines were circulated for discussion in workplaces and neighbourhoods; the government reported some 160,000 meetings and millions of comments, and around two thirds of the draft's paragraphs were

modified (Mesa-Lago and Pérez-López, 2013). Whatever one makes of the mechanism, this was the widest formal consultation the Cuban state had ever run, and Cubans engaged with it seriously because they believed something would come of it.

9.2 The removal of humiliations

The first measures were not economic reforms at all. They were the repeal of prohibitions whose only function had been to remind Cubans of their position.

In 2008 Cubans were permitted to buy mobile telephones, computers, DVD players and microwave ovens, and to stay in tourist hotels — ending, twelve years late, the situation in which a Cuban could not enter a building on Cuban soil that a foreigner could. In 2011 they were permitted to buy and sell their own houses and cars, which had been prohibited since 1960 and had produced an elaborate grey market of swaps, fictional marriages and bribes.

And in October 2012, effective January 2013, the exit permit was abolished. The *tarjeta blanca* — the white card, the permission-to-leave that had cost money, taken months, required a letter of invitation, and been refused at discretion — was gone, and a Cuban could travel on a passport like anyone else. Doctors and some other categories remained restricted in practice, and the eight-year bar on returning was reduced to two. But the principle changed: after fifty-four years, leaving Cuba temporarily stopped being a political act.

None of this cost the state anything. That is precisely the point, and Chapter 24 makes it an operating principle: the first phase of any reform programme should begin with the prohibitions that cost nothing to remove, because they are free, they are immediate, and they buy the credibility that the expensive measures require. Raúl Castro's most popular acts were all of this kind.

9.3 The guidelines, and why they stopped

The Sixth Party Congress of April 2011 — the first in fourteen years — adopted 313 Guidelines for Economic and Social Policy. The programme they described was coherent: shrink the state payroll, expand non-state employment, decentralise enterprise decisions, separate state administration from enterprise management, unify the currency, and move from universal subsidy toward targeted assistance. The announced first step was the transfer of half a million workers off the state payroll within six months, and eventually more than a million.

The instruments were real. Self-employment was expanded to 178 and then 201 categories and the number of licence-holders rose from about 157,000 in 2010 to over half a million by 2015 — roughly a fifth of the employed workforce (Ritter and Henken, 2015). Land was granted in usufruct under Decree-Law 259 of 2008, initially 13 hectares, and under Decree-Law 300 of 2012 up to 67, to anyone willing to farm it; something over 1.7 million hectares of idle land were handed out. Non-agricultural cooperatives were authorised on a pilot basis in 2013. The payroll reduction was never executed at anything like the announced speed, because there was nowhere for the workers to go.

The reforms stalled, and the reason they stalled is the most useful thing in this chapter for Part III, so it is worth stating precisely rather than attributing to a general reluctance.

A Cuban private business between 2011 and 2021 had no legal personality. It was a licensed individual, not a firm. It therefore could not sign a contract as an entity, could not be sued or sue, could not own assets separately from its owner, and could not, in most cases, be sold or inherited as a going concern.

It had no wholesale market. A restaurant bought its ingredients in the same retail shops as households, at retail prices, competing with households for the same scarce goods — which is both economically absurd and the source of the enduring popular complaint that the private sector caused the shortages.

It had no import rights. Everything had to be bought domestically or carried in personal luggage by *mulas*, travellers making repeated trips with suitcases, which is how a substantial share of Cuban private-sector inputs actually arrived for a decade.

It had no credit. Bank lending to the private sector was announced, hedged with conditions, and never materialised at scale.

It had no access to foreign exchange at any usable rate.

And it operated by enumeration, not by right: an activity was legal if it appeared on a list the state could shorten at any time, and the list was written in the vocabulary of trades — *barber*, *button-coverer*, *party clown* — with the professions excluded. A software developer had no category. A design studio had no category.

The result is that Cuba got the private sector its rules were designed to produce: several hundred thousand micro-operators in food, transport, lodging and personal services, and essentially nothing in manufacturing, technology or anything requiring an entity, a contract and a supply chain. Chapter 21 treats this diagnosis as its starting point, because almost every constraint in the list above is legal and internal, could be removed without a dollar of foreign investment, and was not removed for a decade.

9.4 The opening

On 17 December 2014 Raúl Castro and Barack Obama announced simultaneously, from Havana and Washington, that their countries would restore diplomatic relations. The immediate exchange was the release of the American contractor Alan Gross and of a Cuban-held intelligence asset, for the three remaining members of the Cuban Five; behind it lay eighteen months of secret talks brokered in Canada and by the Vatican (LeoGrande and Kornbluh, 2014).

Embassies reopened in July 2015. Obama visited Havana in March 2016 — the first sitting American president since Coolidge in 1928 — and gave a speech, broadcast live on Cuban television, in which he said the embargo would end and addressed Cubans directly about their own future. Regular flights resumed; cruise ships docked; Airbnb entered the market in 2015 and had tens of thousands of

Cuban listings within two years; American visitors rose from around 91,000 in 2014 to over 600,000 in 2017, on top of a much larger flow of Cuban Americans.

For the Cuban private sector this was the best eighteen months it has ever had. The demand went overwhelmingly to the non-state economy, because that is where Americans on people-to-people licences were required to spend: private restaurants, private rooms, private taxis, private guides. Money entered households rather than the state, capital equipment came in with the visiting relatives, and for the first time since 1959 there was a visible, growing, domestically-owned Cuban middle class making its living from foreigners without the state as intermediary.

That is also, precisely, why it was stopped.

9.5 The brake

The Seventh Party Congress of April 2016 was where the guidelines slowed. The tone of the documents shifted from expansion of the non-state sector to its regulation, and the phrase that circulated was the warning against the concentration of wealth and property. In August 2017 the state suspended the issue of new licences in a set of the most successful categories — private restaurants, room rentals, and others — pending a review that lasted more than a year and ended in a tighter regime: one licence per person, caps on the size of restaurants, and stricter inspection.

Two further things happened in the same short period. Fidel Castro died on 25 November 2016. And on 8 November 2016 Donald Trump was elected, and in June 2017 issued a policy memorandum that reversed a substantial part of the opening — restricting individual people-to-people travel, prohibiting transactions with a list of military-linked entities, and beginning the sequence of measures that Chapter 10 takes up.

It is tempting, and common, to attribute the closing entirely to Washington. The sequence does not support it. The licence freeze of August 2017 was a Cuban decision, taken about Cuban businesses, before the American measures had bitten. Havana and Washington

arrived at the same conclusion for opposite reasons: that the growth of an independent Cuban private sector was politically consequential. One of them wanted it and stopped subsidising it; the other did not want it and stopped permitting it.

Raúl Castro handed the presidency to Miguel Díaz-Canel in April 2018, remaining First Secretary of the party until 2021. He left a country with a larger private sector, a freer citizenry and a functioning relationship with the United States, and with every structural problem identified in 2011 — the dual currency, the state payroll, enterprise autonomy, food production — diagnosed, announced, and unresolved.

The ledger: Raúl's decade, 2006–2018

Achieved. The repeal of prohibitions that had no purpose but humiliation: hotels, phones, computers, the sale of houses and cars, and above all the exit permit, which after fifty-four years made leaving Cuba temporarily an ordinary act. A private sector taken from 150,000 to over half a million people, about a fifth of the workforce. Idle land distributed in usufruct on a large scale. A national consultation the population took seriously. Diplomatic relations with the United States restored, embassies reopened, and eighteen months in which American demand flowed directly into Cuban households.

Cost. A reform programme whose own instruments were designed so that it could not succeed: no legal personality, no wholesale market, no import rights, no credit, no foreign exchange, and legality by enumerated list. The result was several hundred thousand micro-operators and nothing above them, which is exactly what those rules produce. The one measure that would have made the economy legible — currency unification — was announced in 2011 and deferred for a decade, and would eventually be attempted at the worst possible moment. And the opening that worked best was braked from the inside, in August 2017, before Washington closed it from the outside.

Chapter 10

The crisis

10.1 A note on writing about the present

This chapter comes closest to the day it was written, and the closer it comes the less certain it gets. Cuban statistical publication thinned after 2020; several series in the *Anuario* stopped appearing or appeared very late; and the figures for 2025 and 2026 that circulate are in most cases estimates by independent economists and demographers rather than published data. Where this chapter gives a recent number it says whose number it is, and the figures for the last two years should be read as provisional and checked against whatever has been published since. The direction of every one of them is not in doubt.

10.2 A new president, and a new constitution

Miguel Díaz-Canel became president in April 2018 and First Secretary of the party in April 2021, when Raúl Castro retired. He is the first Cuban leader since 1959 not named Castro and the first who

did not fight in the insurrection, and neither fact has translated into much independent authority.

His first substantial act was a constitution. The 2019 text, approved in a February referendum, is a more liberal document than the 1976 one it replaced and this should be said before the qualification. It recognises private property and foreign investment as components of the economy rather than as tolerated exceptions; it restores the presumption of innocence and habeas corpus; it limits the president to two five-year terms and imposes an age limit on first election; it re-creates the office of prime minister; it strengthens the language on non-discrimination. Against that, it retains the Communist Party as the superior leading force of society and the state, and it retains the irrevocability of socialism that had been inserted in 2002 to bury the Varela Project. It is a reform of the state's administration and not of its politics.

The draft had contained a definition of marriage that would have permitted same-sex marriage; it was withdrawn after organised opposition from evangelical churches and referred to a future family code. That code was put to a separate referendum in September 2022 and passed with about two thirds of the vote, making Cuba one of the few countries in the world to legalise same-sex marriage by popular vote. It is a genuine liberal advance and it was delivered by referendum, which is worth noting in a chapter that is otherwise unrelieved.

10.3 The re-tightening

Between 2017 and 2020 the United States imposed a long series of measures — the figure usually cited is 243 — that reversed the opening and went beyond it. Cruise ship calls were prohibited in June 2019. Individual people-to-people travel was ended. Remittances were capped and then, in late 2020, effectively cut off when Western Union closed its more than four hundred Cuban offices after the US Treasury designated the military-linked processor through which they ran. Title III of Helms–Burton was allowed to come into force in May 2019 for the first time since 1996, exposing any company

using confiscated property to suit in American courts. And in January 2021, in the last week of the Trump administration, Cuba was returned to the list of state sponsors of terrorism, a designation whose practical effect is less in its own prohibitions than in the behaviour it induces in banks, which de-risk by refusing Cuban business entirely.

The Biden administration eased parts of this from 2022 — remittance caps lifted, group travel restored, consular processing resumed in Havana — and in January 2025 removed the terrorism designation, which was reinstated within days of the change of administration. The net effect over the decade is that the external environment tightened, that the tightening was concentrated on exactly the flows that reached Cuban households rather than the Cuban state — remittances, independent travel, private accommodation — and that Cuban policy toward its own private sector, described in the previous chapter, had begun tightening first.

Meanwhile the Venezuelan subsidy shrank with Venezuela. Oil deliveries that had been near 100,000 barrels a day fell to a fraction of that, with the shortfall covered irregularly by Russia, Mexico and spot purchases the country could not reliably pay for.

10.4 Two years with the borders shut

Tourism had reached about 4.7 million visitors in 2018. It fell to roughly 1.1 million in 2020 and to something under 400,000 in 2021. For an economy in which tourism was the second or third source of foreign exchange and the first source of demand for the private sector, closing the borders for two years was equivalent to closing the private economy. Arrivals recovered to around 2.4 million by 2023 and then stalled below that, at roughly half the pre-pandemic level, while the Caribbean as a whole recovered fully — a divergence that has more to do with blackouts, hotels without water and empty shops than with American policy.

Against this, the pandemic produced the one unambiguous Cuban success of the period. Cuba's biotechnology sector developed its own COVID-19 vaccines — Abdala at the Centre for Genetic Engineering

and Biotechnology, Soberana 02 and Soberana Plus at the Finlay Institute — ran its own trials, reported efficacies above 90 per cent, manufactured at national scale, and vaccinated over 90 per cent of the population with domestically produced product, including children from the age of two earlier than most of the world. No other country in Latin America did this. Cuba did it while unable to import syringes reliably.

The qualification belongs with it: the vaccines were not submitted through, or did not complete, the international prequalification and regulatory pathways that would have allowed them to be sold at scale abroad, which is the same failure Chapter 19 identifies across the whole sector — excellent science, no regulatory conversion, and therefore very little money. Cuba vaccinated itself and could not sell the result.

10.5 Tarea Ordenamiento

On 1 January 2021 — in the middle of a pandemic, with the borders closed, tourism at zero and the economy having contracted by about 11 per cent the previous year — the government executed the currency unification it had been preparing since 2011.

The convertible peso was abolished with a six-month conversion window. The official exchange rate for enterprises went from one peso to the dollar to twenty-four, a devaluation of 2,300 per cent applied at a stroke to every balance sheet in the country. Wages and pensions were raised several-fold to compensate, the minimum wage going to 2,100 pesos. Subsidies were cut and many regulated prices raised.

The direction was right and had been right for twenty years. Chapter 7 explained why: with two rates twenty-five-fold apart, no Cuban enterprise account meant anything, and nothing could be reformed because nothing could be measured. The execution was catastrophic, and the reasons are specific enough to be instructive.

It was done with no reserves. A fixed rate is a promise to sell dollars at that rate, and Cuba had no dollars, so the promise was not kept and a parallel rate appeared immediately.

It was done with no supply response. Wages were multiplied in an economy where the quantity of goods was falling, which is the definition of an inflation.

It was done with the private sector still repressed — the licence freeze of 2017 was still in effect, and the small and medium enterprise law was eight months away — so the one part of the economy that could have responded to higher prices with more output was not permitted to.

And it was done at the bottom of the worst external shock since 1993, when the correct sequence is to stabilise first and unify afterwards.

The results followed. Official consumer price inflation was around 77 per cent in 2021 and remained in the high double digits for the following two years, with independent estimates considerably higher for the goods people actually buy. The informal exchange rate, which the reform was meant to eliminate, detached from the official one and kept going: past 100 pesos to the dollar in 2022, past 200 in 2023, past 300 in 2024, and beyond that since. Real wages and pensions ended lower than before the reform that had multiplied them. Cuban savings were destroyed for the second time in thirty years.

Chapter 15 is written against this chapter. The lesson is not that unification was wrong. It is that a monetary reform without reserves, without a supply response and without a stabilisation programme in front of it is not a reform but a devaluation of the population's savings, and that the same state had demonstrated in 1994–96, with fewer resources and greater competence, how to do it in the right order.

10.6 11 July 2021

On the morning of Sunday 11 July 2021 people came into the street in San Antonio de los Baños, a town of forty thousand south-west of Havana. It was filmed and put on Facebook. Within hours there were demonstrations in dozens of towns and cities across the island — the counts run from about forty to over sixty locations — including

Havana, Santiago, Camagüey, Holguín, Cienfuegos, Matanzas and Palma Soriano.

Nothing like it had happened since 1959. The *maleconazo* of 1994 was one city on one afternoon. This was national, simultaneous, and organised by nobody — which is exactly what made it uncontrollable and, for the government, incomprehensible. The immediate causes were blackouts in a July heatwave, empty pharmacies during the worst COVID wave Cuba had, food queues, and the price shock of the *ordenamiento*. The demonstrators were overwhelmingly young, poor, and disproportionately from the Black and mixed-race neighbourhoods that Chapter 7 identified as the households without remittances — which is to say, from the constituency the revolution has always counted as its own.

The response was mobile internet shut off nationally within hours; the president's televised instruction, the *orden de combate*, calling revolutionaries into the street; deployment of special forces and of civilian brigades; and mass arrests. One protester, Diubis Laurencio Tejeda, was shot dead in La Güinera in Havana. Cubalex and Justicia 11J, the two monitoring groups that did the counting because nobody official would (Justicia 11J and Cubalex, 2022), documented 1,481 people deprived of liberty in connection with the protests, of whom 57 were under eighteen. Around 600 were prosecuted; at least 141 were charged with sedition; and sentences ran from four years to thirty. Three years on, several hundred were still in prison. The prosecutions were not of an organisation, because there was no organisation; they were of individuals who had walked in the street.

This is the central political fact of the period and Part III has to be answerable to it. A government whose claim to legitimacy rests on having lifted the poor and the Black out of the republic's neglect responded to a demonstration by the poor and the Black with twenty-year sentences. Whatever the design in Part III proposes about reconciliation, amnesty and the sequencing of political opening, 11 July 2021 and its prisoners are the case it has to be able to handle.

10.7 The state's own retreat

In August 2021, six weeks after the protests, the government did the thing it had refused to do for a decade: Decree-Laws 46 to 49 created small and medium enterprises with legal personality. A Cuban private business could at last be a firm — own assets, sign contracts, hire, and import and export through a state intermediary. Some eleven or twelve thousand were approved over the following three years, the large majority private, and they rapidly became a significant share of the country's imports of food and consumer goods.

The reform was real and it was also, transparently, a surrender: the state legalised private importers because it could no longer stock its own shops. The same logic produced the partial re-dollarisation of retail from 2024 — new state shops selling in dollars only, to capture remittances — which reintroduced, three years after abolishing it, a two-currency retail economy with the currency of the poor on the wrong side of it.

And the enterprises arrived into a policy environment that turned on them almost at once. From 2024 came price caps on private traders, restrictions on what could be imported, higher taxes, and a public argument from the leadership that the new firms were the cause of inflation rather than a response to it. The pattern of 1996 and of 2017 repeated on a two-year cycle.

10.8 The lights

Cuba's electricity system reached the end of its life in 2024.

The generating fleet is a small number of large oil-fired thermal units, most built between the 1960s and the 1980s, running past their design lives on heavy domestic crude they were not built for, without maintenance, without spares, and frequently without fuel. Through 2023 and 2024 daily deficits of a third or more of demand were routine, and blackouts of twelve to twenty hours were normal outside Havana and increasingly normal inside it.

The scale of the retreat is visible in the output. Cuba generated

on the order of 20 terawatt-hours in 2018; in 2023 it generated 15.3, of which 3.7 — close to a quarter — was lost in transmission and distribution before reaching anybody. Consumption per head fell from about 1,856 kilowatt-hours in 2018 to 1,387 in 2023.

On 18 October 2024 the Antonio Guiteras plant, the largest single unit in the country, tripped, and the entire national grid collapsed. Cuba went dark — eleven million people, every province, no exceptions. Restoration took several days and failed twice during the attempt. It happened again in November after Hurricane Rafael, and again in December, and again in 2025. A national blackout is not a shortage; it is the failure of the system as a system, and once a grid has collapsed it collapses more easily.

The government's answer is solar, financed and built by China: a programme announced around 2024 of dozens of photovoltaic parks with a first target near a thousand megawatts, of which several hundred megawatts were reported energised during 2025. The direction is right — Chapter 16 argues it is the only direction available — but photovoltaic capacity without storage does not address a demand peak that occurs after sunset, and it does not by itself restart a thermal fleet that must still cover the night.

Everything else in this chapter is downstream of the lights. Without stable power there is no water pumping, no refrigeration, no food processing, no tourism worth the name, no hospital that can plan a list, no remote work, and no industry. Chapter 12 therefore treats electricity as the binding constraint on the whole design and puts it before every other sector.

10.9 The exodus

The largest emigration in Cuban history took place between 2021 and 2025, and it dwarfs the ones this book has already described.

The route ran through Nicaragua, which removed visa requirements for Cubans in November 2021, and then overland to the United States border. In the American fiscal year 2022 US authorities recorded something over 220,000 encounters with Cuban nationals, and about 200,000 the following year; a humanitarian

parole programme opened in January 2023 admitted tens of thousands more by air; and substantial numbers went to Spain under a 2022 descent-based nationality law, to Mexico, Uruguay, Serbia and elsewhere. Reasonable estimates of total departures over 2021–2025 run from about one million to something approaching two, against a population that had been about 11.2 million.

The official statistics office put the resident population at 9.7 million in 2024 — a fall of 1.4 million in four years, and the first time it has been below ten million since the 1980s — with the 2024 census postponed. Independent Cuban demographers, working from birth, death and border data, put the true resident figure lower still, one study at around eight million, and Albizu-Campos estimates roughly 1.79 million departures over 2021–24 alone (Oficina Nacional de Estadística e Información, 2024a). The uncertainty is large and the fact is not: Cuba has lost something like a tenth to a fifth of its people in five years, and they were disproportionately of working and child-bearing age.

This lands on a demography that was already among the oldest in the hemisphere. Fertility has been below replacement since 1978 — there is no cohort in the pipeline — and by the mid-2020s the total fertility rate was around 1.2 to 1.4, among the lowest in the world, with more than a quarter of the population over sixty. A pension system, a health system and a tax base all depend on a ratio of workers to dependants, and that ratio has now moved further and faster in Cuba than in any country not at war.

Chapter 14 treats this as the hardest number in the book, and Chapter 18 draws the conclusion that follows from it: a country that has lost this many working-age people cannot be designed as though labour were abundant. Every proposal in Part III is labour-constrained.

10.10 Where the country stands

In 2026 Cuba is poorer, older, smaller, darker and hungrier than at any point since 1994, and by several measures than at any point since 1959.

Output has contracted in most years since 2019 and the cumulative fall is large. Sugar, which was 8.5 million tonnes in 1970 and 4 million as recently as the 2000s, produced a harvest measured in the low hundreds of thousands of tonnes in the mid-2020s — less than in the nineteenth century — and Cuba now imports sugar. The rationed basket is delivered late and incomplete, and in 2023 the government asked the World Food Programme for assistance with powdered milk for children, which it had never previously done. Medicines are absent from pharmacies at rates the health ministry itself has acknowledged. Doctors and nurses are emigrating, and so are teachers. The blackouts continue.

Against that: the schools are open, the clinics are staffed however thinly, the vaccination programme still runs, violent crime remains low by regional standards, and the state has not lost control of the territory. The floor that held in 1993 is thinner and it is still there. It is the thing Part III is written to save.

The ledger: the crisis, 2018–2026

Achieved. A constitution that recognises private property and restores habeas corpus, and a family code legalising same-sex marriage passed by referendum. Three domestically developed COVID-19 vaccines and a population vaccinated with them, alone in Latin America. Private firms with legal personality at last, after a decade of refusal. A currency unification that, whatever its execution, ended a twenty-six-year accounting fiction. Solar capacity begun. And a state that has kept schools open and clinics staffed through a contraction it has not been able to stop.

Cost. A monetary reform executed with no reserves, no supply response and the private sector still shut, which destroyed the population's savings for the second time in thirty years. Twelve-to-twenty-hour blackouts and four national grid collapses. Sugar production below nineteenth-century levels and an appeal to the World Food Programme for children's milk. The largest emigration in Cuban history — between one and two million people in five years, from a population of eleven — onto the oldest demography in the hemisphere. And, on 11 July 2021, sentences of up to thirty years, including of minors, handed to poor and mostly Black Cubans for walking in the

street to say there was no food and no electricity.

Chapter 11

The balance sheet

This chapter does not narrate. The previous ten have done that. This one audits: for each domain, what sixty-seven years produced, stated once, with both sides, and the causal question asked rather than assumed. It is the hinge of the book, because it is where the record turns into the constraint set that Part III has to design within.

11.1 Health

What the record shows. Cuba took a country with the best health indicators in Latin America and a rural half with none of them, closed the rural–urban gap, and held the national outcomes for sixty years through two economic collapses. Life expectancy went from around 62 in 1958 to the high seventies, converging with the United States on a fraction of the spending. Infant mortality went from around 37 per thousand to single digits and stayed there — including through 1990–94, when it continued falling while national output fell by a third. Polio, measles, diphtheria, rubella and neonatal tetanus were eliminated. The delivery model, a polyclinic and then a family doctor responsible for a defined population, is the mechanism, and it has been copied deliberately by other countries because it works.

What has happened since 2019. The system is being hollowed out

from three directions at once. Pharmaceutical supply has collapsed: the domestic industry cannot obtain inputs and the state cannot pay for imports, and pharmacies are empty at rates the health ministry itself acknowledges. Physical plant has degraded past the point where outcome indicators can be maintained by staff effort — hospitals without reliable water, power or sterilisation. And the staff are leaving, in the general emigration and through the medical missions. Cuba's physician numbers remain high on paper because the paper counts doctors on mission abroad.

The honest reading. The achievement is real, was not bought by measurement games, and is the single most valuable thing in this ledger. It is also, in 2026, being lost in real time, and it will not survive another five years of the present trajectory. That is the argument for Chapter 14 and the reason it is placed early: the asset is depreciating faster than the plan can be written.

11.2 Education

What the record shows. Illiteracy eliminated in a year and kept eliminated. Universal primary and secondary schooling. Attainment measured by an outside body two standard deviations above the Latin American mean, with the poorest Cuban quartile outperforming the regional average — which is the number that matters, because it says the system compresses rather than reproduces advantage. Higher education free and expanded by an order of magnitude. Women's attainment ahead of men's since the 1980s.

The cost side, and the erosion. The curriculum was a political formation system, entire disciplines were closed or reduced to catechism, and Cuba consequently has excellent engineers and physicians and very few people trained to think about how a market, a court or a company works — which is a direct constraint on Part III's implementability, not a matter of taste. And the system is eroding now: teachers are leaving for the private sector and for abroad, enrolment in teacher training has fallen, and an informal private tutoring economy has appeared to fill the gap, which is the classic first stage of a two-tier system.

The honest reading. The best asset Cuba has, the one Part III's second track and biotechnology sector are both built out of, and the one whose depreciation is least visible year to year, because a school system's decline shows up in outcomes a decade later.

11.3 Food

What the record shows. A country with good soil, adequate water, a year-round growing season and a trained agronomic profession imports something in the range of two thirds to four fifths of what it eats, and cannot pay for it. A substantial share of arable land is idle or under *marabú* scrub. The sugar industry that was three quarters of exports as recently as 1989 produced a harvest in the low hundreds of thousands of tonnes in the mid-2020s and the country imports sugar.

The causes, in order of size. State procurement monopoly setting prices below cost; tenure that cannot be sold, mortgaged or reliably inherited; no input market; no functioning transport or cold chain; and a currency in which nobody wants to be paid. Every one of these is domestic and legal.

And the money went elsewhere. In 2024, tourism-related construction took 37.4 per cent of national investment and agriculture 2.7 per cent — a ratio of fourteen to one, in a country asking the World Food Programme for children's milk. Whatever else is disputed about Cuban economic policy, this allocation was chosen.

The honest reading. This is the clearest single case in the book where the embargo explains very little. Cuba may buy food from anyone in the world and does buy it from the United States for cash. The reason it does not grow its own is the design of the domestic agricultural system, which Chapter 20 takes apart.

11.4 Energy

What the record shows. Sixty years of electricity generation from imported hydrocarbons supplied on political terms — Soviet, then Venezuelan — burnt in large central thermal plants, most of which

are now past their design lives. When the political supply faltered in 1990 the country went dark; when it faltered again after 2019 the country went dark, and in October 2024 the grid failed as a system, nationally, for the first time.

The honest reading. Energy is where Cuba's three external dependencies converge, and it is the binding constraint on everything else — water, refrigeration, food processing, health, tourism, industry, connectivity. It is also the one domain where the technology has changed enough since 2010 that the answer available now is not the answer that was available then. Chapter 16 accordingly comes before every sector chapter in Part III.

11.5 Demography

What the record shows. Fertility fell below replacement in 1978 and has never returned; by the mid-2020s it was around 1.2 to 1.4, among the lowest in the world. More than a quarter of the population is over sixty. And between one and two million people, disproportionately of working and child-bearing age, left between 2021 and 2025 from a population of eleven million.

The honest reading. This is the hardest constraint in the book and the least reversible. There is no cohort in the pipeline, so no policy adopted in 2026 changes the working-age population before the 2040s except by migration. Every proposal in Part III is therefore labour-constrained rather than capital-constrained, and a pension system at European replacement rates is not financeable on this dependency ratio at any plausible tax rate — which is why Chapter 14 has to say so rather than assume it away.

11.6 Rights

What the record shows, without inflation and without excuse. No independent press since 1961. No legal political organisation other than one. No independent trade union. No court that has ever ruled against the executive on a political matter. Summary executions in 1959 with no published count. Labour camps for believers,

homosexuals and nonconformists, 1965–68. A cultural purge by administrative reassignment, 1971–76. Emigration punished by confiscation and forced labour for two decades, and mobs organised against departing neighbours in 1980. Seventy-five people sentenced to up to twenty-eight years in 2003 and three shot in the same month. Thirty-seven people drowned off Havana in 1994 with no independent investigation. Hundreds prosecuted after 11 July 2021, with sentences of four to thirty years, including of minors.

What should also be said. The scale is not that of the region's worst dictatorships and inflating it is a disservice: Cuba did not run death squads, did not disappear people in the tens of thousands, and the 20,000-dead figure attributed to Batista is itself a fabrication that has been recycled for sixty years by people who should know better. Violent crime is low, the police do not kill people at Latin American rates, and Cubans move around their own cities at night in a way that citizens of most countries in the hemisphere do not. Sixty years of one-party rule produced a repressive state, not a murderous one, and the distinction matters when Chapter 13 designs the transition, because the settlement a country needs after Argentina is not the settlement it needs after this.

11.7 The embargo question

This is the question everything else is argued through, and the honest answer has two halves that are usually published separately.

What the embargo does cost. It is not nothing and the dismissive version is wrong. Shipping: a vessel that calls at a Cuban port cannot enter a United States port for 180 days, which removes Cuba from the efficient Caribbean routings and raises freight on everything. Finance: the extraterritorial enforcement — of which the multi-billion-dollar penalties imposed on European banks are the demonstration — means international banks refuse Cuban business as a matter of policy rather than of law, so a Cuban transaction is expensive, slow, or impossible even where it is legal. Medical and industrial goods: any product with more than a de minimis share of American content is caught, which in practice covers much modern

medical equipment. Credit: Cuba is excluded from the IMF, the World Bank and the Inter-American Development Bank, which is not merely a loss of lending but a loss of the technical assistance and the seal of approval that other countries use to attract private capital. And Helms–Burton converted all of it from an executive instrument into a statute, so it cannot be traded away by a president.

What the embargo does not explain. Cuba trades freely with the rest of the world. The European Union, China, Canada, Brazil, Russia, Mexico, Spain, Vietnam and Venezuela have not embargoed anything, and Cuba has bought from all of them. The binding constraint on Cuban imports is not permission; it is that Cuba cannot pay, and cannot pay because it has been in default since 1986, froze foreign firms' accounts in 2009, and has arrears with almost every supplier who has extended it credit. That is a credit history and it has no foreign author. Nor does the embargo explain the domestic prohibitions: the abolition of small business in 1968, the refusal of legal personality to private firms until 2021, the absence of a wholesale market, the state procurement monopoly in agriculture, the ban on Cubans entering hotels until 2008, the exit permit until 2013, the licensing of activity by enumerated list, or the employment agency that takes the currency spread from workers in foreign ventures. Every one of those was written in Havana, and every one of them can be repealed in Havana.

How to hold both. The formulation this book uses is that the embargo sets the cost of Cuban failure and the Cuban system sets its probability. A well-designed Cuban economy under embargo would be poorer than the same economy without one — meaningfully poorer, and the difference is measured in percentage points of growth a year, not in the fantastical cumulative totals either side publishes. But it would not be an economy that cannot light its cities or feed itself, because several countries in the region manage both with worse endowments and no external assistance. The reason this book's Part III concentrates on domestic institutions is not that the embargo is unimportant. It is that the embargo is the one variable Cubans do not control, and a design that waits on it is not a design.

11.8 The pattern

Across sixty-seven years the same sequence occurs four times, and naming it is the finding Part III has to be built against.

Duress produces reform. 1980, 1993–95, 2008–2013, 2021: in each case a material emergency forced the state to legalise something it had prohibited — farmers’ markets, dollars and self-employment, usufruct land and private house sales, private firms with legal personality.

The reform works. Food appears; the currency stabilises; the private sector grows; the shops restock. In every instance the measured result was positive and visible within months.

The pressure lifts, and the reform is withdrawn. 1986, 1996–2004, 2016–17, 2024–25: markets closed, licences frozen, taxes raised, categories deleted, price caps imposed, and the reformers criticised for concentration of wealth.

The withdrawal arrives just before it is least affordable. Rectification was launched in 1986 and the Soviet Union collapsed in 1990. The licence freeze came in August 2017, fourteen months before the American re-tightening. The attack on the new private importers began in 2024, in the middle of the worst shortages since 1993.

The reason for the pattern is not economic and pretending otherwise wastes everyone’s time. Each reform creates people whose income does not come from the state, and that is the thing the system is organised to prevent. The reforms are therefore always framed as concessions, always revocable, and always revoked — and because they are known to be revocable, nobody invests behind them, which guarantees that they underperform, which supplies the evidence for revoking them.

That is the finding, and it converts directly into the design requirement that opens Part III: *the binding problem in Cuba is not which policies to adopt but how to make them irrevocable*. Every measure this book proposes has been tried in Cuba in some form. Most of them worked. All of them were reversed. A design whose only content is a better list of policies has not understood the last thirty years, and Chapter 13 is placed second in Part III for that reason: without a credible commitment mechanism, nothing that follows it is worth writing down.

The ledger: 1959–2026

Achieved. A health system that closed the rural–urban gap, converged with rich-country outcomes on poor-country money, and held its indicators through a 35 per cent contraction. An education system whose measured results beat its region by two standard deviations and compress rather than reproduce advantage. Illiteracy ended in a year. Legal and enforced racial desegregation. Universal social security, near-universal home ownership, and one of the lowest violent crime rates in the hemisphere. A biotechnology sector that developed and deployed its own vaccines when far richer countries could not. A decisive contribution to ending South African rule in Namibia. And a state that has never lost the capacity to reach every block of its territory, which is why it absorbed 1993 without famine.

Cost. An export basket unchanged in structure from 1958 to 1989 and destroyed since. Three successive external patrons and three identical failures to diversify under them. Sugar below nineteenth-century output and food imported at two thirds of consumption in a fertile country. A grid that failed as a system in 2024. A default unresolved since 1986. Six decades without a press, a union, a party or a court that was not the state's. Camps, purges, exile as punishment, mobs organised against neighbours, and thirty-year sentences in 2021 for walking in the street. Fertility below replacement since 1978 and between one and two million people gone in five years. And a reform record in which everything that worked was later withdrawn, which is the reason none of it compounded.

Part III

The prospect

Chapter 12

Premises

A design is a set of choices made subject to constraints, and most bad designs for Cuba are bad because they quietly relax a constraint that does not relax. This chapter fixes the constraint set. Each item is a finding of Parts I and II rather than an assumption, each is stated in the form that matters for the chapters ahead, and each is something the design must work *with* rather than around.

12.1 The population is the binding scarcity

Cuba has fewer than ten million residents and is losing more. Fertility has been below replacement since 1978, so there is no cohort in the pipeline; more than a quarter of the population is over sixty; and between one and two million people, disproportionately of working and child-bearing age, left between 2021 and 2025.

The consequence is a reversal of the assumption most development planning makes. Cuba is not a country with surplus labour looking for capital to employ it. It is a country with a shrinking, ageing workforce, and every proposal in Part III has to be read as competing for the same scarce people. A hotel that opens draws staff from a hospital. A remote-work sector that grows draws engineers from

the state. This is why the book does not propose labour-intensive import-substitution, why Chapter 14 treats the pension arithmetic as its hardest problem, and why Chapter 18 treats immigration and return migration as fiscal policy rather than as sentiment.

It also sets the clock. No policy adopted in 2026 changes the working-age population before the 2040s except through migration, and the emigration is self-reinforcing: each departure removes a reason for the next person to stay and adds a household abroad that can finance their leaving. The design's first phase is not judged on growth. It is judged on whether the outflow slows.

12.2 The state is insolvent and has no access to credit

Cuba has been in default on its convertible-currency debt since 1986. The 2015 Paris Club arrangement, which forgave a large part of accumulated arrears and rescheduled the rest, broke down within four years when Cuba stopped paying under it. Russia wrote off ninety per cent of the Soviet-era debt in 2014. Suppliers have unpaid invoices going back to 2009, when the state froze foreign companies' accounts. Cuba is not a member of the International Monetary Fund, the World Bank or the Inter-American Development Bank, and so has access neither to their lending nor to the technical assistance and implicit certification that other countries use to attract private capital.

Two things follow. Nothing in Part III can be financed by sovereign borrowing on ordinary terms, because there are no such terms available; the financing has to come from equity, from the diaspora, from concessional and bilateral sources, and above all from cash flow. And the sequence in Chapter 15 has to put debt normalisation early, not because settling it releases money — it does not — but because until it is settled every other transaction in the country is priced as though the counterparty defaults, which it has.

12.3 Nothing works until the electricity does

This is the reason Chapter 16 precedes every sector chapter.

Electricity is not one sector among several in Cuba; it is the input to all of them. Water is pumped. Food is refrigerated. Hospitals cannot run lists without it. Tourism cannot be sold without it — a hotel without power or water does not merely disappoint a guest, it destroys the review that produces the next guest. Remote work is impossible without it. Manufacturing is impossible without it. And a grid that has collapsed nationally four times collapses more easily each time.

The design premise is therefore blunt: any plan whose sector chapters would work only if the power were reliable is a plan that has to deliver reliable power first, and its first eighteen months belong to that and to very little else.

12.4 The welfare floor does not come down

This is stated in the preface and repeated here as a premise because it binds everything after it.

The design retains a universal welfare state at Western European standard — health, education, pensions, disability, income support — and the rest of Part III is arranged around being able to afford it. The reasoning is in Chapter 14. What belongs here is the cost, admitted in advance rather than discovered in a later chapter: a European benefit structure requires a tax burden well above the Latin American norm, which is a real competitive disadvantage against every other country in the region bidding for the same investment. This book resolves that trade-off in favour of the welfare state and compensates on the investment side with certainty, speed and rule of law rather than with low rates. Chapter 21 is written on that basis and must not quietly assume the opposite.

The floor is also, in 2026, thinner than it has ever been and

depreciating. Retaining it is not a matter of leaving something alone. It requires spending that does not currently exist, on staff who are currently leaving.

12.5 The diaspora is an asset with a grievance

Something over two million people of Cuban origin live abroad, most within three hours' flight, holding capital, professional skills, business networks and market access that no policy can otherwise buy. Their remittances are already among the largest sources of foreign exchange in the country. They are the single largest potential source of the investment Chapter 21 needs, and they are the natural first market for Chapter 17's visitors and Chapter 18's clients.

They also hold claims and they hold memories, and the memories are specific rather than diffuse: confiscated houses, the inventory taken at the door, the years of agricultural labour that were the price of an exit application, the children of Peter Pan, the mobs outside the door in 1980, and the relatives who did not get out. A design that treats the diaspora as a chequebook will fail, and a design that treats every claim as valid and every asset as returnable will fail differently and worse. Chapter 21 proposes a bounded, non-restitutionary settlement, and Chapter 13 proposes a truth-telling mechanism with no prosecutorial power, precisely because these two things have to be settled together and neither can be settled by generosity in one direction only.

12.6 The incumbents are in the room

This is the premise most reform proposals for Cuba omit, and omitting it is what makes them unserious.

Any transition from the present arrangement is designed and executed by, or at minimum with the acquiescence of, three groups that currently hold power: the Communist Party apparatus, the Revolutionary Armed Forces, and the military-linked business con-

glomerate GAESA, which controls a large share of the country's hard-currency earnings through tourism, retail and logistics — how large is not published, which is itself the point.

These are not abstractions to be legislated away. They hold the assets, the guns, the foreign-currency accounts and the administrative capacity, and they have watched what happened to the Soviet nomenklatura, to Ceaușescu, and to Gaddafi. A design that offers them nothing is a design they will not permit, and the historical record is unambiguous that they can prevent it. A design that offers them everything is a design that produces an oligarchy with a flag, which Chapter 22 argues is the most likely failure mode and the one that is effectively irreversible once it happens.

The book therefore treats the incumbents as a party to be bargained with explicitly: guarantees of personal security, pensions and property in exchange for the surrender of discretionary control, with the bargain written down, sequenced, and made verifiable. Chapters 13, 21 and 22 each carry part of it.

12.7 The United States is a variable, not a constant

The embargo may be relaxed, tightened, or left where it is, and Cuban policy does not control which. Helms–Burton converted it from an executive instrument into a statute, so its removal requires an act of Congress rather than a decision by a president, and the domestic American politics of that act have been immovable for thirty years and are not obviously about to move.

The design premise is symmetric and it is the only defensible one: *assume nothing*. Nothing in Part III may be contingent on the embargo ending, because it may not; and nothing in Part III may be built on the assumption that it will persist, because a sudden normalisation is also possible and a country with no plan for it will be overrun by the resulting inflow — of capital, of visitors, of claims — exactly as it was overrun in miniature in 2015–16. Each chapter that touches the American relationship therefore states what it does under both branches.

12.8 Race and region decide who is exposed

Cuba's inequality has a colour and a compass bearing, and both were made by policy.

The remittance economy flows overwhelmingly to households whose relatives left in the 1960s, who were overwhelmingly white. The tourism sector's foreign-currency jobs have been allocated in part by appearance. Meanwhile the five eastern provinces — the poorest, the most agricultural, the most Afro-Cuban — have the worst housing, the worst infrastructure and the longest blackouts, and are the origin of most internal migration to Havana, which the state has restricted by decree since 1997.

Every liberalising measure in Part III advantages the household with capital, with a relative abroad, with a house in a tourist district, with a computer. That is not an argument against the measures; it is an argument for knowing in advance who is on the other side of them, and for the compensating instruments — the universal floor of Chapter 14, the regional allocation rules in Chapters 16 and 17, and the deliberately broad-based structure of Chapter 21's privatisation — to be designed in from the start rather than added when the disparity becomes visible. Cuba has already run the experiment of declaring the racial question solved and then not measuring it for thirty years. The design's first requirement in this domain is to publish the data.

12.9 What this book does not assume

Finally, the negative premises, stated so that a reader can hold the argument to them.

No shock therapy. The book does not propose simultaneous liberalisation of all prices, rapid mass privatisation, or the withdrawal of the social state in advance of the institutions that would replace it. The historical record of that package is bad everywhere and would be worse here, for the demographic reason in the first section: a country whose young have already left does not survive a decade of

transitional depression.

No restitution by expulsion. No Cuban is removed from a dwelling to return it to a prior owner. Chapter 21 settles claims in money and paper, within a cap.

No assumption about Washington. As above, in both directions.

No assumption of regime collapse, and none of regime permanence. The design is written so that its early phases are executable by the present government if it chooses, and its later phases by whatever government follows. That is deliberate. A plan that requires a particular political rupture before its first measure can be taken is a plan that does nothing until the rupture, and Cubans have been waiting for the rupture since 1991.

No claim of certainty. Chapter 24 closes with a list of indicators by which the design can be judged to be failing. A proposal that cannot be falsified is not a proposal.

Chapter 13

The polity

13.1 The problem is irrevocability, not policy

Chapter 11 ended on a finding, and this chapter is its consequence.

Almost every measure this book proposes has already been tried in Cuba. Private restaurants: tried, 1993, worked, restricted 2017. Farmers' markets: tried, 1980, worked, abolished 1986, tried again 1994. Dollars: legalised 1993, withdrawn 2004. Land in usufruct: 2008 and 2012. Private firms with legal personality: 2021, then price-capped and blamed for inflation by 2024. The Cuban state has not been short of good ideas. It has been unable to keep any of them for longer than the emergency that produced them.

The consequence is not merely that reforms get reversed. It is that they underperform *while in force*, because everyone can see they are revocable. A Cuban who is deciding whether to invest three years' savings in equipment is making a bet on the durability of a decree, and the historical base rate on that bet is poor. So capital is not committed, businesses stay small and liquid, nobody trains anyone, nobody builds anything that cannot be moved — and the resulting mediocrity is then produced as evidence that the private sector does not work, which justifies the next restriction.

This is a solved problem in institutional economics and the solution has a name (North and Weingast, 1989). What a state needs, when it wants people to invest behind its promises, is a mechanism by which it can *disable its own future discretion* in a way outsiders can verify. Not a better promise — a promise it cannot cheaply break. Everything in this chapter is an instrument of that kind, and everything in the rest of Part III depends on at least some of them existing. A chapter of excellent energy policy in a country where policy is revocable at will is a chapter about what could have been.

Design 13.1. Commitment before content

- (1) ORDERING No liberalising measure in Part III is adopted without at least one commitment device attached to it: a constitutional entrenchment, a justiciable right, a treaty obligation, an independent regulator with secured tenure and budget, or a published rule with an automatic trigger. A measure adopted by decree alone is understood by everyone, correctly, as temporary.
- (2) RIGHTS NOT LISTS Economic activity is permitted by right, subject to published prohibitions, and never by enumerated list of allowed categories. The list is what made the Cuban private sector a collection of trades with no software developers in it, and it is also the instrument by which the sector is shrunk without any law being repealed.
- (3) ASYMMETRY Instruments are designed so that expanding a freedom is administratively cheap and withdrawing it is expensive — requiring legislation, publication, notice periods and compensation — rather than the reverse, which is the present arrangement.
- (4) STANDING Every entitlement created in Part III is enforceable by the person who holds it, in a court, against the state. An entitlement nobody can sue over is a policy, and policies are what this chapter is about not relying on.

13.2 Sequence: law before ballots

The most consequential choice in this chapter is the order, and the book takes a position that will not please everybody.

Rule of law first; competitive national elections later, and on a published schedule.

The reasoning is empirical rather than principled. The post-communist transitions that went well — Estonia, Poland, Slovenia, the Czech lands — built courts, property registries, competition authorities and independent central banks alongside or ahead of their electoral opening. The ones that went badly held elections into an institutional vacuum, where the winner inherited undivided control of unpriced assets and no court capable of restraining anybody, and the resulting concentration then captured the electoral process itself. Russia's 1996 election was won by an incumbent financed by the men to whom he had sold the economy; the sequence, not the ballot, was the problem (Åslund, 2013).

Cuba's own history says the same thing from the other side. Chapter 1 argued that the decisive feature of January 1959 was that Batista's coup had already destroyed every institution capable of arbitrating between contenders, so that the only body with legitimacy was the guerrilla movement and there was nothing to slow its consolidation. A Cuban transition that produces one legitimate actor and no others will end where 1959 ended, whatever its intentions. The institutions have to exist before the contest, or the contest decides who owns them.

Two things must be said against this position, because it is the position under which authoritarian governments always promise elections later.

First, "later" has to be a date. A sequencing argument without a published calendar is an indefinite postponement with a rationale, and the Cuban government has been offering exactly that since 1959. The commitment device here is that the calendar is entrenched at the beginning, along with the institutions, and is itself justiciable.

Second, some political opening is a *precondition* for the institutions rather than a reward for them. A judiciary cannot be made independent in secret; an anti-corruption body cannot function with-

out a press to publish what it finds; a property registry is only trusted if its contents can be disputed in public. So the sequence is not law-then-politics but a specific ordering *within* the political opening: association, press, information and courts early, because the institutions need them; competitive national elections on a fixed later date, because the assets need pricing and the parties need building first.

Design 13.2. The political calendar

- (1) IMMEDIATE Legal association, independent press and publishing, independent trade unions, and the right to receive and impart information across borders. These are preconditions for every institution below and cost the state nothing to concede.
- (2) YEAR ONE Judicial reform (Design 13.3), an administrative court in which a citizen or firm can sue the state, publication of the state's accounts, and a party law setting the terms on which political organisations register — deliberately permissive, with public financing and disclosure.
- (3) YEARS TWO AND THREE Competitive municipal elections, held first and twice, because municipalities are where parties learn to be parties, where the electoral administration learns its job, and where a mistake is survivable.
- (4) YEAR FIVE Competitive elections to the national assembly, on a date fixed in the constitutional amendment that begins the process, with an independent electoral commission constituted in year one and no incumbent majority on it.
- (5) NON-POSTPONEMENT The dates are entrenched and their postponement requires a supermajority plus judicial confirmation of a stated emergency, with automatic expiry. A calendar a government can move at will is not a calendar.

13.3 Courts

Cuba has had no court that ruled against the executive on a political matter since 1959, and the practice of subordinating the judiciary to

political direction was established in the first weeks of that year, when acquitted defendants were retried on the leader's public insistence. Sixty-seven years of that produces a bench with technical competence in ordinary civil and criminal work and no institutional memory whatever of independence.

The reform therefore cannot be a decree declaring judges independent. It has to change the four things that actually determine whether a judge can rule against the government: who appoints them, how long they serve, who pays them, and who can remove them.

Design 13.3. Judicial independence

- (1) **APPOINTMENT** Judges are appointed by a judicial council whose membership is majority-judicial and whose non-judicial members are appointed by qualified majority of the assembly, not by the executive. Vacancies are advertised; criteria and shortlists are published.
- (2) **TENURE** Appointment to retirement age, removable only for incapacity or proven misconduct, by a process that is itself judicial and published. The present arrangement, in which judges serve at the pleasure of the assembly that elects them, is abolished.
- (3) **BUDGET** The judiciary's budget is a fixed minimum share of state spending, set in the constitution, disbursed on the judiciary's own authority and audited publicly. A judiciary whose electricity bill is paid at the executive's discretion is not independent, and in Cuba this is not a metaphor.
- (4) **ADMINISTRATIVE COURT** A specialised administrative jurisdiction is created in which any person or firm may sue any organ of the state over a licence refused, a permit delayed, a good seized, a contract broken or a regulation misapplied, with the state bearing costs when it loses. This is the single most important court in Part III: it is where the commitment devices of Design 13.1 become real, and it is the mechanism by which Chapter 22's removal of discretion is enforced.
- (5) **PUBLICATION** All judgments are published in full, in a free public database, within thirty days. A body of published reasoning is what turns a set of decisions into a law, and it is also the cheapest

available check on the decisions themselves.

- (6) **COMMERCIAL CAPACITY** Commercial, insolvency and administrative benches are staffed by judges trained for them — a specific, funded, decade-long programme, since Chapter 3 established that Cuba trained very few lawyers to think about how a market works. This is a training constraint, not a legislative one, and it binds.

13.4 What has to be legal

The following is a floor rather than an ideal, stated as the minimum on which the rest of Part III depends. Each item is included because something later fails without it, and the reason is given.

Design 13.4. The minimum conditions

- (1) **ASSOCIATION** The right to form and join organisations — parties, unions, professional bodies, cooperatives, churches, neighbourhood groups — by registration on published criteria, with refusal appealable to the administrative court. Required by: every collective actor in Part III, including the farmers' associations of Chapter 20 and the professional bodies that must accredit Chapter 19's regulators.
- (2) **PRESS** Independent publishing and broadcasting, by registration, with no prior restraint and with defamation as a civil rather than a criminal matter. Required by: Chapter 22, whose entire enforcement model is that somebody publishes what the auditor finds. Law 88 of 1999 and the criminal provisions on enemy propaganda are repealed.
- (3) **INFORMATION** A statutory right of access to state-held information, with a presumption of disclosure, a short statutory deadline, a narrow and enumerated exemption list, and appeal to the administrative court. Required by: open contracting, asset declarations, the published cadastre, the electoral register, and the environmental data of Chapter 23.
- (4) **MOVEMENT** The right to leave and to return, entrenched, with the remaining professional exit restrictions repealed. Required by:

Chapter 19's reform of medical missions, and by the simple fact that a state which can prevent departure never has to earn its citizens' consent to stay.

- (5) COMMUNICATION No interruption of internet or telephone service on political grounds, with any interruption requiring judicial authorisation, publication and a stated duration. Required by: Chapter 18, whose entire proposition is that a foreign client can rely on a Cuban supplier being reachable. The national shutdown of 11 July 2021 would, under this rule, be unlawful — and that is the point of writing it down.

13.5 The bargain with the incumbents

Chapter 12 established that the party, the armed forces and GAESA are parties to any transition rather than obstacles to be removed from one. This section states the bargain, because a bargain that is left implicit is one that gets renegotiated at every step.

The exchange is: *security, standing and property, in return for the surrender of discretion*. The incumbents keep their liberty, their pensions, their homes, a legal route into politics, and — within limits and through a transparent process — an interest in the assets they now run. They give up the power to decide cases, allocate licences, set exchange rates, direct prosecutions and control foreign currency.

This is not generosity and it should not be presented as such. It is the price of a transition that does not go through a collapse, and the alternative histories are available: the transitions that offered the outgoing elite nothing were the violent ones, and the ones that offered them everything produced oligarchies. The design's job is to be specific about where the line falls.

Design 13.5. Amnesty, and where it stops

- (1) COVERED Amnesty for the fact of having held office, belonged to the party, served in the security organs, administered the economy, enforced the laws then in force, or participated in the political

system as it existed. Nobody is prosecuted for having been part of it.

- (2) **COVERED** Amnesty for economic offences below a published threshold, and for offences arising from the ordinary conduct of a state enterprise under the rules then applying.
- (3) **NOT COVERED** Killing, torture, enforced disappearance, sexual violence, and orders to commit them. These are not amnestied, at any rank, and no statute of limitations runs on them. A transition that amnesties torture teaches that torture is a policy option.
- (4) **NOT COVERED** Large-scale personal enrichment through the abuse of office, above the threshold in clause 2, where the asset can be traced. The remedy here is recovery of the asset rather than imprisonment, which is both more achievable and more useful.
- (5) **CONDITIONAL** Amnesty under clauses 1 and 2 is conditional on full disclosure to the commission in Design 13.6 and on the surrender of undeclared foreign assets. It is revocable for a materially false declaration, and only for that.
- (6) **SYMMETRIC** The amnesty runs in both directions. Everyone convicted for political expression, association, unauthorised departure, or the events of 11 July 2021 is released and their conviction vacated. This is not a concession traded against the others; it is a precondition, and it happens first.

Design 13.6. Truth without prosecution

- (1) **MANDATE** A commission with a fixed term of three years, a duty to take testimony from any person, powers to compel documents from the state, and a duty to publish. It has no power to prosecute, to punish, or to name a person as criminally liable.
- (2) **SCOPE** The whole period, both directions, and both kinds of harm: the executions and prison terms, the camps, the purges, the confiscations and the repudiation mobs; and also the deaths caused by the exile organisations, the sabotage campaigns, and the Bay of Pigs. A commission that examines one side produces a document that half the country will not read.
- (3) **RECORDS** The state archives on these matters are transferred to

the commission and thereafter to a public archive, with a fixed and short closure period for genuine personal privacy and nothing else.

- (4) **RESTITUTION OF STANDING** The commission may recommend, and the state must act on, the restoration of professional standing, pensions and academic records to people removed under *parametración* between 1971 and 1976 and to their equivalents in later periods. Much of that harm was done by administrative letter and can be undone the same way, at almost no cost, and it is the cheapest act of repair available.
- (5) **NO HIERARCHY OF VICTIMS** The commission does not rank suffering, adjudicate between narratives, or produce a verdict on the revolution. It produces a record.

13.6 Where power sits

One structural point, easy to omit and expensive to omit.

Cuban centralism is older than the revolution. The republic was run from Havana, the colony was run from Havana, and the 1976 constitution's *Poder Popular* gave municipalities elections without giving them money. A municipality that cannot raise or retain revenue cannot maintain a road, a clinic or a water pump, and so the entire country's problems arrive at the same desk in the capital, where they queue.

This matters for Part III specifically because so much of what it proposes is local: rooftop generation and distribution in Chapter 16, licensing and food procurement in Chapter 17, the cadastre and building control in Chapter 21, coastal setbacks in Chapter 23, and the property tax in Chapter 14, which is the natural local revenue and the instrument that makes local government answerable to local residents rather than to the ministry.

Design 13.7. Municipal capacity

- (1) **OWN REVENUE** Municipalities retain the property tax and a fixed

share of the accommodation levy and of business licence fees raised in their territory, set by formula and not by negotiation.

- (2) **EQUALISATION** A transfer formula, published and rule-based, redistributes toward the eastern provinces on measured indicators of need. Without this, Chapter 12's regional constraint becomes a regional divergence: tourism revenue accrues where the tourists are, and the tourists are not in Guantánamo.
- (3) **COMPETENCE** Municipalities hold and exercise building control, local licensing, local roads, water and waste, and primary-care and school premises, with the service standards set nationally and the delivery local.
- (4) **ACCOUNTS** Every municipality publishes its budget, its execution and its contracts, in a common machine-readable format, quarterly. This is also Chapter 22's cheapest and most effective instrument, because local theft is visible to local people if the numbers are available to them.
- (5) **INTERNAL MOVEMENT** Decree 217 of 1997 and the associated restrictions on residence in Havana are repealed. Restricting internal migration does not develop the east; it just makes being from the east a legal disability.

13.7 What this chapter does not settle

Two things are left open, deliberately, and a reader should know which.

The first is the form of the state — presidential or parliamentary, unicameral or bicameral, the electoral formula. These matter, they are extensively studied elsewhere, and nothing in this book's economics turns on them. They belong to whoever writes the constitution and should be settled by Cubans in a constituent process rather than proposed by an economist in a chapter.

The second is who begins. The design in this chapter is executable by the present government — every clause of it could be enacted by the National Assembly as presently constituted, and several of them, notably the amnesty in both directions and the release

of communications from political interruption, would cost it very little and buy it a great deal. It is also executable by a successor government. It is not executable by nobody, and Chapter 24 returns to the question of which coalition carries which phase.

Chapter 14

The social contract

14.1 The commitment

Cuba keeps a universal welfare state at Western European standard — health, education, pensions, disability, child support, income support — and the rest of Part III is arranged around being able to afford it.

That is the book's one non-negotiable, and this chapter is where it is argued rather than asserted. Four reasons, in descending order of how much weight they carry.

It is the thing worth keeping. Chapter 11 audited sixty-seven years and found one unambiguous asset. The health outcomes are measured, not claimed; the education results were tested by an outside body and beat the region by two standard deviations, with the poorest quartile above the regional average. Nothing else in the ledger is in that class.

It is what makes the rest politically possible. A population asked to accept markets, foreign owners, price liberalisation and privatisation will accept them if the floor holds. It will not if the floor goes, and the evidence is not theoretical: the post-communist transitions that withdrew social protection early had their economic reforms reversed by their own electorates within a decade (Roland, 2000), which is the most expensive way to discover that consent is an input. Cuba is being asked to run a transition with a population that has

been told for sixty-seven years that this is what the sacrifice bought. Take it away in year two and there is no year five.

It is the protection for the people the transition will hurt. Chapter 12 identified who is exposed: households without remittances, which are disproportionately Black; the eastern provinces; the old; the rural. Every liberalising measure in this book advantages the household with capital and a relative abroad. The universal floor is the instrument that stops that from becoming a permanent stratification, and it is the only instrument that works without an administrative capacity Cuba does not have.

It is an input, not only a transfer. The education system is what Chapter 18 and Chapter 19 propose to monetise. It cannot be defunded for a decade and restarted, because the teachers leave and the cohorts pass through. Spending on it is not consumption of the growth model's proceeds; it is the growth model's raw material.

14.2 What changes is who pays

Here is the chapter's central move, and the sentence that connects Part III back to Parts I and II.

Cuba's welfare state has never been financed by Cubans.

It was financed between 1972 and 1989 by Soviet terms of trade — sugar bought above the world price, oil supplied below it, and a petroleum rent Cuba was permitted to re-export. It was financed from 2004 to about 2015 by Venezuelan oil and by the medical-services contracts that oil relationship generated. In the intervals it was financed by monetary emission and by not maintaining anything, which is why the hospitals have no water and Havana loses buildings to collapse.

An entitlement funded by an external rent lasts exactly as long as the rent. Cuba has now discovered this twice, in 1990 and again after 2015, and on both occasions the welfare state was saved only by extraordinary effort and at the cost of its physical plant. The European systems this book takes as its standard are financed by taxing a domestic base — which is why they survive changes of government, of commodity price, and of patron, and why no European

government has ever had to choose between paying pensions and importing fuel.

So the proposal is: same benefit, different plumbing. A European benefit structure on a European tax structure. Rent-financed becomes tax-financed.

This is not a smaller undertaking than it sounds. Cuba has no modern tax system — it abolished most taxation in the 1960s on the grounds that a state which owns everything need not tax itself, and reintroduced taxes only from 1994 and only partially. It has no culture of filing, no significant enforcement history, no withholding infrastructure outside the state payroll, and a population that has never in living memory paid income tax. Building a revenue administration that can collect a value-added tax has taken five to eight years everywhere it has been attempted. That is the single longest lead time in this book and it is why Chapter 24 starts it in the first phase, when it yields nothing.

14.3 The arithmetic

The numbers below are orders of magnitude with their assumptions stated, not forecasts. They are here because a commitment of this kind is meaningless without a rough answer to what it costs and whether it can be paid for.

What the benefit costs. A European-standard universal system runs at roughly 25 to 30 per cent of national income in social spending: public health around 7 to 8 per cent, education around 5, pensions around 10 to 12, other social protection the balance. Total general government spending in EU member states sits in the 40 to 50 per cent range and tax revenue in the 35 to 45 per cent range.

What Cuba currently does. Cuban published figures put social spending at a very high share of output and revenue likewise, but — per the note on sources — most of that is an artefact of converting a state-owned economy at an administrative exchange rate, and the ratios do not mean what the same ratios mean elsewhere. What can be said without the conversion problem is physical: Cuba currently delivers something close to universal coverage at a real resource cost

far below European levels, because it pays its doctors, nurses and teachers a fraction of a European wage and has stopped maintaining its buildings and replacing its equipment. The system is being subsidised by its own staff and its own capital stock.

Where the gap is. That is the honest form of the problem. Cuba does not need to buy European coverage — it has coverage. It needs to buy European *inputs*: wages that stop the staff leaving, drugs that are actually in the pharmacy, equipment that works, buildings that do not flood, and a pension that someone can live on. Those are the expensive parts and they are exactly the parts currently unfunded.

Whether it is affordable now. No. At Cuban 2026 output a European benefit level is not financeable at any tax rate, and a book that pretended otherwise would be worthless. What is proposed instead is a stated trajectory: protect the *floor* immediately and let the *ceiling* converge as the base grows, with the convergence given a target and a date rather than left aspirational, and with the tax system built now so that it can collect the revenue when the base exists. The design below is written to be affordable in its first phase and to scale.

14.4 Which European model

Three families, and the choice matters more than it looks.

The *Nordic* model is universal, financed largely from general revenue with a high broad-based consumption tax, service-heavy rather than cash-heavy, and depends on a high employment rate. The *Bismarckian* model of Germany and France is contributory and payroll-financed, so coverage follows formal employment. The *Southern European* pattern is universal in statute and patchy in delivery, financed by a narrow base alongside large informality.

Cuba would drift into the third by default. It is therefore the outcome to design against, and the design choices below are made specifically to avoid it.

The book proposes a *Nordic benefit structure* — universal, service-heavy, not means-tested — financed by a *broad consumption and*

property base rather than by payroll.

Universality rather than means-testing, for two reasons that are about Cuba rather than about principle. Cuba already has universality and dismantling it to build a targeting apparatus would destroy a working system to construct one it has no capacity to run. And every means test is a discretionary decision taken by an official about a citizen, which is precisely the surface Chapter 22 spends its length trying to eliminate. In a country where *resolver* is a verb, a benefit that depends on an official's assessment is a benefit that depends on knowing the official.

Against payroll financing as the main instrument, for two more. Coverage would track formal employment in an economy where most new jobs will start informal, which reproduces the Southern European outcome directly. And a high payroll wedge is the most reliable known way to kill small enterprise, which is the sector Chapter 21 needs to grow and which Cuba has spent fifty years failing to have.

Design 14.1. The benefit structure

- (1) **UNIVERSAL IN KIND** Health care and education free at the point of use to every resident, without means test, contribution condition or nationality qualification beyond residence. This is the existing Cuban settlement and it is retained unchanged.
- (2) **UNIVERSAL IN CASH** A universal flat basic pension at a level set to prevent poverty, payable to every resident above the retirement age regardless of contribution record; a universal child benefit; and disability support on assessed need rather than on contribution.
- (3) **EARNINGS REPLACEMENT SEPARATELY** Above the flat tier, an earnings-related pension on individual accounts, funded or notional, with the contribution visible on the payslip. Separating the two tiers is what allows the flat tier to be adequate without being unaffordable.
- (4) **UNEMPLOYMENT** Time-limited unemployment insurance with an active component, introduced in the phase in which state employment is reduced and not before — because it is the instrument that makes that reduction politically possible, and introducing

it late is what turned the 2011 payroll plan into a plan nobody executed.

- (5) **LEGISLATED, NOT ADMINISTERED** Every benefit level is set by legislation and uprated by a published formula. No benefit is variable at administrative discretion. This is the difference between an entitlement and a favour, and per Design 13.1 it is also a commitment device.

14.5 The tax system

This is the operational core of the chapter. Chapter 15 owns stabilisation, the currency and the deficit; this chapter owns what is taxed and at what rate. The two must agree on one number — the revenue share the welfare commitment requires — and Chapter 15 is where it is shown to be financeable without inflation.

The design principle throughout is that a weak administration collects a simple tax and fails at a clever one. Cuba is building a revenue service from almost nothing, in a cash economy, with a demoralised civil service and no filing culture. Every sophistication is a place where the system leaks or where an official acquires discretion.

Design 14.2. The revenue structure

- (1) **BROAD BASE, LOW RATES, NO EXEMPTIONS** Stated first because it governs everything below. Exemptions are how tax systems die: each is small, each is defensible, each creates a boundary to be argued over, and the argument is conducted between a taxpayer and an official with discretion. No sectoral holidays, no negotiated rates, no special regimes for named investors.
- (2) **VAT AS THE WORKHORSE** A single-rate value-added tax, one rate and no reduced rates, with a registration threshold high enough that micro-enterprise stays outside it. Food is *not* zero-rated: zero-rating is regressive per peso of revenue forgone — the rich buy more food than the poor in absolute terms — and the distributional repair belongs on the spending side, through clause 2 of Design 14.1, where it can be aimed.

- (3) **INCOME TAX, WITHHELD AND SIMPLE** Progressive, few brackets, a high exempt threshold so that most workers file nothing, and withholding at source wherever a source exists. The objective in the first decade is coverage and habit, not yield.
- (4) **PROPERTY TAX** An annual tax on residential and commercial property, assessed on area and location by formula rather than by valuation judgment, collected locally and retained locally per Design 13.7. This is the easiest tax to administer, the hardest to evade, and the natural local revenue. It is also the instrument that makes the 1960 Urban Reform's gift pay for the services around it: Cuba has near-universal home ownership and a collapsing housing stock, and the two facts are the same fact.
- (5) **PRESUMPTIVE REGIME FOR MICRO-ENTERPRISE** A single small payment based on turnover, replacing income tax, VAT and social contributions together, on the Latin American *monotributo* model, with a published graduation path into the ordinary regime. Get this wrong and the private sector stays informal permanently, which is the Southern European failure this chapter is designed against.
- (6) **MOBILE BASES TAXED LIGHTLY AND SIMPLY** Remote-work and nomad income under Chapter 18 pays a low flat rate with no deductions. A high rate on a base that can leave collects nothing; a low rate on a base that stays collects something and is worth complying with.
- (7) **RENTS WHERE THEY EXIST** An accommodation levy on visitor nights, lease and concession revenue from state land and coastline, and nickel royalties. See Design 14.3 for what may be done with the proceeds, which is the part that matters.
- (8) **WHAT IS NOT DONE** No wealth tax — unadministrable here and it drives out exactly the diaspora capital Chapter 21 is trying to attract. No exit tax on emigrants, which is a tax on the grievance this book is trying to settle. No high statutory corporate rate with negotiated exemptions, which is a corruption machine with a tax attached. No earmarking beyond Design 14.7, because earmarking fragments a budget and then binds it in the wrong year.

14.6 The rent trap

Parts I and II show the same failure three times: the American sugar quota, the Soviet protocols, the Venezuelan oil. A fourth is available — a tourism boom, a normalisation windfall, a nickel price spike — and if it arrives it will be spent on recurrent social commitments, because that is what every government in this position has done, including this one.

The welfare state must be financed from a broad domestic base *because* a rent-financed one has already died twice in living memory. That is not a preference for taxation on ideological grounds. It is the specific institutional lesson of Chapters 4 and 8.

Design 14.3. The rent rule

- (1) SEPARATION Revenue from concessions, royalties, privatisation proceeds, debt relief and any windfall is paid into a stabilisation and investment fund and is not available to the recurrent budget.
- (2) PERMITTED USES The fund finances capital spending, the pension transition of Design 14.4, and countercyclical support after a hurricane or an external shock. It does not finance wages, benefits or subsidies.
- (3) ANCHOR The recurrent budget balances over the cycle on non-rent revenue alone, with the deficit rule published, monitored by an independent fiscal council, and reported on annually to the assembly.
- (4) ESCAPE The rule may be suspended only for a declared natural disaster or external shock, by supermajority, with an automatic expiry and a published return path. A fiscal rule with a discretionary escape hatch is not a rule, and a fiscal rule with no escape hatch at all does not survive a Category 5 hurricane — which Cuba will have.

14.7 The pension arithmetic

This is the hardest number in the book and the chapter should not pretend otherwise.

Cuban fertility has been below replacement since 1978, is now around 1.2 to 1.4, and there is no cohort in the pipeline. More than a quarter of the population is over sixty. And between one and two million people, disproportionately of working age, left between 2021 and 2025. A pay-as-you-go pension at European replacement rates is not financeable on that dependency ratio at any plausible tax rate, and no amount of design changes that.

What design can do is decide who bears the gap and make the arithmetic visible in advance rather than in a crisis.

Design 14.4. Pensions

- (1) **TWO TIERS** The universal flat pension of Design 14.1, tax-financed, set to prevent poverty and uprated by formula; plus an earnings-related tier on individual accounts. The flat tier is what protects the people Chapter 12 identified as exposed. The earnings tier is what stops the flat tier from having to be generous.
- (2) **RETIREMENT AGE INDEXED** The pension age is indexed to life expectancy by a formula legislated once and applied automatically thereafter. Legislating the formula is politically possible; legislating each increase separately is not, which is why countries that chose the second route do not have the increases.
- (3) **IMMIGRATION IS PENSION POLICY** The cheapest instrument for a dependency ratio is more working-age people. Return migration and immigration are therefore treated in Chapter 18 as fiscal instruments and not as a cultural question, and the residence regime there is designed with this chapter's arithmetic in view.
- (4) **PORTABILITY** Bilateral social-security agreements with the countries holding the diaspora — the United States, Spain, Mexico, Italy — so that contributions made abroad count, returnees are not penalised, and Cubans abroad can contribute voluntarily. This converts a diaspora into a contributor base, which no other instrument does.
- (5) **HONEST ACCOUNTING** An independent actuarial report to the assembly every year, published, stating the projected gap and what closes it. Some of this gap does not close, and a published number is how a country decides which part it will meet by later retirement, which by lower replacement rates, which by higher

taxes and which by migration. Cuba's pension deficit is currently neither published nor projected.

14.8 The staff

A welfare state is people, and this is where the current arrangement is failing fastest. Cuban doctors, nurses and teachers left for the tourism sector in the 1990s and are leaving the country in the 2020s. No financing plan works if the staff are gone, and no plan that treats staffing as free has been costed.

Design 14.5. Public pay

- (1) **BENCHMARK** Public professional pay is set against the domestic private and tourism wage, published, and reviewed annually by an independent body. It is not set against a historic peso scale, which is how a surgeon came to earn less than a bellhop and how the whole society learned that its education system was a joke.
- (2) **CONSISTENCY** Chapter 17 and Chapter 19 are bound by this clause. A tourism strategy that bids up service wages without a corresponding public pay settlement is a strategy for emptying the hospitals, and Cuba has run that experiment.
- (3) **COSTED** The public wage bill under clause 1 is the largest single line in this chapter's cost and is stated as such in Chapter 24's financing table rather than treated as an administrative detail.
- (4) **RETENTION BEFORE RECRUITMENT** In the first phase, spending goes to retaining existing staff rather than to training replacements, because a trained nurse who leaves takes four years to replace and the outflow compounds.

14.9 Delivery, and where the line falls

Universal coverage does not require state monopoly provision, and pretending it does costs Cuba capacity it cannot spare. But a two-

tier system in the wrong place destroys the thing being protected, because once the professional and political classes leave the public system it stops being defended in the budget. So the line has to be drawn explicitly rather than left to practice.

Design 14.6. Public and private provision

- (1) **PRIVATE PROVISION PERMITTED** In dentistry, pharmacy, optical services, elective and cosmetic procedures, eldercare, rehabilitation, and non-compulsory tertiary and vocational education. In these, private and cooperative providers operate alongside the public system, are licensed and inspected, and may contract to it.
- (2) **PUBLIC PROVISION PROTECTED** Primary care, emergency care, maternity, public health and immunisation, and compulsory schooling remain universal, public and free at the point of use, with no parallel private tier permitted to substitute for them. These are the services whose universality produces the outcomes in Chapter 11, and they are the ones a two-tier settlement hollows out first.
- (3) **NO QUEUE-JUMPING** No public facility sells priority access. Health tourism under Chapter 19 operates in designated facilities with ring-fenced staff, and its revenue accrues to the public system — and if that ring-fence cannot be maintained, the programme stops, because Chapter 12 established that staff are the binding scarcity.
- (4) **COOPERATIVES FIRST** Where provision is opened, cooperative and professional-partnership forms are the default and are given the licensing preference, because they distribute the surplus to the practitioners and are the form least likely to concentrate.

14.10 Making it stick

Everything above is a policy until it is entrenched, and Chapter 13 established what happens to Cuban policies.

Design 14.7. The floor

- (1) **CONSTITUTIONAL MINIMUM** A floor on combined health and education spending as a share of general government expenditure, set in the constitution, below which a budget may not be presented. This is the one earmark Design 14.2.8 permits, and it is permitted because it is the commitment device the whole chapter rests on.
- (2) **SEPARATED ACCOUNTS** Social insurance accounts are held separately from the general budget, published in full, and may not be borrowed from.
- (3) **ANNUAL ACTUARIAL REPORT** Per Design 14.4.5, to the assembly, published, independent.
- (4) **JUSTICIABLE** The rights to health care and to compulsory schooling are enforceable by an individual in the administrative court of Design 13.3. Not the level of provision — courts should not set budgets — but the fact of access, and the non-discrimination of it.
- (5) **PUBLISHED OUTCOMES** Health and education outcomes are published annually by province and, per Chapter 12, disaggregated by race. Cuba declared the racial question solved in 1962 and then made it unmeasurable for thirty years, and entered its market opening with no idea where it stood. The first requirement of this design in that domain is to publish the data.

14.11 What this chapter costs the rest of the book

Stated plainly, because a design that hides its trade-offs is a brochure.

A European welfare state means a tax burden well above the Latin American norm. Cuba will be bidding for investment against countries in the same region with lower rates, weaker labour protection and no comparable social spending, and it will lose some of that bidding. Chapter 21 therefore cannot compete on rates and must compete on the other things investors price: certainty, speed of approval, enforceable contracts, an honest customs service, a court

that can be lost in fairly, and a workforce that can read. Those are harder to build than a tax holiday and they are also, unlike a tax holiday, worth having for their own sake.

It also means that the growth chapters have to work. A welfare state at this standard requires a tax base that does not currently exist, which is to say it requires Chapters 16 through 21 to succeed. This chapter is not an alternative to the growth strategy. It is the reason for it.

Chapter 15

Money

15.1 What Ordenamiento got wrong

Chapter 10 described the currency unification of January 2021. This chapter is written against it, so it is worth restating the failure precisely, because the lesson is about sequence and not about direction.

Unification was right and had been right for twenty years. With the household rate twenty-five-fold away from the enterprise rate, no Cuban firm's accounts meant anything, and an economy whose accounts are fiction cannot be reformed even by people who want to reform it. What went wrong was everything around it.

It was done with no reserves, and a fixed rate is a promise to sell dollars at that rate. Cuba had no dollars, so the promise was not kept and a parallel market reopened within weeks — and once a parallel rate exists, the official rate stops being a price and becomes an entitlement, allocated by whoever controls access to it.

It was done with no supply response. Wages and pensions were multiplied several-fold in an economy where the quantity of goods was falling. That is not a distributional decision; it is the definition of an inflation.

It was done with the private sector still shut. The 2017 licence freeze was in force and the law giving private firms legal personality was eight months away, so the one part of the economy capable of

responding to higher prices with more output was legally prevented from doing so.

And it was done at the bottom of the worst external shock since 1993, when the correct order is to stabilise first and unify afterwards.

The most damning comparison is internal. The same state, in 1994–96, with fewer resources and a worse starting position, cut the deficit from around thirty per cent of output to two, brought the informal exchange rate from a hundred and fifty to twenty-five, and stabilised the peso without external assistance — and it did those things *before* it did anything else. Chapter 7 tells that story. Cuba has demonstrated that it knows how to do this in the right order. In 2021 it did not.

15.2 The monetary regime

The book takes a position here, and it is the most contestable recommendation in Part III.

Cuba should officially and unilaterally dollarise, as its entry regime, with a stated route out.

The argument is not that dollarisation is optimal in general. It is not. It is that the alternatives all require something Cuba does not have.

A *managed float with an independent central bank* requires a central bank whose commitments are believed. Cuba's has destroyed its citizens' savings twice in thirty years — in 1993–94 and again in 2021 — and its ability to promise anything about the value of money is, as a matter of observed fact, zero. Credibility of that kind is rebuilt over a decade of demonstrated behaviour, and the design needs a functioning currency in month one.

A *currency board*, on the Baltic or Bulgarian model, keeps a national currency fully backed by reserves. It is the right answer for a country that has reserves. Cuba has none, cannot borrow them, and would have to acquire them before the board could open — which is a five-year detour at best.

A *crawling peg* requires reserves to defend and discretion to adjust, and discretion over an exchange rate in a country with

Cuba's institutional history is a rent, not a policy. Chapter 22 will note that the single largest corruption surface in the Cuban economy for thirty years has been access to foreign currency at the official rate.

Dollarisation is the regime that requires no reserves, no credibility and no discretion, which is exactly the combination Cuba's constraint set specifies. It also happens to be where the country is already going: households price in dollars, remittances arrive in dollars, and since 2024 the state's own shops have been selling in dollars. The proposal is to make official and universal what is already partial and inequitable — because a partly dollarised economy is the worst of all arrangements, dividing the population into those with access to hard currency and those paid in something else, which is precisely the inversion Chapter 7 described and which is happening again.

The costs are real and must be stated. Cuba gives up monetary policy, gives up the exchange rate as an adjustment mechanism, gives up seigniorage, and gives up a lender of last resort. The first two matter less than they would elsewhere: Cuba's export earners — tourism, professional services, nickel, and the remote services of Chapter 18 — price in dollars already, so a devaluation would not buy the competitiveness it buys a manufacturing exporter. Seigniorage on a currency nobody wants to hold is small. The lender of last resort is the genuine loss, and it binds hardest in exactly the scenario Cuba faces regularly — a Category 5 hurricane — which is why Design 14.3's stabilisation fund and the contingent credit line below are not optional extras but the other half of this regime choice.

Ecuador, El Salvador and Panama are the comparators. The Ecuadorean case is the instructive one: dollarisation was adopted in 2000 in the middle of a banking collapse, was expected by most economists to fail, and has survived twenty-five years and several governments that disliked it — because it turned out to be enormously popular with people who had lost their savings, and no government has been able to reverse it. That is a commitment device in the sense of Chapter 13: a policy that becomes hard to undo because the population will not permit it.

Design 15.1. The monetary regime

- (1) **ADOPTION** The United States dollar is adopted as legal tender and unit of account. The Cuban peso is retired on a published schedule at a single conversion rate applied to all balances, wages, prices, contracts and public accounts simultaneously.
- (2) **THE CONVERSION RATE** Set at or close to the prevailing informal market rate, not at an administrative rate. Setting it too strong is what destroys savings; setting it at the market rate merely recognises a loss that has already occurred, which is the difference between a reform and a confiscation.
- (3) **SINGLE RATE, IMMEDIATELY** There is no transitional dual rate, no preferential rate for state enterprises, no allocation of currency at a privileged rate to anybody, for any purpose, including food and fuel imports. Every such arrangement in Cuban history has become a rent.
- (4) **WAGE AND PENSION PROTECTION** Public wages and pensions are converted at the same rate and then re-set under Design 14.5, with a floor guaranteeing that no pensioner's real income falls at conversion. This is the clause that makes the regime survivable, and it is costed in Chapter 24.
- (5) **THE EXIT** A published set of conditions — reserves, fiscal record, inflation, an established supervisory authority — on which a national currency may later be reintroduced, together with the procedure. Naming the exit is what distinguishes a considered choice from a surrender, and it costs nothing to write.

Design 15.2. What replaces the central bank

- (1) **FUNCTION** Under dollarisation the central bank does not issue money. It becomes a bank supervisor, a payments authority and the manager of the stabilisation fund, with tenure, budget and reporting on the model of Design 13.3.
- (2) **NO FISCAL FINANCING** The prohibition on lending to the government is absolute and constitutional. This is the single most important monetary clause in the book, because monetary financing of the budget is how Cuba paid for its welfare state in the

years when neither Moscow nor Caracas was paying, and it is what Chapter 14 is designed to replace.

- (3) **LIQUIDITY** A liquidity facility funded from the stabilisation fund and, if obtainable, a contingent credit line with a multilateral or bilateral partner — the substitute for the lender of last resort that clause 1 gives up. Obtaining it is a first-order objective of the debt normalisation below, not an afterthought.
- (4) **BANKING SUPERVISION** Capital, liquidity and disclosure standards on international norms, applied to state and private banks identically, with a resolution regime that exists before it is needed.

15.3 Debt, arrears and claims

Cuba owes money to almost everyone and has not paid most of them. The stock is several distinct problems that are conventionally handled separately and that this book proposes to handle in one operation.

Paris Club official debt. The 2015 arrangement forgave a large share of accumulated arrears and rescheduled the remainder over eighteen years. Cuba began missing payments within three years and the deal is in default.

Russian claims. Ninety per cent of the Soviet-era debt was written off in 2014; the remainder was converted into investment commitments, largely unrealised.

Chinese and other bilateral claims. Substantially undisclosed, which is itself a problem: a restructuring cannot be negotiated against an unknown liability, and the first act of any process is to publish the stock.

Supplier arrears. Unpaid invoices to foreign trading partners, in some cases running to 2009. These are commercially the most damaging, because the creditors are the same firms who would otherwise supply the imports the recovery needs.

Frozen accounts. Balances of foreign companies blocked from 2009 onward. Legally these are not debt; commercially they are the reason nobody trusts a Cuban account.

Expropriation claims. The Foreign Claims Settlement Commission registered 8,821 reports of seized assets and certified 5,913 of them, at a principal of \$1.9 billion. At the 6 per cent simple interest the governing statute specifies, that is now valued at something over \$9 billion (Foreign Claims Settlement Commission of the United States, 1972). Only about 913 of the certified claims — some 15 per cent — are thought to meet the conditions for a Title III suit. Alongside them sits a far larger and uncertified body of claims by Cuban Americans and by Cubans who never left.

Design 15.3. One restructuring, not six

- (1) **PUBLISH FIRST** A complete, audited public statement of every external obligation — official, bilateral, commercial, arrears, frozen balances and contingent — before any negotiation opens. No restructuring has ever succeeded against an undisclosed stock, and the disclosure is also the first act of the transparency regime in Chapter 22.
- (2) **SINGLE PROCESS** One comparable-treatment negotiation covering all classes of claim, including the expropriation claims. Settling the sovereign debt while leaving the expropriation claims outstanding leaves every foreign investor exposed to Title III litigation and therefore leaves the country uninvestable; settling the claims outside a general restructuring gives one class of creditor priority and collapses the rest.
- (3) **INSTRUMENTS** Long-dated, low-coupon bonds with a growth-linked component, so that creditors participate in a recovery they are helping to finance and the schedule does not bind in the years when the plan needs cash. Not a haircut disguised as a maturity extension — state the reduction and negotiate it.
- (4) **CLAIMS SETTLED IN PAPER, WITHIN A CAP** Per Chapter 21, the expropriation claims are settled by compensation, not restitution, in the same instruments, within an aggregate cap that is announced in advance. No Cuban is removed from a dwelling.
- (5) **MULTILATERAL MEMBERSHIP** Accession to the International Monetary Fund and the multilateral development banks is pursued in parallel and is an objective of the restructuring rather than a consequence of it, because their technical assistance and their

contingent lending are what clause 3 of Design 15.2 requires.

- (6) **BEHAVIOUR BEFORE TERMS** The frozen balances of Chapter 8 are released and current supplier invoices are paid on time from the first month, before and irrespective of any negotiation. A credit history is repaired by conduct over years, not by a signed agreement, and this is the cheapest credibility available.

15.4 The budget

Chapter 14 owns what is taxed. This chapter owns the deficit, how it is financed, and the anchor that keeps the arrangement from unwinding.

Design 15.4. Fiscal architecture

- (1) **ONE BUDGET** All state revenue and spending, including that of state enterprises and of the military-linked conglomerates, appears in a single published budget on international classification standards. At present a large share of Cuban public finance is off-budget, which means nobody — including the government — knows the fiscal position.
- (2) **NO MONETARY FINANCING** Per Design 15.2.2, and under dollarisation it is mechanically impossible as well as prohibited, which is one of the regime's principal attractions.
- (3) **ANCHOR** The recurrent balance rule of Design 14.3.3, with an independent fiscal council reporting annually to the assembly and publishing its own forecasts.
- (4) **ENTERPRISE HARD BUDGETS** State enterprises face a hard budget constraint with a published insolvency procedure. Subsidy to a loss-making enterprise, where it continues, is an explicit and visible budget line, not a bank overdraft that never gets called.
- (5) **SEQUENCED** The budget is brought into balance before the tax system is capable of financing the welfare commitment, which means the first phase runs on expenditure discipline and the rent rule, and the convergence to Chapter 14's benefit levels waits on

the base. Saying which of those comes first is the whole content of the word “sequence”.

15.5 The ration book

The *libreta* was introduced in March 1962 as a temporary measure and is sixty-four years old. It is a universal in-kind subsidy, badly targeted — the minister and the pensioner receive the same allocation — expensive in foreign exchange, and increasingly not delivered. It is also, for a substantial number of elderly and eastern households, the difference between eating and not.

The transition is a genuine design problem rather than a matter of political will, and getting it wrong starves people.

Design 15.5. From ration to income

- (1) NOTHING IS WITHDRAWN BEFORE THE REPLACEMENT WORKS The universal cash benefits of Design 14.1 must be paid, in full, reliably, through a payment system that reaches every municipality, for a minimum of two consecutive years before any ration item is removed.
- (2) REPLACE, DO NOT MEANS-TEST The *libreta* is converted into cash paid to every household, not into a targeted benefit. Universality is retained for the reason in Chapter 14: a means test is a discretionary decision and Cuba cannot administer one honestly.
- (3) INDEXED The cash amount is indexed to a published basket price, so that the conversion is not a one-off transfer whose value inflation removes — which is what happened to the wage increases of January 2021.
- (4) SEQUENCE BY ITEM Items are converted individually, beginning with those whose market supply is adequate and ending with those whose supply depends on Chapter 20 succeeding. Rice, oil and milk go last.
- (5) REVERSIBLE If the replacement fails — if supply collapses or the payment system does not reach a province — the ration is restored

for that item and that province, automatically, on a published trigger. This is the one place in Part III where reversibility is a feature.

15.6 Payments

The least glamorous section in Part III and among the most binding.

A tourist cannot pay in a Cuban restaurant with an ordinary card. A remote worker in Havana cannot receive money from a client in Madrid into an account they control. A private firm cannot pay a foreign supplier. A pensioner in Baracoa cannot receive a transfer without a physical office. Chapter 17's value-per-visitor strategy, Chapter 18's entire proposition, and Design 15.5's conversion all fail without payment rails, and none of them can be built until the rails exist.

Design 15.6. Payment rails

- (1) **CORRESPONDENT BANKING** Re-establishing correspondent relationships with international banks is a first-order objective and the principal practical reason for Design 15.3's clause 6 and for the anti-money-laundering standards in clause 4 below. Under the embargo this is difficult and, for non-United States institutions, not impossible.
- (2) **CARDS AND ACCEPTANCE** Card acceptance at any licensed business, with the merchant fee capped and the settlement period fixed in regulation. A private restaurant that can take a card is worth several times one that cannot.
- (3) **THE RIGHT TO HOLD AND BE PAID** Any resident may hold a foreign-currency account, receive international transfers into it, and spend from it without administrative permission. This is the clause Chapter 18 depends on entirely, and it is currently the binding constraint on Cuban service exports — more so than internet speed and far more so than visas.
- (4) **STANDARDS** Anti-money-laundering and beneficial-ownership

standards to international norms, supervised by the authority in Design 15.2. These are not a concession to foreigners; they are the price of the correspondent relationships in clause 1, and they are the same register that Chapter 22 needs for domestic reasons.

- (5) REACH A basic transaction account available to every resident, including by mobile, at every municipality. Financial inclusion is a precondition for Design 15.5 and for the property tax of Design 14.2.4, and without it both are administratively impossible in the east.

15.7 The first five hundred days

The order, compressed. Chapter 24 places this inside the whole programme; here is the monetary sequence alone.

Days 1–100. Publish the complete debt and arrears stock. Pay current supplier invoices. Release the frozen balances. Announce the dollarisation date and the conversion rate. Legislate the no-monetary-financing rule and the recurrent balance rule. Open the debt negotiation and the IMF accession conversation.

Days 100–250. Convert. Single rate, all balances at once, wages and pensions protected at the floor. Begin card acceptance and the basic account rollout.

Days 250–500. Stand up the revenue administration — registration, taxpayer numbers, withholding on the state payroll, the presumptive regime for micro-enterprise — knowing that it yields little for years. Begin the VAT build. Establish the fiscal council and publish its first report. Begin the libreta conversion on the two or three items whose market supply is adequate.

What is deliberately *not* in the first five hundred days: raising benefit levels to European standards, mass privatisation, and the withdrawal of any ration item whose replacement is not already being paid. Each of those is a later phase, and attempting them here is how the sequence fails.

Chapter 16

Sun

16.1 Why this chapter comes first

Chapter 12 named electricity the binding constraint. It is worth being concrete about what that means, because it is not a figure of speech.

Cuban water is pumped, so no power means no water. Cuban food is refrigerated in a climate where meat spoils in hours, so no power means the cold chain fails and the food is lost between the field and the plate. Hospitals cannot run a surgical list against a schedule of outages. A hotel without power or water does not merely disappoint a guest; it produces the review that removes the next fifty guests. A remote worker with no electricity has no job. A bakery with no electricity does not bake. And a grid that has failed as a system four times fails more easily each time, because every black start damages equipment and every restoration is attempted with a fleet in worse condition than the last.

So the ordering in Part III is not a preference. Every sector chapter after this one is conditional on it.

16.2 Where the grid is

Cuban generation is a small number of large oil-fired thermal units, most commissioned between the 1960s and the 1980s, running past their design lives on heavy, high-sulphur domestic crude they were not designed to burn, without spares and frequently without fuel. Alongside them sit the distributed diesel and fuel-oil sets installed in the Energy Revolution of 2006, now twenty years old, and a set of floating Turkish power barges hired at hard-currency rates the country struggles to pay.

In a normal pre-crisis year the system generated on the order of 20 terawatt-hours and peaked around 3,000 to 3,300 megawatts. By 2023 gross generation was down to 15.3 terawatt-hours, and consumption per head had fallen from about 1,856 kilowatt-hours in 2018 to 1,387 — a quarter less electricity per person in five years, which is not conservation but rationing. Through 2023–25 the daily shortfall routinely ran a third of demand or more, with blackouts of twelve to twenty hours outside Havana, and the national grid collapsed entirely in October, November and December of 2024 and again in 2025.

The fuel is the recurring cost that matters. At the fleet's thermal efficiency — somewhere near 30 per cent — 20 terawatt-hours of electricity requires roughly six million tonnes of fuel oil equivalent a year. Cuba produces perhaps half of that domestically, as heavy crude suitable for little else, and must buy the rest — which is why the electricity system, the balance of payments and the political relationship with whoever is supplying the oil have been the same problem since 1960.

16.3 The resource, and the arithmetic

Cuba sits between 20 and 23 degrees north with high, even insolation and very little seasonal variation: global horizontal irradiance around 5.0 to 5.5 kilowatt-hours per square metre per day, which for a well-built fixed-tilt photovoltaic system in a hot, humid climate yields something in the range of 1,500 to 1,700 kilowatt-hours per

installed kilowatt-peak per year after temperature derating and system losses (Solargis and World Bank Group, 2020). The absence of a winter is worth more than the peak irradiance: a Cuban array in January produces most of what it produces in June, which removes the seasonal storage problem that dominates temperate-latitude planning.

What it does not remove is the daily one. Cuban demand peaks between about six and ten in the evening, when air conditioning, cooking and lighting coincide and the sun has set. *This is a storage problem at least as much as a generation problem* (Fig. 16.1a), and any programme that installs photovoltaics without storage will find that it has displaced midday fuel, which is the cheapest fuel to displace, and left the evening — the hours that determine whether the grid holds — exactly as it was.

Here is the calculation, with its assumptions on the page so that a reader can change them.

Target: half of a 20 TWh year from PV	10 TWh/yr
Specific yield (assumed, conservative)	1,600 kWh/kWp/yr
Capacity required	≈ 6 GWp
Land at 1.5 ha/MWp	≈ 90 km ²
share of Cuban territory	≈ 0.08%
Capital at \$0.80/Wp	≈ \$5 bn
Evening storage, 4 h × 2 GW	8 GWh
Storage at \$200/kWh	≈ \$1.6 bn
Total capital, over a decade	\$6–8 bn
Fuel displaced at 30% efficiency	≈ 3 Mt/yr
Its value at \$400–500/t	\$1.2–1.5 bn/yr
Simple payback	5–6 years

Every figure above is an order of magnitude; each is marked unverified in the manuscript and listed in `verify.md`. Three things are deliberately excluded and would raise the cost: transmission and distribution reinforcement, which Cuba needs anyway and which is a large number; the thermal fleet, which must be kept running for firm capacity through the whole transition and therefore still needs fuel, spares and maintenance; and the cost of capital, which for a defaulted sovereign is the hardest input in the table and the

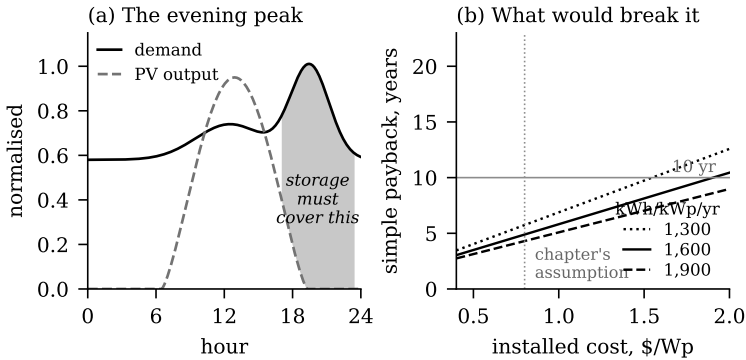


Figure 16.1: (a) Why the storage is not optional. Cuban demand peaks after sunset, so photovoltaic capacity on its own displaces midday fuel — the cheapest fuel to displace — and leaves the hours that decide whether the grid holds exactly as they were. The load shape here is schematic; obtaining Cuba’s actual hourly demand curve is the first thing that should be done to this chapter, and `verify.md` records it. (b) The sensitivity the case actually turns on. Below roughly \$1.50 per watt-peak the programme pays for itself inside a decade across the whole plausible range of Cuban yields; above it, the argument weakens and would need to be remade on grounds other than fuel saving. Generated by `figs/solar.py`.

reason Chapter 15’s debt normalisation is a precondition rather than a parallel exercise.

But the shape of the answer survives all of that, and the shape is the point. The land requirement is a rounding error — less than a tenth of one per cent of the country, against something between one and two million hectares currently under *marabú* scrub. The capital cost is large but is of the same order as a decade of the fuel bill it eliminates. And the recurring saving is in foreign exchange, which is the scarcest thing Cuba has, converting the country’s largest recurring hard-currency cost into a domestic asset with no fuel input at all.

That is the case, and it is a strong one. It was not a strong one in 2010, when modules cost five times as much and batteries fifteen. The reason to do this now is that the arithmetic changed.

16.4 Hurricanes

This is the design constraint that most solar advocacy for the Caribbean omits, and omitting it is how a country spends five billion dollars and loses it in an afternoon.

Cuba is hit by major hurricanes regularly and will be hit by stronger ones. The lesson available from Puerto Rico after María in 2017 is specific and usable: photovoltaic installations failed at very different rates depending on how they were engineered, and the failures were overwhelmingly mounting failures — racking, fasteners, tracker stow — rather than module failures. Systems built to genuine high-wind specification largely survived; systems built to mainland-United States standard specification largely did not.

Design 16.1. Building for the storm

- (1) **SPECIFICATION** All installations are engineered to Category 5 wind loading as a minimum, with mounting, fastening and foundation design certified to that standard, and the certification published. This adds materially to capital cost and is not optional; the alternative is to pay for the array twice.
- (2) **DISTRIBUTION** Capacity is distributed across provinces and across many sites rather than concentrated in a few large parks. A hurricane is a spatially correlated risk, and the entire point of moving off a fleet of six large thermal plants is not to rebuild that concentration in photovoltaics.
- (3) **ROOFTOPS COUNT DOUBLE** Distributed rooftop generation on hospitals, pumping stations, bakeries, telecoms sites and cold stores is resilient in a way central capacity is not: it survives transmission damage, which is what actually fails in a Cuban hurricane, and it keeps the critical load running while the grid is restored.
- (4) **SPARES HELD, NOT ORDERED** A national inventory of modules,

inverters, racking components and cable is held in country, in hardened storage, sized against a modelled event. Post-storm procurement by a country with no credit takes months.

- (5) **SELF-INSURANCE** Commercial insurance for Cuban assets is expensive and in many markets unobtainable. The stabilisation fund of Design 14.3 carries an explicit reserve against grid and generation losses, and the reserve is sized and published.

16.5 Who is allowed to own a generator

The technology is the easy part. The reason Cuba has almost no distributed generation despite a decade of favourable economics is legal: the national electricity utility is a monopoly, private generation is permitted narrowly and by permission, there is no bankable framework under which an independent producer can sell power, and until very recently a household or private business had no right to install a panel and use what it produced.

The solar parks now being installed with Chinese financing are the right technology inside the wrong institution: capacity procured centrally, owned by the state, connected to a grid the state runs, financed by a bilateral relationship of exactly the kind Chapters 4 and 8 warned about. They will help. They will not, by themselves, solve this, and they do nothing at all about the evening peak.

Design 16.2. Generation rights

- (1) **ROOFTOP BY RIGHT** Any owner or lawful occupier may install generation and storage on their own premises and consume what it produces, without permission, subject only to published safety and interconnection standards. Not by licence, not by category, not by application — by right, per Design 13.1.2.
- (2) **NET BILLING** Surplus exported to the grid is purchased at a published tariff set by the regulator, paid in the currency of Design 15.1, with settlement periods fixed in regulation. Net *billing* rather than net metering, because the value of exported

midday power is genuinely lower than the value of consumed evening power and pretending otherwise builds a subsidy that later has to be withdrawn.

- (3) **INDEPENDENT PRODUCERS** A standard-form power purchase agreement, published, non-negotiable in its core terms, with international arbitration and a payment guarantee. A one-off negotiated PPA in a country with Cuba's payment record is not bankable; a standard instrument with a guarantee behind it can be.
- (4) **REGULATOR** An independent energy regulator with the tenure, budget and publication obligations of Design 13.3, separate from the utility it regulates and from the ministry that owns the utility. At present the operator, the owner, the planner and the regulator are the same entity, which is the same conflict of interest Chapter 19 identifies in pharmaceuticals.
- (5) **UNBUNDLING, EVENTUALLY** Transmission and system operation separated from generation, with open, non-discriminatory access to the network. This is a later phase — it is hard, and doing it badly is worse than not doing it — but it is stated here because the interconnection standards written in phase one determine whether it is possible in phase three.
- (6) **IMPORT FREEDOM** Panels, inverters, batteries, cable and mounting hardware enter free of duty and without licence, for anybody. Cuba currently has a population willing to buy its own solar equipment with remittance money and a customs regime that makes doing so an ordeal. This clause costs the state nothing and is the single fastest megawatt available.

16.6 Biomass, and what is left of sugar

Bagasse — the fibre left after cane is crushed — is the one dispatchable renewable Cuba already knows how to burn, and the 2020 plant at Ciro Redondo demonstrated that a modern bagasse unit can export power to the grid rather than merely running its own mill.

The problem is that the resource has gone. Chapter 10 recorded

a harvest in the low hundreds of thousands of tonnes; there is no bagasse because there is no cane. So biomass in Cuba is hostage to a decision that belongs to Chapter 20, and this chapter should not plan around a feedstock whose existence depends on reviving an industry that mostly is not coming back.

What is available regardless is *marabú*. The thorny scrub that has taken over between one and two million hectares of Cuban farmland is a dense hardwood, it already supports a small charcoal export, and clearing it serves two purposes at once — it recovers agricultural land and it produces fuel. It is not a large power resource and should not be sold as one. It is a genuine transitional one, and it is the rare case where the disposal cost of a problem is negative.

16.7 The order of operations

The last section, and the most important, because a five-billion-dollar programme takes a decade and Cuba's problem is now.

Before anything is built. The cheapest megawatt is the one not consumed, and the second cheapest is the one not lost on the way. In 2023 Cuba generated 15.3 terawatt-hours and lost 3.7 of them — close to a quarter — in transmission and distribution before they reached a customer. That is an extraordinary figure by any standard, and it means that repairing the network delivers usable energy at a fraction of the cost per kilowatt-hour of building new capacity to generate power that will then be lost in the same wires.

Alongside it, demand. Cuba has done this before and done it well: the 2006 programme replaced nine million incandescent bulbs and millions of inefficient appliances and cut consumption measurably. Two decades on the appliance stock is old again and the same intervention is available at lower cost with better technology. Efficient lighting, refrigeration and air conditioning, industrial motor replacement, and loss reduction are all faster, cheaper and less capital-hungry than generation, and they can begin in month one. A programme that goes straight to building solar parks while a quarter of the output leaks out of the network is buying generation it will not be able to deliver.

Critical loads first. Before utility-scale anything, put photo-voltaics and batteries on the loads whose failure is catastrophic rather than merely expensive: hospitals, water pumping stations, cold stores, bakeries, telecoms. These are small, fast, high-value, hurricane-resilient, and they buy the programme public support that a distant solar park does not.

Keep the thermal fleet alive. This is unglamorous and unavoidable. The existing plants must provide firm capacity for the whole of the transition, which means fuel, spares and maintenance have to be funded now, at the same time as the new capacity is being financed. A plan that starves the old fleet to pay for the new one produces a worse blackout in year three.

Then capacity, with storage, distributed. Per the arithmetic above and Designs 16.1 and 16.2, over roughly a decade, with the storage procured alongside the generation and not after it.

Design 16.3. The energy sequence

- (1) MONTHS 0–6 Import liberalisation for equipment (Design 16.2.6); rooftop rights by decree; efficiency programme launched; critical-load installations begun; fuel and spares for the existing fleet funded as a first call on the budget.
- (2) MONTHS 6–24 Regulator established; standard PPA published; net billing tariff set; interconnection standards issued; first independent producer tenders, competitively bid with published results per Chapter 22.
- (3) YEARS 2–10 Roughly 6 GWp and 8 GWh built out, distributed across provinces, with the eastern provinces weighted first — they have the worst supply, the most land and the least political voice, and per Design 13.7.2 the design does not let the benefits accrue where the tourists are.
- (4) THROUGHOUT No preferential allocation of capacity, fuel, currency or grid access to any state or military-linked entity. Energy is where Cuban discretion has been most valuable and therefore most corrupting, and Chapter 22 treats generation procurement as a priority publication target.

Chapter 17

Visitors

17.1 The industry Cuba built, and what it cost

Cuba rebuilt tourism from almost nothing after 1990 — 340,000 visitors in that year, 4.7 million in 2018 — and by any conventional measure that is a success. The trouble is what happened next, and it is the clearest case study in this book of investment allocated by power rather than by return.

Through the decade to 2024 the Cuban state, principally through the military-linked conglomerate GAESA and its hotel arm, built hotel rooms at a sustained and increasing rate while occupancy collapsed.

The 2024 figures are worth setting out in full, and in shares rather than in pesos, because shares are immune to the exchange-rate problem of the note on sources. Of total national investment that year, business services, real estate and rentals took 24.9 billion pesos and hotels and restaurants a further 11.9 billion — together 37.4 per cent of everything the country invested. Agriculture, livestock and forestry took 2.6 billion, or 2.7 per cent. Public health and social assistance took 2.0 billion, less than a fifth of what went into hotels and restaurants alone. As the economist Pedro Monreal put

it, agricultural investment was fourteen times smaller than tourism's (Oficina Nacional de Estadística e Información, 2024b).

That is the allocation of a country importing most of its food, unable to light its cities, appealing to the World Food Programme for children's milk, and running its existing hotels at occupancy rates reported around a quarter. New hotels opened into a market with no guests while the pumping stations that supply those same hotels with water failed for want of parts.

No market signal produced that, and no market signal could have. It was produced by an institution that could allocate capital to itself, without publishing a return, without competing for the funds, and without anyone in a position to say no. Chapter 22 uses this as its worked example, because it demonstrates that Cuba's investment problem is not primarily a shortage of capital. It is that the capital which exists is allocated by whoever controls it, to whatever they control.

Two other features of the industry as built matter for what follows.

It was an *enclave*. The dominant form is the all-inclusive beach resort at Varadero and the northern cays, physically separated from the country around it, supplied largely with imported food, staffed by workers paid in pesos through a state agency that keeps the currency spread, and — until 2008 — legally closed to Cubans. The Caribbean-wide evidence on this model is that the share of the visitor's spending which stays in the host country is low, commonly cited in the range of a fifth to a third for all-inclusives, with the remainder accruing to the foreign tour operator, the foreign airline and the imported supply chain. Cuba adopted the model that leaks most.

And it carries a memory. Chapter 1 described the Havana of the 1950s: casinos, organised crime, a pleasure district for foreigners in a country whose own people were excluded. The revolution's founding claim included abolishing that. Any proposal to make tourism an anchor industry has to answer the objection that it recreates it, and the answer cannot be that times have changed. It has to be a design in which Cubans own the businesses, are not excluded from the spaces, and capture the income.

17.2 The strategy: value per visitor, not visitors

The book proposes an explicit reversal of the metric the industry has been run on.

Cuba should not pursue arrival numbers. It should pursue *net foreign exchange retained per visitor*, and should be willing to accept fewer visitors to raise it.

The reasoning is partly environmental and partly arithmetic. Chapter 23 sets out what mass tourism does to reefs, wetlands and colonial fabric, and Cuba's comparative advantage is precisely the thing that volume destroys. But the arithmetic is sufficient on its own: a visitor in a private room eating in private restaurants and buying from private guides leaves several times as much in Cuban hands as a visitor in an all-inclusive, even when the second one paid more for the holiday. On a labour-constrained island with a failing grid, adding visitors is expensive — each one needs water, power, food and staff, all of which are scarce — while adding value per visitor is nearly free.

This also resolves the tension with Chapter 12's labour constraint. Cuba does not have the people for a Cancún. It does have the people for something better paid.

Design 17.1. The visitor economy

- (1) **THE METRIC** The published objective of tourism policy is net foreign exchange retained per visitor-night, reported annually with its components. No arrival target is set or published. What a state measures is what it builds, and Cuba built rooms because it counted arrivals.
- (2) **PRIVATE BY DEFAULT** Accommodation, food service, guiding, transport and retail to visitors are open to private and cooperative providers by right, under Design 13.1.2, with the state's role confined to standards, safety and inspection. The *casa particular* and the *paladar* are the default form of the Cuban tourism industry, not its tolerated margin.
- (3) **DIRECT EMPLOYMENT AND DIRECT PAYMENT** The state employ-

ment agency arrangement is abolished for this sector as for all others (Chapter 21): hotels hire their own staff and pay them directly. The currency spread taken from tourism workers is the industry's founding injustice and it is also why the service is indifferent.

- (4) **LOCAL PROCUREMENT** Accommodation businesses above a size threshold must publish the share of food purchased domestically, and the accommodation levy of Design 14.2.7 is set at two rates, the lower applying above a published domestic-procurement threshold. This is the instrument that connects Chapter 20 to this chapter: Cuban agriculture's most reliable potential customer is standing in a hotel two hours away, buying imported tomatoes.
- (5) **NO NEW STATE HOTELS** A moratorium on state hotel construction until system-wide occupancy exceeds a published threshold for two consecutive years. The capital released is redirected under Chapter 24's financing table. This is the single largest reallocation of investment available to Cuba and it requires no foreign money at all.
- (6) **OPEN TO CUBANS** No facility, beach, cay or district is closed to Cuban residents, in law or in practice, and the prohibition is justiciable under Design 13.3.4.

17.3 What Cuba actually has

The asset is not beach. Every country in the Caribbean has beach, several have better, and competing on it means competing on price with places that have better airports and working electricity.

What Cuba has that the competition does not is a combination that cannot be built: nine UNESCO World Heritage sites; four intact colonial cities; a musical culture that is a destination in itself; some of the best-preserved coral in the hemisphere at the Jardines de la Reina; the largest wetland in the insular Caribbean at Zapata, with the endemic bird list that goes with it; five thousand seven hundred kilometres of coast, much of it undeveloped; and — unusually and importantly — a very low violent crime rate, which is a decisive advantage in a region where several competitors have the opposite

reputation.

The products that follow from that list are diving, birding and nature tourism, cultural and musical tourism, architectural and heritage tourism, medical tourism under Chapter 19, and long-stay visitors under Chapter 18 — all of which are higher-value, lower-volume, less seasonal and less environmentally destructive than beach resorts, and all of which are currently underdeveloped or absent.

17.3.1 The Old Havana precedent

Cuba has already built the best heritage institution in the Caribbean and then dismantled it, and the episode is instructive in both directions.

The Office of the City Historian of Havana, under Eusebio Leal, was given in 1993 something no other Cuban body had: the right to run commercial enterprises in Old Havana — hotels, restaurants, shops — and to *retain and reinvest its own revenue* in restoring the district. It worked. Old Havana's restoration is the most successful physical project of the Cuban period, it was self-financing, it employed and trained local residents, it restored housing alongside monuments rather than displacing people, and it produced a district that is now the single strongest tourism asset in the country.

In 2016 its commercial arm was transferred to the military conglomerate, and the Office's independent revenue base with it.

Both halves belong in the design. The mechanism — a heritage authority that keeps its own earnings and is accountable for the fabric — is correct and should be extended to Trinidad, Cienfuegos, Camagüey and Santiago. And the reversal is the reason it has to be entrenched under Design 13.1 rather than granted, because it was granted, it worked for twenty-three years, and it was taken away.

Design 17.2. Heritage authorities

- (1) **FORM** A statutory heritage authority for each historic city, with jurisdiction over planning consent, building standards and public space in its district.
- (2) **REVENUE** It retains a defined share of the accommodation levy

and of concession income raised in its district, by statute and not by grant, and this share may be altered only by legislation.

- (3) **OBLIGATION** A published, audited restoration programme, and a binding requirement that a fixed share of restored floorspace remains residential and that existing residents have a right to rehousing within the district. Heritage tourism that empties a district of its inhabitants produces a stage set and loses the thing visitors came for.
- (4) **INDEPENDENCE** Its commercial arm may not be transferred to any other entity without legislation. Written because this is exactly what happened.

17.4 Carrying capacity

Chapter 23 makes the environmental argument at length. What belongs here is that it is an economic argument, not a sentimental one.

The Cuban reefs, the Zapata wetland and the cays are the physical stock that the high-value products above are sold from. A reef that is bleached, dynamited by anchors or silted by coastal construction stops producing revenue permanently, and the revenue it was producing was the highest per visitor in the industry. Diving tourism at the Jardines de la Reina works precisely because the fishing was excluded and the sharks are still there; it is worth more per visitor-day than any beach resort in the country and it exists only because someone drew a line on a map and enforced it.

Design 17.3. Limits

- (1) **SITE LIMITS** Published daily and annual visitor limits for reef, cay and protected-area sites, set by the environmental regulator of Chapter 23 and not by the tourism ministry, with access allocated by published auction or ballot rather than by administrative permission.
- (2) **COASTAL SETBACK** A statutory construction setback from the

shoreline, applied without exception, including to existing state and military-linked operators. Chapter 23 sets the distance; this clause is about the absence of exceptions, which is where every Caribbean setback rule has failed.

- (3) **NO NEW CAY CAUSEWAYS** The *pedraplenes* that connect the northern cays to the mainland have measurably damaged water circulation and the ecosystems behind them. No further ones are built, and remediation of the existing ones is assessed.
- (4) **PRICE THE SCARCITY** Where a site is at its limit, the access price rises. This is the mechanism that converts an environmental constraint into revenue under Design 17.1.1 rather than into a queue.

17.5 Dutch disease, and the wage problem

A successful tourism sector earning foreign exchange will, under Chapter 15's dollarised regime, raise domestic prices and wages in the sectors that serve it and squeeze everything that competes with imports — which is agriculture, which is the sector Chapter 20 needs to grow. This is the standard resource-boom pathology and it is aggravated here by the labour constraint: there is no reserve of workers to expand into the boom sector, so it draws them from the others.

Cuba has already run this experiment. In the 1990s the tourism wage drew doctors into taxi driving and surgeons into renting rooms, which is where the inverted pyramid of Chapter 7 came from.

Design 17.4. Not repeating the 1990s

- (1) **PUBLIC PAY IS THE CONTROL** Design 14.5 binds here. Public professional pay is benchmarked against the tourism and private wage and moves with it. A tourism strategy adopted without the corresponding public pay settlement is a strategy for emptying the hospitals, and this book will not propose one.

- (2) **REVENUE TO THE FUND** The accommodation levy and concession revenue are rent under Design 14.3 and go to the stabilisation and investment fund, not to the recurrent budget — which both dampens the domestic demand impulse and stops a tourism boom from financing recurrent commitments that a bust cannot sustain.
- (3) **LINKAGE OVER EXPANSION** Policy effort goes into raising the domestic content of each visitor's spending — food, furniture, textiles, crafts, guiding — rather than into raising visitor numbers. Linkage is the instrument that turns a boom sector into a diversified one, and it is the reason Design 17.1.4 is written as a tax differential rather than as an exhortation.
- (4) **REGIONAL WEIGHTING** Tourism development incentives are weighted toward the eastern provinces, per Design 13.7.2. Left alone, the industry concentrates in Havana, Varadero and the northern cays, and the equalisation formula then has to do all the work.

Chapter 18

The second track

18.1 The claim, stated carefully

This is the most speculative chapter in the book and it should announce that before it argues anything.

The claim is that Cuba can build a second export sector, alongside tourism, out of educated people selling services across a border without leaving the island — software, engineering, design, back-office work, data work, Spanish-language content, remote professional services — and that this sector could plausibly reach the same order of magnitude as tourism within a decade.

The claim is not that Cuba becomes an artificial-intelligence power. It will not. Training large models requires electricity Cuba does not have, capital it cannot raise, and hardware it cannot legally buy. Any proposal that has Cuba building data centres is not a proposal about Cuba. What Cuba can plausibly do is supply *labour* into the global services market that the current technology wave has expanded — the annotation, evaluation, localisation, integration, tooling and ordinary software engineering that sit around the models rather than inside them — and it can do so in Spanish, in the New York time zone, with a workforce that can read.

The arithmetic is worth putting on the page because it is the reason the chapter exists. A hundred thousand Cubans — about

two per cent of the labour force — earning an average of eighteen thousand dollars a year in remote services would generate roughly \$1.8 billion annually in foreign exchange. That is the same order as Cuban tourism earnings in a good year, from a sector requiring no hotels, no beach, no water supply, no fuel and almost no capital. The per-worker figure is well below what a comparable worker earns in a high-income country and well above anything available in Cuba today, which is precisely why the market would clear.

18.2 Why it has not happened

Cuba has had educated, underemployed, English-capable technical workers for thirty years, and the global market for remote services has existed for twenty. The sector did not appear. The reasons are specific, and identifying the binding one is the whole content of this chapter.

It is not talent. Cuban technical education is the asset Chapter 3 described, and Cuban programmers working abroad are competitive.

It is not, primarily, the embargo. American clients are constrained, but Spain, Mexico, Argentina, Canada, Germany and the entire Spanish-speaking world are not, and none of them buys Cuban services at scale either.

It is not, mainly, connectivity — though connectivity is a real constraint and the next section is about it. Plenty of countries export services on worse bandwidth than Havana has had since 2018.

The binding constraint is that a Cuban cannot lawfully and reliably contract with a foreign client and be paid into an account they control. Until 2021 a Cuban private operator had no legal personality and so could not sign a contract as an entity. The permitted-activity list was written in the vocabulary of trades and had no category for a software developer or a design studio. Foreign-currency accounts were restricted. International card and transfer rails did not reach individuals. And professionals trained at state expense were, for much of the period, prohibited from practising their profession privately at all.

The result is that Cubans who do this work do it informally,

through a relative's foreign account, or through an intermediary in Miami or Madrid who takes a cut and holds the client relationship — or, increasingly, they emigrate, which converts a potential export sector into a permanent loss.

This is the good news in the chapter. The binding constraint is legal, it is domestic, it costs the state nothing to remove, and it does not require anyone's permission.

Design 18.1. The right to sell your own work

- (1) **THE CORE RIGHT** Any resident may contract directly with a foreign client, in any currency, without licence, intermediary or prior authorisation, and may receive payment into an account in their own name and spend from it. This is the single most important clause in Part III's growth strategy and it is a repeal, not a programme.
- (2) **PROFESSIONS INCLUDED** Explicitly extended to professions trained at state expense — engineers, physicians, architects, lawyers, scientists. The exclusion of professionals from self-employment since 1993 is the reason Cuba's private sector is a collection of trades, and it is the reason its most valuable workers had only two options, the state or the airport.
- (3) **NO ENUMERATED LIST** Per Design 13.1.2, service exports are permitted by right. Any list of permitted activities in this domain would be obsolete before it was published, which is a general argument against lists and a decisive one here.
- (4) **A SIMPLE TAX THAT IS WORTH PAYING** Income from cross-border services pays the low flat rate of Design 14.2.6, filed annually, with no deductions and no separate social contribution. A mobile base taxed heavily is not taxed; a mobile base taxed lightly and simply is registered, counted, and eventually becomes a constituency for the system that registered it.
- (5) **EQUIPMENT** Computers, phones, networking hardware and their parts enter free of duty and licence, per Design 16.2.6. Cuba currently has citizens buying their own tools with remittance money and a customs regime that treats it as smuggling.
- (6) **CLASSIFICATION** Cross-border services are classified and counted as exports in the national accounts and the balance of payments,

which sounds clerical and is not: what is not measured is not defended in the budget, and this sector will need defending against the ministries whose staff it draws.

18.3 Connectivity

Cuba came online late and expensively. Mobile data arrived in December 2018; international capacity has run principally through a single submarine cable laid from Venezuela in 2011, which is both a bandwidth constraint and a single point of failure; the state telecommunications company is a monopoly; and the price of data as a share of Cuban income has been among the highest in the world, which is what the protests over the 2025 tariff changes were about.

No second track exists on that basis. A foreign client contracting with a Cuban supplier is buying reliability, and reliability is exactly what a single cable, a monopoly operator, a fragile grid and a state that switched off mobile data nationally on 11 July 2021 do not provide.

Design 18.2. Connectivity as infrastructure

- (1) **REDUNDANCY** At least two additional international routes, on diverse physical paths and diverse landing points, procured on published competitive terms. One cable is not a network.
- (2) **OPEN WHOLESALE ACCESS** Whatever the ownership of the international and backbone capacity, access to it is sold to any licensed retail provider on published, non-discriminatory terms, regulated by an independent authority on the model of Design 16.2.4. Retail competition is what moves the price; full privatisation of the incumbent is not required and should not be attempted early.
- (3) **NO POLITICAL INTERRUPTION** Per Design 13.4.5. This is restated here because it is a commercial clause as much as a civil-liberties one: a country whose government has demonstrated that it will switch off the internet during a political event cannot sell service-level agreements, and every potential client knows it.

- (4) **PRICE TRANSPARENCY** Published wholesale and retail tariffs, and a published affordability metric — data cost as a share of median income — reported annually.
- (5) **POWER** Telecommunications sites are critical loads under Design 16.3.1 and are among the first to receive photovoltaics and storage. A network that goes down with the grid is not a network either.

18.4 The residence regime

Only after the two designs above does a visa policy make any sense, and the ordering matters because the international discussion of this subject has it backwards.

The digital-nomad visa is a modest instrument. The honest reading of the international experience — Estonia, Portugal, Barbados, Croatia, Costa Rica — is that these schemes attract a few thousand people, generate useful but small consumption spending, and do not by themselves build an industry. Barbados's Welcome Stamp was a competent response to a pandemic-era collapse in tourism, not an economic transformation. Estonia's e-Residency, the most cited of them, has enrolled a large number of people and produced a much smaller number of real businesses paying real Estonian tax.

What actually built service exports in the comparator countries was not visas for foreigners. It was *residents being allowed and able to sell their work abroad*, plus predictable law and a payment system — which is Design 18.1. Uruguay's software export sector, the most relevant Latin American case, was built on domestic firms and domestic engineers, not on inbound nomads.

So the residence regime here is proposed for three narrower and more defensible reasons: it brings in people who train and hire locally; it brings in demand for the long-stay accommodation that Chapter 17 wants to grow instead of hotel rooms; and — per Design 14.4.3 — more working-age residents is the cheapest available instrument for a dependency ratio.

Design 18.3. Residence for mobile workers

- (1) **THE PERMIT** A renewable residence permit for a person with foreign income from remote work or a foreign business, granted on published objective criteria — income threshold, health cover, clean record — with a statutory decision period and deemed grant if the period passes. No discretion, no interview, no sponsor.
- (2) **TAX CERTAINTY** A published flat rate on locally-remitted income for a fixed number of years, legislated rather than administered, with the treatment of foreign income stated in advance. Uncertainty deters this population far more than rates do.
- (3) **ORDINARY RIGHTS** The right to rent, to open an account, to import personal equipment, to bring family, and to use the health system on the same terms as residents while paying the same taxes. A permit that does not allow a bank account is not a permit.
- (4) **A ROUTE TO STAYING** A defined path to permanent residence and, for those who want it, naturalisation. The objective is residents, not visitors; a scheme that resets every year selects for people who do not put down roots.
- (5) **LOCAL LINKAGE** A reduced rate for permit-holders who employ or formally train Cuban residents, verified simply and audited. This is what converts an inbound flow into a domestic industry, and it is the difference between this proposal and a beach with better wifi.

18.5 The diaspora, and not taxing the door

Two million people abroad, most of them one flight away, many with capital and professional standing, and a substantial number who would work with Cuba if working with Cuba were possible. They are simultaneously the natural first clients for this sector, the natural first investors under Chapter 21, and the largest single reservoir of the working-age people Chapter 12 says the country has lost.

Design 18.4. Circulation, not retention

- (1) **NO EXIT TAX** Cuba does not tax, fine, or impose a training levy on emigrants. The instrument is popular with governments losing skilled workers, it collects almost nothing, it is trivially avoided by simply not returning, and it converts a temporary departure into a permanent one. It is also, in Cuba specifically, a tax on the grievance this book is trying to settle.
- (2) **RETURN WITHOUT PENALTY** Full restoration of residence, property rights, professional registration and pension entitlement for returning Cubans, automatic and without discretionary review, including for those who left irregularly or abandoned a mission.
- (3) **DUAL NATIONALITY** Recognised, with the right to hold property, to invest, to work, to vote in local elections after a residence qualification, and to inherit. Chapter 21 depends on this; so does the pension portability of Design 14.4.4.
- (4) **VOLUNTARY CONTRIBUTION** Cubans abroad may contribute voluntarily to the Cuban social insurance system and accrue entitlement, per Design 14.4.4. This converts a diaspora into a contributor base, which no other instrument does.
- (5) **FACULTY AND CLIENTS** Programmes to bring diaspora academics and professionals back on short teaching, clinical and advisory terms, with no political test, administered by universities rather than by ministries.

18.6 The universities

The sector proposed here needs a pipeline, and Chapter 3 identified the specific gap: Cuba produces excellent engineers, physicians and mathematicians and very few people trained in the disciplines an institutional transition and a services economy actually run on — commercial law, accounting, finance, management, applied economics, and the empirical social sciences.

Design 18.5. Academic capacity

- (1) **AUTONOMY** Universities appoint their own staff, set their own curricula and admit their own students on published academic criteria. Political record is not an admission or appointment criterion, and the prohibition is justiciable.
- (2) **THE MISSING DISCIPLINES** A funded programme to build law, accounting, finance, management and applied economics faculties, with foreign partnership, visiting appointments from the diaspora, and professional accreditation recognised internationally. This is a decade-long constraint on everything in Part III — Design 13.3.6 needs commercial judges, and Chapter 14 needs tax administrators — and it binds from year one.
- (3) **FOREIGN PARTNERSHIP** Joint programmes, dual degrees and branch arrangements with foreign universities, permitted by right and regulated for quality only.
- (4) **PUBLISH AND BE READ** Cuban academic work published openly, and Cuban academics free to publish abroad, collaborate and travel. The dismantling of the Centre for the Study of the Americas in 1996 taught Cuban social science what happens to research that analyses policy, and that lesson has to be publicly unlearned before anyone will do the work this book's later phases need.

18.7 What this chapter cannot promise

Three honest limits.

The sector is small until the electricity is fixed. Everything here is downstream of Chapter 16, and a remote worker in a province with sixteen-hour blackouts cannot take a call.

The sector competes for exactly the people the state most needs. Every engineer who moves to remote work at eighteen thousand dollars a year is an engineer not in the ministry, the utility or the university at a state wage — which is why Design 14.5 is a precondition for this chapter and not a separate topic.

And the sector is a rival to emigration, not a complement to it. Its real justification is that it offers a person the income they left for

without requiring them to leave. If it works, the outflow slows; if the outflow does not slow, it is not working, and Chapter 24 makes net migration one of the indicators by which this book can be judged to be failing.

Chapter 19

Life sciences

19.1 The one sector that already owns something

Everywhere else in Part III the problem is building a capability. Here the capability exists, is forty years old, has products, has a reputation, and has never been organised to sell anything.

Cuban biotechnology began in the early 1980s as a deliberate state decision to build a research capacity in a poor country, and it produced, against every reasonable expectation, a genuine one. The institutions — the Centre for Genetic Engineering and Biotechnology, the Finlay Institute, the Centre of Molecular Immunology, and the rest, consolidated since 2012 into the BioCubaFarma holding — have delivered products that were not derivative.

The meningococcal B vaccine developed at the Finlay Institute in the 1980s was the first of its kind anywhere. The *Haemophilus influenzae* type b vaccine licensed in 2004 used a fully synthetic antigen, the first synthetic-antigen vaccine in routine human use. Heberprot-P, a recombinant epidermal growth factor injected into diabetic foot ulcers, addresses a condition that causes a large number of the world's amputations and has no close equivalent elsewhere. CIMAvax-EGF, a therapeutic lung cancer vaccine, has been suf-

ficiently interesting to American oncologists that a United States cancer centre pursued trials of it through the licensing exceptions. And in 2021 the sector developed, tested, manufactured and deployed its own COVID-19 vaccines, vaccinating over ninety per cent of the population — the only country in Latin America to do so, and it did it while unable to reliably import syringes.

Why did this work when nothing else did? Three reasons, and they are worth naming because the reform must not destroy them.

It was a *closed loop*. Research, clinical development, manufacture and deployment sat inside one system with one customer, the national health service, so a product went from bench to population without an intervening market and without a reimbursement negotiation. That is a genuine structural advantage for developing products aimed at a whole population.

It had *patient capital*. The state funded it for a decade before results, which no shareholder would have done, and did not close it during the Special Period.

And it had *no profit constraint on target selection*. It worked on meningitis B, on diabetic foot, on lung cancer in smokers — conditions of interest to Cuba and to poor countries generally rather than to the markets that set global pharmaceutical priorities.

19.2 Why it earns almost nothing

And yet. Cuban pharmaceutical and biotechnology exports are, by any published measure, a small fraction of what a portfolio like this should be worth — of the order of a few hundred million dollars a year, against a global market in the trillions and against Cuban medical *services* exports that were at their peak twenty or thirty times larger.

The reasons are not scientific. They are regulatory, contractual and financial, and each is fixable.

No regulatory conversion. Almost nothing in the portfolio holds approval from the United States Food and Drug Administration or the European Medicines Agency, and much of the clinical evidence was not generated to the standards those agencies require — not

because the trials were dishonest, but because they were designed for Cuban registration and published in Cuban journals. A product without a stringent-authority approval cannot be sold in the markets that pay, cannot be licensed to a partner who sells there, and cannot be priced at anything like its therapeutic value. The COVID vaccines are the clearest case: Cuba developed vaccines that worked, and could not sell them, because they did not complete the international prequalification pathway.

No distribution partner. Value in pharmaceuticals is captured through regulatory affairs, distribution, salesforce and reimbursement negotiation — capabilities Cuba does not have and would take twenty years to build. The answer is to partner, and Cuba has largely not partnered, in part because of the next two reasons.

No capital. Bringing one product through international trials costs more than the whole sector's annual budget, and a defaulted sovereign cannot finance it.

An ownership structure nobody will contract with. A foreign pharmaceutical company entering a licensing deal is buying a stream of obligations enforceable over fifteen years. Its counterparty in Cuba is a state holding company, in a jurisdiction with no independent courts, subject to a statute that exposes anyone trafficking in confiscated property to litigation in the United States, with a sovereign that froze foreign firms' accounts in 2009. The science is not the risk.

And the regulator is the producer. The Cuban medicines regulator, CECMED, is a state body regulating products made by state bodies, within a single system answerable to one government. Whatever the integrity of its individual staff — and its technical reputation is not bad — it is structurally not independent, and structural independence is precisely what a foreign regulator or a foreign partner is looking for. This is the same conflict of interest Chapter 16 identified in electricity, where the operator, owner, planner and regulator are one entity.

19.3 The design

The strategy is to convert an existing scientific asset into an investable, regulated, exporting sector without destroying the closed loop that made it work. The order matters: regulator first, because nothing else is possible until a foreign counterparty can rely on something.

Design 19.1. An independent medicines regulator

- (1) **SEPARATION** The medicines regulator is constituted as an independent authority with the tenure, budget and publication guarantees of Design 13.3, institutionally separate from BioCubaFarma, from the health ministry as owner, and from any producing entity.
- (2) **FEE-FUNDED AND PUBLISHED** Funded by application fees at published rates, with all assessment reports, approvals, refusals and inspection findings published, as the European and American agencies publish theirs. Publication is what makes a regulator's judgments checkable by outsiders, which is the entire asset being built.
- (3) **TARGET: STRINGENT-AUTHORITY STATUS** An explicit, dated programme to reach World Health Organization listed-authority standing, with the assessment criteria, the gap analysis and the progress published annually. This is a multi-year technical project with a definite endpoint, and it is the single highest-return institutional investment available to Cuba in any sector.
- (4) **RELIANCE IN THE INTERIM** Automatic recognition of approvals by stringent foreign regulators for products Cuba imports, so that scarce assessor capacity is spent on Cuban products seeking export rather than on re-reviewing medicines already approved elsewhere. This also, immediately, shortens the queue for imported medicines that are currently absent from pharmacies.
- (5) **TRIALS TO INTERNATIONAL STANDARD** All Cuban clinical trials conducted and registered to international good-clinical-practice standards, in international registries, with independent monitoring and ethics review. Cuban trial data is currently discounted abroad for reasons of procedure rather than of substance, and this clause is what makes forty years of accumulated science sellable.

Design 19.2. Capital and partnership

- (1) **PRODUCT-LEVEL JOINT VENTURES** Partnership at the level of the individual product — one asset, one partner, defined territories, defined milestones — rather than at the level of the institution. This raises capital and buys regulatory and distribution capability without selling the research base.
- (2) **KEEP THE PLATFORM, LICENSE THE PRODUCT** Cuba retains ownership of its research institutions, its manufacturing and its underlying technology platforms, and licenses rights to specific products in specific markets. The comparators are Singapore and Ireland — countries that built pharmaceutical sectors by hosting and partnering rather than by originating — and not Switzerland, which is not an available model for a country of nine million with no capital.
- (3) **ENFORCEABLE TERMS** Contracts under foreign law with international arbitration, per Chapter 21. A licensing agreement justiciable only in Cuban courts is worth a fraction of the same agreement justiciable in London or Singapore, and the discount is larger than anything Cuba could negotiate on the royalty.
- (4) **DIASPORA SCIENTISTS** Cuban scientists abroad are the natural bridge to foreign partners, regulators and capital, and Design 18.4.5 applies directly. Several of the people best placed to take a Cuban product through an FDA submission are Cubans who left.
- (5) **THE DOMESTIC SUPPLY FIRST** Nothing in this design permits export of a medicine that is unavailable in Cuban pharmacies. Domestic supply is the first call on production, stated as a rule because the revenue pressure will run the other way.

19.4 The medical missions, reformed

Cuba's largest export for over a decade was the labour of its health workers, and Chapter 8 reported both halves of that: a real service really delivered in places no other country would staff, financed by paying the workers a fraction of the contract value and, in several

destinations, holding their passports.

The programme should be kept. It is valuable, it is genuinely good for the receiving countries, and it is one of the few things Cuba sells that nobody else sells. But the labour arrangement has to change, and the argument for changing it is not only ethical. It is that the industry's binding constraint is now the supply of willing doctors — Cuban health workers are leaving the country altogether, and a programme that treats them as conscripts loses them to emigration, which yields Cuba nothing at all.

Design 19.3. Medical service export

- (1) **INDIVIDUAL CONTRACTS** Each health worker contracts individually, with the full contract value disclosed to them and a published minimum share payable to the worker. The state may retain a share — there is a legitimate public claim on the return to a publicly funded training — and the share is published, legislated, and the same for everyone.
- (2) **NO CONTROL OF DOCUMENTS OR MOVEMENT** Passports are held by their owners. Restrictions on residence, association and relationships are abolished. The penalty for leaving a mission is contractual, not a bar on returning to one's own country, and Design 18.4.2 applies.
- (3) **VOLUNTARY** Participation is voluntary, refusal carries no professional consequence, and this is auditable.
- (4) **COMPLIANT WITH LABOUR STANDARDS** The programme is brought into conformity with international labour conventions, and the conformity is externally verified and published. This is what converts a programme under permanent international criticism into an exportable, expandable and defensible service.
- (5) **DOMESTIC STAFFING FLOOR** A published minimum staffing level for the Cuban health service, below which missions are not staffed. Per Chapter 12, people are the binding scarcity, and the domestic system is the asset the whole book is written to protect.

19.5 Two further lines of business

Clinical trials as an export. Cuba has attributes that contract research organisations pay for: a population with universal health coverage and complete medical records, high follow-up and retention because people do not move or lose their insurance, a physician-dense primary care network organised by defined population, and a trial cost base far below North America or Europe. Under Design 19.1.5 and an independent ethics framework, running phases of international trials in Cuba is a genuine, capital-light, high-skill export that also brings the domestic system access to investigational treatments. It requires scrupulous consent standards — a population that has limited access to medicines is a population that can be induced to enrol — and the ethics review has to be independent of both the sponsor and the state, which is a stronger requirement than most middle-income trial hosts apply.

Health tourism. Cuba has treated foreign patients for decades and could do far more of it, in cardiology, orthopaedics, ophthalmology, oncology, neurological rehabilitation and addiction treatment, at prices well below the United States and with outcomes that stand comparison. The constraint is Chapter 12's again: every foreign patient consumes a Cuban clinician's time. Design 14.6.3 already binds — designated facilities, ring-fenced staff, revenue to the public system, and the programme stops if the ring-fence cannot be held. This chapter adds nothing to that except the observation that health tourism is the most valuable visitor per bed-night available under Chapter 17's metric, and the least environmentally damaging.

19.6 What is being protected

A closing caution, because the reforms above are all in the direction of markets and partnership and that direction has a known failure mode.

The Cuban biotechnology sector works because it is a closed loop with a public customer, patient capital and no obligation to select targets by market size. Open it to partnership incorrectly and it

becomes a contract manufacturer for somebody else's pipeline, with its research agenda set in New Jersey and its public-health function abandoned — at which point Cuba has sold its one originating capability for a manufacturing margin.

The clauses that guard against that are Design 19.2.2, which keeps the platform and licenses only products; Design 19.2.5, which puts the domestic pharmacy first; and the public research budget, which Chapter 14's floor protects. They are the whole of the protection, they are not self-enforcing, and a future government under fiscal pressure will be offered a great deal of money to weaken them.

Chapter 20

Land and food

20.1 The failure, stated exactly

A tropical island with deep soils, adequate rainfall, a year-round growing season, no frost, a trained agronomic profession and a hundred and fifty years of commercial agriculture imports somewhere between two thirds and four fifths of what it eats, cannot pay for the imports, and has between one and two million hectares of farmland under *marabú* scrub.

This is the most complete policy failure in the book. It is more complete than the energy failure, because energy at least has a physical explanation — the plants are old and the fuel is foreign. Nothing physical explains this one. Cuba's neighbours with worse soil, worse rainfall and less education feed themselves.

It is also the clearest case where the embargo explains very little. Cuba may buy agricultural inputs from anyone in the world, and does buy food from the United States itself, for cash, under the exemption that has stood since 2000. The constraint is not permission to import fertiliser. It is that Cuban farmers have been given every incentive not to grow anything.

20.2 Why

Four mechanisms, in order of how much they explain.

Acopio. The state procurement monopoly requires producers to sell a defined share of output to the state at a state-set price, historically below the cost of production and paid late, in a currency that was losing value while the farmer waited. A farmer facing that has three rational responses: produce the minimum the quota requires, divert everything else to the informal market, or stop. All three are observed, and all three are then prosecuted as indiscipline.

Tenure. The usufruct grants of 2008 and 2012 handed out large areas of idle land, and they handed out the weakest possible interest in it. The land cannot be sold. It cannot be mortgaged. Its inheritance is conditional. The term is limited and renewal is discretionary. Buildings erected on it may belong to the state. Nobody plants a fruit tree that yields in year seven on a tenure that might end in year five, and nobody can borrow against land a bank cannot take. The result is that the grants produced short-cycle crops on the best parcels and scrub on the rest.

Inputs and equipment. No functioning market in fertiliser, seed, veterinary supplies, fencing, irrigation pipe, tyres or spares. What exists is allocated administratively, which means it is allocated to state farms and to whoever knows the person allocating.

Everything after the harvest. No transport, no refrigeration, no processing, no packaging. Cuban post-harvest losses are very large — estimates of a fifth to a third of what is grown, and higher for perishables — and the loss happens between a field where the food exists and a city where people are hungry. A cold chain is not agriculture policy in the usual sense; in Cuba it is the largest single available increase in food supply, and it requires electricity, which is why this chapter follows Chapter 16.

Every one of these four is domestic, legal and reversible.

20.3 The design

Design 20.1. Tenure

- (1) **CONVERT USUFRUCT TO OWNERSHIP** Existing usufruct holders who have worked their land for a qualifying period receive full, registered, transferable freehold or a long lease of not less than ninety-nine years, at nominal consideration. Not a longer usufruct — an interest that can be sold, mortgaged, inherited and defended in court.
- (2) **REGISTERED** Every parcel entered on the public cadastre and registry of Chapter 21, with boundaries surveyed and title guaranteed by the state. An unregistered right is a right that has to be re-argued with each official.
- (3) **MORTGAGEABLE** Agricultural land may secure a loan and may be taken by a creditor on default. This is the clause that makes rural credit possible, it is the clause that will be resisted hardest, and the fear behind the resistance — that farmers lose their land — is met by clause 4 rather than by refusing to allow borrowing at all.
- (4) **ANTI-CONCENTRATION** A published ceiling on holdings by a single beneficial owner, and a right of first refusal for the sitting occupier and for neighbouring smallholders on any sale. Cuba's land question was settled in 1959 against latifundia and this design does not reopen it; the objective is a market in family-scale and cooperative holdings, not the reassembly of the sugar estates.
- (5) **FOREIGN OWNERSHIP** Foreigners may lease agricultural land, and may not own it. Chapter 21 takes a permissive line on foreign investment generally, and this is the deliberate exception, for reasons that are political rather than economic and that any Cuban government would face.
- (6) **USE IT** Land held out of production beyond a defined period is subject to a rising land tax under Design 14.2.4 rather than to confiscation. A tax is administrable, is not discretionary, and does not require an official to decide whether a farm is being farmed.

Design 20.2. Markets

- (1) **END ACOPIO** The state procurement monopoly is abolished. Producers sell to whom they choose at prices they agree. The state continues to buy — for schools, hospitals, the armed forces and the strategic reserve — as one buyer among others, at market prices, paying on published terms and on time.
- (2) **INPUT IMPORTS OPENED** Any producer, cooperative or trading firm may import fertiliser, seed, veterinary products, machinery, spares and irrigation equipment, free of licence and of duty, per Design 13.1.2.
- (3) **WHOLESALE MARKETS** Physical wholesale markets in each province, open to any buyer and seller, with published prices. The absence of a wholesale tier is why a Cuban restaurant buys its vegetables in a retail shop in competition with households, and why the private sector gets blamed for shortages it is standing in the queue for.
- (4) **RURAL CREDIT** Agricultural lending against the registered, mortgageable title of Design 20.1, through banks and credit cooperatives, with a partial public guarantee in the first phase to establish the instrument. Cuba currently has neither the collateral nor the lenders.
- (5) **NO PRICE CONTROLS** Retail price caps on food are prohibited. They were imposed on the private importers from 2024, they produced exactly the withdrawal of supply that price caps always produce, and the distributional problem they were addressing is met on the spending side by Design 15.5 instead.

20.4 Food security is not self-sufficiency

A distinction the Cuban policy tradition has consistently confused, at considerable cost.

Self-sufficiency means producing domestically what one consumes. Food security means that people can reliably obtain food. These are different objectives with different instruments, and pursuing the first is a poor way to achieve the second on a small island.

Cuba should not try to grow its own wheat — it cannot, at any price. It should not try to compete in bulk maize or soy against continental producers with a tenth of its transport cost per tonne. Those are imports, permanently, and the security question about them is whether the country can pay, which is a question about export earnings and reserves rather than about agronomy.

What Cuba can produce competitively is a well-defined list: roots and tubers, plantain, vegetables and salads, tropical fruit, beans, eggs, poultry, pork, dairy at a modest scale, fish and aquaculture, and the high-value export crops it has always had — tobacco, coffee, cacao, honey, citrus, rum. Several of these are worth far more per hectare than anything the country currently grows, and Cuban coffee and cacao in particular sell at premiums in specialty markets that Cuba has never organised itself to supply.

Design 20.3. What the state is actually for

- (1) **RESERVES, NOT TARGETS** Food security is delivered by a strategic reserve of staples sized in months of consumption, published, rotated, and financed from the stabilisation fund — not by production targets for crops the country grows badly.
- (2) **BUY WHERE IT IS CHEAP** Import policy is set on cost, not on the political identity of the seller. Cuba currently buys food from the United States for cash while describing it as a hostile power, which is awkward and correct.
- (3) **PUBLIC GOODS** The state's agricultural role is research and extension, plant and animal health and quarantine, cadastre and registry, rural roads, irrigation infrastructure, and market price information. These are the things private actors will not provide and the things the Cuban state has stopped doing while it was busy setting prices.
- (4) **EXPORT PROMOTION** Certification, traceability, phytosanitary compliance and origin protection for the export crops — which is what standing between a Cuban cacao farmer and a European specialty buyer, and what no individual farmer can do alone.

20.5 Sugar

The honest answer is that it does not come back, and a design that plans on its return is planning on nostalgia.

Cuban sugar was eight and a half million tonnes in 1970 and is now measured in the low hundreds of thousands. The mills are gone or cannibalised, the railways that served them are gone, the cane fields are under scrub, the workforce has aged out, and the world price is set by Brazilian and Indian producers operating at a scale and cost Cuba cannot approach. Rebuilding the industry would require capital that has better uses in every other chapter of this book.

What is worth keeping is smaller and more specific. The remaining efficient mills in the best cane districts, run as an integrated sugar-ethanol-bagasse-power business rather than as raw sugar exporters, on a scale of perhaps a fifth of the 1989 industry. The rum, which is a high-value branded export with a genuine international market and the one part of the sugar complex that has never stopped earning. And the land, which is the real asset: a million hectares of former cane land, flat, road-served and now idle, is exactly what Chapter 16 needs for photovoltaics and what Design 20.1 needs for redistribution.

The social question is the difficult part and it should not be waved away. The sugar towns — and there are hundreds of them, built around a single mill, with no other employer — were devastated by the closures of 2002 and have never recovered. Whatever is decided about the industry, those communities need an answer, and the honest one is that it is not sugar. It is the eastern weighting of Design 16.3.3 and Design 17.4.4, the equalisation formula of Design 13.7.2, and land.

20.6 What the 1990s did and did not prove

The organic and urban agriculture that emerged from the Special Period is the most internationally celebrated thing in Cuban agriculture

and the most over-claimed.

What it demonstrably did: put fresh vegetables into cities at a moment when the transport system had failed, using raised beds on vacant urban lots, biological pest control and composting; and it did so on a scale that supplied a meaningful share of Havana's fresh vegetable consumption. That is a real achievement, it was achieved by Cuban agronomists under extreme constraint, and the technique transfers.

What it did not do: feed the country. Cuba's food import dependence rose across the whole period rather than falling. Urban horticulture is good at leafy vegetables and herbs on small plots close to consumers, and it is not a substitute for field crops, protein or calories. And the low-input methods were adopted because there were no inputs, not because they outperformed; where fertiliser has been available since, farmers have used it.

The design position is therefore neither the romantic one nor the dismissive one. Keep the urban horticulture, which works and is cheap, and extend it under Design 13.7. Keep the biological pest control and the agronomic extension capacity, which are genuinely good. And do not build national food policy on a legend, because Cubans have been hungry throughout the entire period during which the legend has been circulating abroad.

Chapter 21

Capital and property

21.1 Why the money has not come

Cuba has been open to foreign investment since 1995 and has received very little of it. The standard explanation is the embargo. The embargo is real and it is not the main reason, and the evidence for that is simply that European, Canadian, Chinese, Brazilian, Russian and Latin American capital faces no such prohibition and has also stayed away.

The actual deterrents, in order of how much investors complain about them:

The employment agency. A foreign venture does not hire its own workers. It contracts them from a state agency, pays the agency in hard currency, and the agency pays the worker in pesos, keeping the difference — historically a difference of more than an order of magnitude. The investor therefore pays a first-world wage bill for a workforce receiving a fraction of it, has no control over selection, cannot reward performance, and knows the arrangement is visible to every employee. It is bad for the investor, worse for the worker, and its only beneficiary is the state's foreign-currency account.

Approval by case. Every project is negotiated and approved individually, at ministerial and often at higher level, on timescales measured in years and criteria that are not published. This is

the definition of a discretionary system, with the consequences Chapter 22 describes.

No dispute resolution anyone trusts. Contracts are justiciable in Cuban courts, which have never ruled against the state on anything material. The absence of credible arbitration is priced into every deal.

No repatriation guarantee that has held. Dividends and capital may be repatriated, in principle. In 2009 the state froze foreign firms' balances and in several cases they remain unpaid. That is not a legal fact but a historical one, and it is the one investors actually price.

Title III. Since 2019, a company using property confiscated from an American national is exposed to suit in United States courts. Since much of the valuable commercial property in Cuba was confiscated from somebody, this is a diffuse and unquantifiable liability attached to almost any large project, and it will remain so until the claims are settled — which is why Design 15.3.2 puts them inside the debt restructuring.

The Mariel special development zone is the demonstration. It has excellent infrastructure, a deep-water container port, a favourable tax regime, and after more than a decade a small number of operating projects (Feinberg, 2016) and a total invested capital far below what was projected. It fixed the tax rate and the customs regime, which were not the problem, and kept the employment agency, the case-by-case approval and the courts, which were.

Design 21.1. The investment regime

- (1) **DIRECT EMPLOYMENT** The state employment agency monopoly is abolished. Foreign-invested firms hire, pay, promote and dismiss their own staff under the ordinary labour law, in the currency of Design 15.1. Applies to every sector, including tourism (Design 17.1.3) and the special zones.
- (2) **APPROVAL BY RIGHT, WITH A CLOCK** Investment is permitted by right, subject to a published negative list of restricted sectors and to ordinary regulatory consents. Where an authorisation is genuinely required, a statutory decision period applies and expiry

of the period is a grant. Deemed consent is the single most effective anti-corruption instrument in this book: an official who cannot delay cannot extract.

- (3) NATIONAL TREATMENT Foreign and domestic investors, including Cuban private firms and returning Cubans, face the same rules, taxes, access to inputs and access to foreign exchange. The current arrangement, in which a foreign joint venture may import and a Cuban private firm may not, is indefensible and is also why Cuban businesses cannot grow.
- (4) ARBITRATION A standing offer of international arbitration under ICSID or UNCITRAL rules, seated abroad, in the investment law itself and in every state contract, plus a network of investment treaties. Cuba cannot borrow credibility; it can rent it, and this is how.
- (5) REPATRIATION A guaranteed right to convert and repatriate profits and capital, entrenched under Design 13.1, with the 2009 balances released first as evidence.
- (6) PUBLISH THE DEALS Every investment authorisation, its terms, its beneficial owners and any incentive granted, published. Per Chapter 22, and it applies to military-linked entities without exception.

21.2 Property at home, which comes first

Here is a point that foreign-investment discussions consistently miss. Cuba's largest property problem is not foreign; it is that Cubans own their homes and cannot do anything with them.

The Urban Reform of 1960 transferred rental housing to its occupants and abolished the residential rental market. It was the revolution's most popular and most durable redistribution and it produced near-universal home ownership. It also produced a housing stock in which nobody had a legal mechanism or a financial incentive to invest, and which has been decaying for sixty-six years. Havana loses buildings to collapse. The housing deficit runs to hundreds of

thousands of units. Cubans hold what is, on paper, the country's single largest asset class, and cannot borrow against it, cannot insure it properly, often cannot prove title to it, and until 2011 could not sell it.

Chapter 14 proposed a property tax as the natural local revenue. That tax and this section are the same project: a cadastre, a registry, clean title, and mortgageability turn a decaying liability into a financeable asset and into the base of local government at the same time.

Design 21.2. Making Cuban property real

- (1) **CADASTRE AND REGISTRY** A complete national cadastre and public property register, surveyed, digitised and open to inspection, with a defined programme and a date. This is a large, dull, multi-year administrative project and it is a precondition for the property tax, for mortgage lending, for Design 20.1, for building control, for insurance, and for the settlement of claims. Nothing else in this chapter works without it.
- (2) **TITLE GUARANTEED** Registered title is guaranteed by the state and defensible in the administrative court of Design 13.3.4, with a compensation fund for registry error. A register nobody can rely on is a filing cabinet.
- (3) **REGULARISATION** A time-limited programme to regularise informal occupation, extensions and self-built housing on published criteria, favouring the sitting occupier. A large share of Cuban housing is legally irregular after six decades of improvisation, and a registry that excludes it excludes the people it was meant to protect.
- (4) **MORTGAGE LAW** A mortgage statute with a workable, judicially supervised enforcement procedure, plus consumer protections and a rehousing obligation on foreclosure of a sole residence.
- (5) **RENTAL MARKET** Residential letting legalised and regulated, with security of tenure and published rent-review rules. Prohibiting rental did not make housing affordable; it made it invisible, informal and unmaintained.

21.3 The claims

Two bodies of claim, of very different size and character, and they have to be settled together.

The *certified United States claims* number 5,913, were certified at a principal of \$1.9 billion, and at the statutory 6 per cent simple interest are now valued at something over \$9 billion (Foreign Claims Settlement Commission of the United States, 1972). They are documented, adjudicated, and held largely by corporations and their successors.

The *uncertified claims* — of Cuban Americans who were not United States nationals at the time of taking, and of Cubans who never left and whose property was taken under the same laws — are far more numerous, far less documented, emotionally far heavier, and potentially far larger in aggregate. They include the houses left in the care of relatives in 1961 by people who expected to be back in months.

Any settlement has to satisfy three constraints simultaneously. It must be credible to the claimants, or Title III litigation continues and the country stays uninvestable. It must be affordable to a bankrupt state. And it must not displace a single Cuban from a home.

The central European experience is the guide, and it is a guide by negative example. Restitution in kind — returning the actual property — was tried at scale, most extensively in eastern Germany, where over two million claims were lodged. It froze the property market for the better part of a decade, because no purchaser and no lender would touch an asset that might be claimed; it consumed enormous administrative and judicial capacity; and it delivered its benefits overwhelmingly to people who had left, at the expense of people who had stayed. Countries that compensated in money and paper instead — Hungary's voucher scheme, and the Baltic states in part — got through it faster and cheaper.

Design 21.3. Settling the claims

- (1) NO RESTITUTION IN KIND No occupied dwelling, no operating farm, no school and no hospital is returned to a prior owner.

Compensation only. This is stated first, absolutely and without exception, because the fear of it is the single biggest source of Cuban domestic resistance to normalisation, and removing that fear is worth more than the assets involved.

- (2) **ONE PROCESS, ONE CAP** A claims commission with a single filing window, a published methodology, and an announced aggregate cap on total compensation, inside the debt restructuring of Design 15.3. Claims are paid pro rata within the cap. A settlement without a cap is not a settlement; it is an open liability.
- (3) **PAID IN PAPER** Compensation in the same long-dated instruments as the restructured debt, with an option to convert into equity in privatisations under Design 21.4 or into investment in Cuba. Converting a claimant into an investor is the best outcome available for both sides.
- (4) **BOTH BODIES OF CLAIM** The uncertified Cuban and Cuban-American claims are admitted to the same process on the same terms as the certified United States claims. A settlement that pays foreign corporations and not Cuban families would be politically impossible inside Cuba and morally indefensible anywhere.
- (5) **FINALITY** Participation extinguishes the claim, in all jurisdictions, including under Title III. Finality is the entire product being purchased.
- (6) **SMALL CLAIMS FAST** A simplified track with a low evidentiary threshold and a flat payment for claims below a threshold, so that the large number of modest family claims are resolved quickly rather than being ground through a process designed for corporate assets.

21.4 Privatisation, and how not to create an oligarchy

This is where transitions are lost, and Cuba has the advantage of thirty-five years of other people's experience.

The lesson of the post-Soviet privatisations is not that privatisation is wrong. It is that *how* and *in what order* determined

everything. Mass voucher schemes, in Russia and Czechoslovakia, dispersed nominal ownership among citizens who promptly sold their vouchers for cash to the few actors with capital and information, concentrating real ownership faster than a straightforward auction would have. Loans-for-shares in Russia handed the largest assets to a handful of people at derisory prices in exchange for financing a government. Poland, which sold small enterprises early and fast, restructured medium ones slowly, and left the largest for last under a functioning competition authority and a functioning press, did better than anyone.

Cuba's specific risk is named in Chapter 12: the incumbents control the assets and the information, and a privatisation conducted before the institutions exist hands them the economy with legal title attached. That outcome is effectively irreversible, and it is the most likely failure mode of everything in this book.

Design 21.4. Sequencing privatisation

- (1) **SMALL AND MANY, FIRST** Retail, catering, repair, transport, small manufacturing and services — thousands of units — sold or leased early, in open auctions, with a preference for the existing workers and for cooperatives. These are the businesses the 1968 Revolutionary Offensive abolished, and restoring them is both economically useful and politically popular.
- (2) **LARGE AND FEW, LAST** Utilities, ports, telecommunications, mining and the military conglomerate's core assets are not sold until the competition authority, the sector regulators, the courts, the press and the beneficial-ownership register are all functioning. Until then they are corporatised, audited and published, but not transferred.
- (3) **AUCTIONS, PUBLISHED** Every sale by open competitive auction with the process, the bidders, the price and the beneficial owners published within thirty days. No negotiated sales, no management buy-outs at book value, no strategic investors selected on unpublished criteria.
- (4) **NO VOUCHERS** For the reasons above. Where wide ownership is the objective it is pursued through worker and cooperative

preference in clause 1 and through the stabilisation fund holding minority stakes, not through paper distributed to people with no means of valuing it.

- (5) **BENEFICIAL OWNERSHIP** No entity may acquire a privatised asset without disclosing its ultimate beneficial owners to a public register. No anonymous vehicles, no nominee structures, no exceptions for anybody.
- (6) **COOLING-OFF** Officials involved in a privatisation, and their households, may not acquire an interest in the asset, work for the acquirer, or advise on the transaction for a published period. Enforced by disclosure and by Chapter 22's asset declarations.
- (7) **PROCEEDS TO THE FUND** Privatisation proceeds are rent under Design 14.3 and go to the stabilisation and investment fund. A government that can spend privatisation receipts on the recurrent budget will privatise for cash-flow reasons and at the wrong price.

21.5 GAESA

The question nobody wants to write down, so it gets written down.

The armed forces' business conglomerate controls a large share of Cuba's hard-currency economy — hotels, retail chains, ports, logistics, remittance processing, import and export — and does not publish accounts. Its size is not known outside itself, which is the first thing that has to change. It is simultaneously the country's most competent commercial organisation, the largest single misallocator of capital in the country per Chapter 17, and the entity with the most to lose from every design in this book.

Three options, honestly assessed.

Expropriate and break up. Politically impossible. The armed forces will not permit it, they can prevent it, and a design that assumes otherwise is not a design for Cuba. Attempting it is how a managed transition becomes a contested one.

Leave it alone. Guarantees the oligarchy outcome. A conglomerate with privileged currency access, no published accounts and no competition, sitting inside a liberalising economy, becomes its

owner.

Normalise it. Convert it into an ordinary state holding company, strip its privileges, make it compete, publish its accounts, and let its constituent businesses be sold off over a decade under Design 21.4. Slower and less satisfying than the first, and it is the only one that is both achievable and survivable.

Design 21.5. Normalising the military economy

- (1) **CIVILIAN OWNERSHIP** The conglomerate's holdings are transferred to a civilian state holding company under the finance ministry. The armed forces retain their budget, their pensions and their defence functions and cease to be a commercial actor. Personnel keep their standing and their pensions under Design 13.5.
- (2) **PUBLISHED AUDITED ACCOUNTS** Full consolidated accounts, audited to international standards by an external auditor, published annually, from the first year. This is the clause that has to come first, because none of the others can be assessed without it.
- (3) **NO PRIVILEGES** No preferential access to foreign currency, to import licences, to land, to grid capacity, to state contracts or to regulatory forbearance. Every clause of Design 21.1 and Design 21.4 applies to it identically.
- (4) **INSIDE THE BUDGET** Its revenues and transfers appear in the single budget of Design 15.4.1.
- (5) **DIVESTED OVER A DECADE** Constituent businesses sold under Design 21.4, small first, on a published schedule, with the proceeds to the stabilisation fund and with the cooling-off rule of Design 21.4.6 applied strictly to serving and former officers.
- (6) **THE BARGAIN IS EXPLICIT** This clause exists to say plainly what Chapter 12 argued: the surrender of commercial discretion is exchanged for security, pensions, property and a legal route into ordinary political and business life. That exchange should be negotiated openly and written down, because a bargain left implicit is one that gets renegotiated at every step, and Cuba cannot afford to renegotiate this one annually.

21.6 The diaspora as the first investor

Finally, the observation that makes the rest of this chapter plausible.

Cuba is not going to attract large amounts of general international capital quickly. It is a small, poor, defaulted, sanctioned market with no track record, and the global competition for investment is intense. What it does have is two million people abroad who know the country, speak the language, have relatives there, and are already sending money in volumes that dwarf foreign direct investment.

Remittances are consumption. The design objective is to convert a share of them into investment, and the instruments are not exotic: dual nationality, the right to own property and businesses on national treatment, a payment system that works (Design 15.6), a registry that gives a Miami investor confidence that the title exists (Design 21.2), a court that a Cuban American can use (Design 13.3.4), and a claims settlement that closes the past rather than leaving it as an open wound (Design 21.3).

None of that requires the embargo to end. Cuban Americans may already travel, send money and, within limits, support private Cuban businesses. The binding constraints are on the Cuban side, and they are the same ones this chapter has been listing.

Chapter 22

Integrity

22.1 The failure mode

Every chapter of Part III has a way of going wrong. This is the one that is common to all of them, and it is the way transitions of this kind actually die.

The mechanism is not mysterious. A transition takes assets that were previously unpriced — land, buildings, licences, spectrum, ports, mineral rights, state enterprises — and prices them, over a period of a few years, in a country whose courts are weak, whose press is new, whose officials are underpaid, and whose rules are being rewritten. During that window, a small number of people are in a position to decide who gets what, and they know the window will close. Everything about the situation rewards taking as much as possible as fast as possible.

The results are on the record. Russia converted a planned economy into a private one in five years and produced, by the end of it, an ownership structure so concentrated and so illegitimate that the political system reorganised itself around protecting it. Ukraine spent twenty-five years with a formally democratic system captured by a handful of industrial groups (Åslund, 2013). Romania's privatisations produced a decade of scandal. Against those, Poland, Estonia and Slovenia did not avoid corruption — nobody does —

but they kept it below the level at which it determines who governs, and they did so with a set of instruments that are well-documented and unglamorous.

Cuba's version of the risk is specific and named in Chapter 12: the incumbents hold the assets, the accounts and the information, and they are the people who would be administering the transition. A liberalisation conducted before the checks exist does not distribute the Cuban economy. It hands it to the people currently holding it, with legal title attached, and that outcome is effectively permanent.

22.2 Two different problems

An important distinction, because Cuba's public discussion and its criminal law run them together and the instruments for each are opposite.

Survival corruption. In an economy of chronic shortage, ordinary Cubans have spent thirty years obtaining what they need through informal channels — *resolver*, *luchar*, the state warehouse worker who diverts a case of cooking oil, the doctor given a gift for an appointment, the inspector who overlooks a thing. This is pervasive, it is how the country has fed itself, and it is a symptom of scarcity and of wages that do not cover subsistence. It is not addressed by enforcement; forty years of enforcement have not addressed it. It is addressed by supply and by wages, which is to say by Chapters 20, 16 and 14. Treating it as the same offence as the next category is both unjust and a distraction, and it has the further cost of making the whole population complicit and therefore uninterested in anti-corruption as an idea.

State capture. A small number of decisions (Hellman, 1998) — who gets the port concession, who gets the currency at the official rate, who buys the hotel chain, whose company wins the solar tender — determine the ownership of the country. This is the category that decides whether Cuba in 2045 is Estonia or Ukraine, and it is entirely about a few hundred people and a few thousand decisions.

Everything in this chapter is aimed at the second. The first is a welfare problem wearing a criminal-law costume.

22.3 Remove the discretion first

The most important thing in this chapter, and the one most often left out of anti-corruption programmes because it does not look like anti-corruption.

Corruption requires a discretionary decision. Where an official has no choice — where the rule is published, the rate is single, the answer is automatic, the deadline is binding and the appeal is real — there is nothing to sell. Most effective anti-corruption is therefore deregulation and simplification rather than enforcement, and it has the additional advantage of being achievable by a government that does not yet have honest courts.

Cuba is an extreme case because the number of discretionary surfaces is extraordinary. Every one of the following has been, and mostly still is, a decision an official takes about a citizen: whether an activity may be licensed; which category it falls in; whether a permit is renewed; whether an import is allowed; what exchange rate applies; whether currency is allocated; whether a building may be extended; whether a plot is granted; whether a business may be inspected today; whether a doctor may travel.

Design 22.1. Discretion, and its removal

- (1) **RULES, NOT PERMISSIONS** Activity is permitted by right subject to published prohibitions (Design 13.1.2). Every remaining licensing regime must be individually justified, published, and reviewed on a sunset schedule.
- (2) **SINGLE RATES** One exchange rate (Design 15.1.3), one VAT rate (Design 14.2.2), one tariff schedule, one set of published fees. Every multiple rate in Cuban history has become a rent, and the largest single corruption surface in the economy for thirty years has been access to foreign currency at the official rate.
- (3) **DEEMED CONSENT** Statutory decision periods on every authorisation, with expiry constituting a grant (Design 21.1.2). An official who cannot delay cannot extract, and this single mechanism does more work than an agency.
- (4) **NO NEGOTIATED TERMS** No sectoral tax holidays, no bespoke

incentive packages, no individually negotiated rates or exemptions (Design 14.2.1). Anything negotiated one-to-one is negotiated by two people in a room.

- (5) **APPEAL THAT WORKS** Every refusal, delay, seizure and penalty appealable to the administrative court of Design 13.3.4, with the state bearing costs when it loses. This is what converts the rules above from statements into constraints.
- (6) **SUNSET** Every regulation carries an expiry date and must be re-enacted. Regulatory accumulation is how discretion returns after a reform, quietly, one decree at a time.

22.4 Publish everything

The second instrument, and the cheapest.

Design 22.2. Transparency by default

- (1) **OPEN CONTRACTING** Every public contract, tender, award and amendment published in a machine-readable open standard, from day one, with the bidders, the evaluation and the price. This includes contracts of state enterprises and of the entities in Design 21.5, without exception. Cuba is starting from zero, which is an advantage: there is no legacy system to migrate.
- (2) **ASSET DECLARATIONS** Annual public declarations of income, assets, liabilities and interests by ministers, deputies, judges, senior officials, enterprise directors, military officers above a rank, and their household members. Verified by sampling against the registries, with a published sanction for false declaration.
- (3) **BENEFICIAL OWNERSHIP REGISTER** A public register of the ultimate natural persons behind every company, trust and vehicle operating in Cuba, with no anonymous ownership permitted and no exceptions (Design 21.4.5).
- (4) **BUDGET AND ACCOUNTS** The single budget of Design 15.4.1 and its execution published quarterly, at national and municipal level (Design 13.7.4), in a common format.

- (5) **REGISTRIES OPEN** Cadastre, property register, company register, licences granted, land allocations, and privatisation results all publicly searchable.
- (6) **PRIORITY TARGETS** Where capacity is scarce, publication effort goes first to the sectors with the largest rents: energy procurement (Design 16.3.4), tourism and hotel investment, port and logistics concessions, mining, and the divestments under Design 21.5.5.

Publication only works if somebody reads it, which is why Design 13.4.2 and 13.4.3 — a free press and a statutory right to information — are in Chapter 13 rather than here. An open contracting portal in a country with no journalists is a database nobody opens. This is the single most important dependency in the chapter and the reason the political sequencing cannot be deferred: *the enforcement mechanism for transparency is publicity, and publicity requires a press.*

22.5 Cash, and the digital state

Corruption's medium is cash and its enabler is the paper transaction that leaves no record. Cuba is an overwhelmingly cash economy, which is one reason Design 15.6's payment rails appear in the money chapter as infrastructure rather than convenience.

Design 22.3. Removing the cash

- (1) **PUBLIC PAYMENTS DIGITAL** All state payments and receipts above a low threshold — wages, pensions, benefits, taxes, fees, supplier payments — electronic and traceable. This also delivers Design 15.5's conversion and Design 14.2's withholding.
- (2) **PROCUREMENT DIGITAL** All public purchasing through a single electronic platform with automatic publication, standard documents, and an audit trail that cannot be edited.
- (3) **CUSTOMS** An electronic single window with published tariffs, risk-based inspection selected by algorithm rather than by officer, and published clearance times. Customs is the highest-yield corruption

surface in every small importing economy, and Cuba is about to become one.

- (4) **INTEROPERABLE REGISTRIES** Tax, property, company, beneficial ownership and asset declarations queryable against each other by the audit and anti-corruption bodies. This is what makes clause 2 of Design 22.2 verifiable rather than ceremonial.

22.6 The institutions, and their honest limits

The conventional anti-corruption programme — an agency, a law, a hotline — comes last here deliberately, because it is the part that fails most often and that reformers reach for first.

Design 22.4. Enforcement

- (1) **AUDITOR-GENERAL** An independent supreme audit institution reporting to the assembly and not to the executive, with its own budget, the power to audit any public body including the entities in Design 21.5, and an obligation to publish every report in full and on time.
- (2) **ANTI-CORRUPTION AGENCY** An independent body with its own budget, its own investigators and its own prosecutors, appointed on the model of Design 13.3.1 and removable only by the process in Design 13.3.2. *With the caveat stated plainly:* agencies of this kind fail without independent courts to prosecute into, and where they have been created ahead of judicial reform they have been captured and used against political opponents, which is worse than not having one. This clause is conditional on Design 13.3 being in force.
- (3) **WHISTLEBLOWERS** Statutory protection for disclosure in the public interest, in the public and private sectors, with reinstatement, compensation and anonymity, and a defence to any charge under the state-secrets provisions.
- (4) **ASSET RECOVERY** A civil-recovery route allowing the state to

recover unexplained assets on the balance of probabilities, with judicial supervision and a right of explanation. Per Design 13.5.4, recovery of the asset is the remedy for historic enrichment rather than imprisonment — more achievable, more useful, and it does not require the state to relitigate the past.

- (5) **PAY THE OFFICIALS** A tax inspector, customs officer, judge or procurement official who cannot live on their salary will supplement it. Public pay under Design 14.5 is an anti-corruption instrument and is the one most likely to be cut when money is short, which is exactly when it matters.

22.7 Measurement

A design that cannot be checked is a wish. Chapter 24 collects the book's indicators; the ones belonging to this chapter are listed here because they are the earliest warning available and because they are cheap.

Design 22.5. What gets published annually

- (1) **CONCENTRATION** The share of privatised assets by value acquired by the largest ten beneficial owners, and the share acquired by persons who held public office or a state enterprise directorship in the preceding five years. If this number is large, the design is failing in the way it is most likely to fail.
- (2) **COMPETITION** The share of public contracts by value awarded through open competitive tender with more than two bidders, and the share awarded directly.
- (3) **DISCRETION** The number of licensing and authorisation regimes in force, and the share of authorisations decided within the statutory period. Both should fall and rise respectively; if the count of regimes is rising, discretion is returning by accumulation.
- (4) **COMPLIANCE** The share of asset declarations filed on time and the number sampled, verified and sanctioned.
- (5) **INDEPENDENT MEASUREMENT** Cuba should join and be assessed

by the international transparency and open-contracting standards, and publish the assessments whatever they say. External measurement is a commitment device in the sense of Design 13.1: it is a judgment the government cannot edit.

22.8 The order

To close, the ordering claim of the chapter, since order is what it is really about.

Discretion is removed *before* assets are sold. Accounts are published *before* the entities holding them are restructured. The press and the right to information are established *before* the transparency portals, because the portals are useless without them. The courts come *before* the anti-corruption agency. And the largest assets are sold *last*, when all of the above exist.

Reverse any of these and the instrument does not merely fail; it becomes an instrument of the thing it was meant to prevent — an anti-corruption agency that prosecutes opponents, an auction that launders a pre-agreed allocation, a transparency portal that publishes into silence. That is the specific way this chapter goes wrong, and it goes wrong through haste rather than through bad intent.

Chapter 23

Nature

23.1 An accident worth keeping

Cuba has the least degraded large-island ecology in the Caribbean, and it has it substantially by accident.

The country retains extensive coral reef systems, including at the Jardines de la Reina — an archipelago off the southern coast, closed to commercial fishing since 1996, where the shark and grouper populations are among the last in the region that resemble what Caribbean reefs looked like before industrial fishing. It retains the Ciénaga de Zapata, the largest wetland in the insular Caribbean, with a bird list including several endemics found nowhere else. It retains Alejandro de Humboldt National Park in the east, one of the most biologically diverse island sites on earth by area, and the mogotes of Viñales. Cuban endemism is high across plants, birds, reptiles and molluscs, as it is on any large old island, and unusually much of it is still there.

The reasons are not virtuous. Cuba stopped being able to afford fertiliser and pesticide in 1990. It stopped being able to build coastal hotels at Caribbean rates. It never developed the cruise-and-condominium coastline that has consumed much of the region. Its citizens could not buy boats or second homes. Poverty is an effective, brutal and temporary conservation policy, and every year

of the recovery this book proposes will make it less effective.

There is also a real institutional element, which should be credited. Cuba protected the Jardines de la Reina deliberately and enforced it. It has a national system of protected areas covering a substantial share of territory, a serious environmental law from 1997, a genuine tradition of field biology, and in Tarea Vida a national climate adaptation programme adopted in 2017 that is scientifically better than most in the region and almost entirely unfunded. The knowledge exists. The money and the enforcement capacity do not.

23.2 The mechanism that destroys it

The pattern is the same everywhere in the Caribbean and it is worth stating as a mechanism rather than as a warning, because the mechanism is what a design can interrupt.

A country decides to grow tourism. Beachfront becomes the most valuable land, so it is built on — and building on it removes the dune, the mangrove and the sea grass, which is what was holding the sand and filtering the runoff. The construction and the new population produce sewage that the treatment infrastructure does not handle, so nutrients enter the water. Nutrients feed algae; algae smother coral. Fishing intensifies to feed the visitors, removing the herbivorous fish that graze the algae off the coral. Boats anchor on the reef. Roads and causeways cut water circulation in the lagoons behind the cays. Within fifteen to twenty-five years the reef that the destination was sold on is dead, the beach it protected is eroding, and the country is defending its coastline with imported rock.

Cuba is at the beginning of this sequence rather than the end, and it has one enormous advantage: it can see the whole curve, in detail, in a dozen jurisdictions within an hour's flight.

23.3 The exposure

Climate risk in Cuba is not a future scenario; it is a schedule.

Hurricanes are the acute risk, and the trend in the intensity of the strongest storms is upward. Cuba's civil defence is genuinely

excellent and its death tolls are among the lowest in the region for storms of comparable strength — this is a real and underappreciated Cuban achievement — but the physical damage is not preventable by evacuation. The 2008 season alone cost something on the order of a fifth of national output.

Sea level is the chronic one. Cuba's own Tarea Vida assessments project the loss of low coastal settlements over the century and identify communities that should not be rebuilt where they stand — which is a politically extraordinary thing for a government to publish, and which it did.

Saline intrusion into coastal aquifers is already measurable and is a water-supply problem for Havana and for the sugar plain.

Coral bleaching follows sea temperature, and the Caribbean has had repeated regional bleaching events. The Cuban reefs are not exempt; what they have is better condition going in, which is the single largest determinant of whether a reef recovers from a bleaching event.

Drought in the eastern provinces — the poorest, per Chapter 12 — has been severe and recurrent, and it interacts with everything in Chapter 20.

23.4 The design

The argument throughout is economic rather than sentimental, because the sentimental version loses to a hotel proposal every time.

The reefs, the wetlands and the cays are the physical stock from which Chapter 17's high-value products are sold. A dive day at the Jardines de la Reina earns more per visitor than any beach room in the country and it exists solely because somebody drew a line and enforced it. That is not a constraint on the tourism strategy; it is the tourism strategy's inventory, and the correct treatment of inventory is not to consume it.

Design 23.1. Protecting the stock

- (1) AN INDEPENDENT REGULATOR An environmental authority with

the tenure, budget and publication guarantees of Design 13.3, independent of the tourism ministry, of the entities in Design 21.5, and of any ministry that owns a development. Consent decisions and their reasons published; refusals published. At present the promoter and the consenting authority are frequently the same state.

- (2) COASTAL SETBACK, WITHOUT EXCEPTION A statutory construction setback from the mean high-water line, applied to every developer including state and military-linked ones, with no variance procedure. Every Caribbean setback rule that has a variance procedure has failed through it, which is why this clause is about the absence of the exception rather than the distance.
- (3) NO NEW CAUSEWAYS No further *pedraplenes* to the cays; existing ones assessed for remediation with culverts to restore circulation. The damage these have done to lagoon systems is documented and is reversible in part.
- (4) SEWAGE BEFORE ROOMS No accommodation development is consented without treatment capacity commissioned and operating. Nutrient loading is the single largest controllable driver of reef loss, and it is an infrastructure decision taken years before the reef dies.
- (5) FISHING No-take zones maintained and extended around reef systems, with enforcement funded from visitor revenue under clause 1 of Design 23.2. The Jardines de la Reina model works and is replicable.
- (6) ENVIRONMENTAL DATA PUBLISHED Reef condition, water quality, protected area extent and enforcement actions published annually, per Design 13.4.3.

Design 23.2. Paying for it

- (1) VISITOR FINANCE A share of the accommodation levy and of protected-area access fees is assigned by statute to the protected-area system and retained by it, on the Design 17.2 model. Conservation that depends on an annual budget negotiation loses the negotiation.

- (2) **PRICE THE ACCESS** Where a site is at its carrying capacity, access is priced and allocated by published auction or ballot (Design 17.3). This converts scarcity into revenue for the thing that is scarce.
- (3) **CARBON AND ECOSYSTEM FINANCE** Mangrove and forest carbon, and payment-for-ecosystem-services arrangements, pursued as a revenue stream. These markets are imperfect and the amounts are modest, but they are hard currency for not doing something, which is unusually well suited to Cuba's position.
- (4) **SCIENCE AS AN EXPORT** Cuban field biology plus intact ecosystems plus foreign research funding is a real and immediate line of business, and it overlaps with Chapter 17's high-value visitor and Chapter 18's academic opening. It requires only that foreign scientists can obtain permits without a political assessment.

Design 23.3. Building for the century

- (1) **FUND TAREA VIDA** The existing national adaptation programme is scientifically sound and unfunded. It is assigned a standing allocation from the stabilisation fund of Design 14.3.2, which is exactly the kind of capital spending that fund exists for.
- (2) **BUILDING CODES** Hurricane-resistant construction standards, enforced through the municipal building control of Design 13.7.3, applied to housing as well as to hotels. Cuba's housing stock is its most storm-vulnerable asset and its rebuilding under Chapter 21 is the once-in-a-century opportunity to raise the standard.
- (3) **MANAGED RETREAT** Where Tarea Vida identifies settlements that cannot be defended, relocation is planned, funded, consented and compensated, and no reconstruction is publicly financed in place. This is politically painful and cheaper than the alternative, and Cuba has already done the analytical work.
- (4) **WATER** Aquifer protection, leak reduction in the distribution network — where losses are very large — and water pricing above a free basic allowance. Cuba currently loses a great deal of pumped water to leaks, which is simultaneously a water problem and, since the water is pumped, an electricity problem under Chapter 16.
- (5) **INSURANCE AND RESERVE** Per Design 16.1.5, an explicit published

reserve against catastrophe losses, since commercial cover for Cuban assets is expensive and in several markets unavailable.

23.5 The trade-off, stated

This chapter constrains Chapter 17 and Chapter 21 and it should say so rather than pretend the objectives are automatically aligned.

Setbacks, sewage requirements, carrying capacity limits and no-take zones reduce the number of hotel rooms that can be built and the speed at which they can be built. In the short run that is less investment, fewer construction jobs and less revenue, and a government under fiscal pressure will be offered a great deal of money to make an exception — which is why Design 23.1.2 has no variance procedure and why the regulator's independence is written in the same terms as the judiciary's.

The case for accepting the constraint is Chapter 17's: Cuba is not competing on volume, cannot win on volume, and is proposing to sell the one thing in the Caribbean that is genuinely scarce. Every other country in the region has already made the opposite trade and can report the result.

Chapter 24

The sequence

24.1 Why order is the whole content

Chapter 11 found that almost every measure proposed in this book has already been tried in Cuba, that most of them worked, and that all of them were reversed. Part III has added institutional machinery to stop the reversing. What remains is the order, and the order is not a presentational matter. Three of the four failures in Part II were failures of sequence rather than of direction.

1993–95 got the sequence right and is the counter-example: stabilise first, then liberalise, then grow. It worked, and it was then undone by politics rather than by economics.

2011 got it wrong by omission. The Lineamientos legalised activity without building any of the things that make activity possible — no legal personality, no wholesale market, no import rights, no credit, no foreign exchange. The reform was a permission slip in an economy with no plumbing, and it produced exactly what such a reform produces.

2021 got it wrong by inversion. The currency was unified before the economy was stabilised, before reserves existed, before supply could respond, and before the private sector was legal. The right measure in the wrong position destroyed the savings it was meant to rationalise.

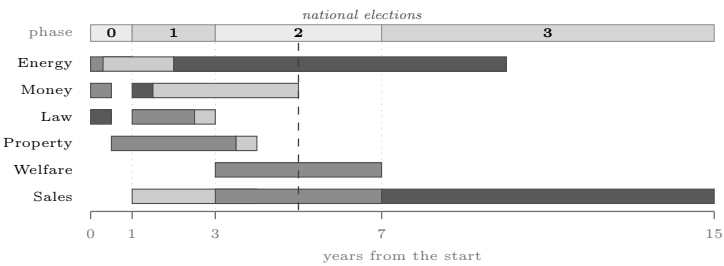


Figure 24.1: The sequence. The header strip marks the four phases; each row is a workstream and each bar a distinct activity within it, darker shading indicating the later activity in that row — not budget, not effort. Energy runs repeals and efficiency, then critical loads, then the build-out; Money publishes the debt stock, converts the currency, then restructures; Law repeals and amnesties, then builds courts, then holds municipal elections; Property registers before it sells; Welfare converges only in Phase 2; Sales go small, then medium, then — after year seven — large. The ordering claims are the content: the build-out cannot start before the regulator and the tariff exist, the currency is not converted before the debt stock is published, the courts precede the first competitive elections, registration precedes any sale, welfare convergence waits on revenue that Phase 1’s tax administration will not produce until Phase 2, and the largest assets are sold last — which is the single clause standing between this design and the outcome described in Chapter 22.

So this chapter is the book’s actual proposal. Everything before it is inputs.

24.2 The four phases

Dates are relative to a start year rather than to a calendar, because the start year is not something this book can set.

Phase 0 — months 0 to 12: stop the bleeding

The organising principle is that Phase 0 contains *only* things that are fast, cheap, and reversible in their costs but not in their benefits. Nothing here requires foreign money, a constitutional amendment, or anyone's permission.

The repeals first, because they are free. The right of any resident to contract with a foreign client and be paid into their own account (Design 18.1). Rooftop generation by right and duty-free import of energy and computing equipment (Designs 16.2.1, 16.2.6, 18.1.5). Activity permitted by right rather than by enumerated list. Professionals allowed to practise privately. The remaining exit restrictions lifted. The amnesty of Design 13.5.6 — everyone convicted for political expression, association or the events of 11 July 2021 released and their convictions vacated — which costs the state nothing and is the single clearest signal available that this time is different. Raúl Castro's most popular measures were all of this kind and the lesson is in Chapter 9: begin with the prohibitions that cost nothing to remove, because they are immediate, they are free, and they buy the credibility the expensive measures will need.

Then the emergencies. Fuel, spares and maintenance for the existing thermal fleet, funded as a first call on the budget — unglamorous, unavoidable, and the thing a solar enthusiast would skip. Demand-side efficiency at national scale, on the 2006 precedent. Photovoltaics and storage on critical loads: hospitals, water pumping, cold stores, bakeries, telecoms. Food imports maintained. Current supplier invoices paid and the 2009 frozen balances released (Design 15.3.6).

And the announcements that start the clocks: the complete debt and arrears stock published; the dollarisation date and conversion rate announced; the political calendar of Design 13.2 entrenched; the cadastre programme begun.

Phase 1 — years 1 to 3: build the plumbing

Money: convert to the dollar at a single rate, all balances at once, wages and pensions protected at the floor (Design 15.1). Legislate the prohibition on monetary financing and the recurrent balance

rule. Open the debt restructuring with the claims inside it.

Law: judicial reform and, above all, the administrative court in which a citizen or a firm can sue the state (Design 13.3.4). Press, association and the right to information. The party law and the electoral commission. Competitive municipal elections at the end of the phase.

Property: cadastre and registry delivered; usufruct converted to real, mortgageable title (Design 20.1); acopio abolished; wholesale markets opened; the residential rental market legalised.

Investment: the single investment law with direct employment, deemed consent, national treatment and arbitration (Design 21.1). Small-scale privatisation begins — thousands of retail, catering, repair and service units, by open auction, worker and cooperative preference.

Energy: the regulator, the standard power purchase agreement, net billing, and the first competitively tendered independent producer capacity.

And the long-lead item that yields nothing in this phase and must start in it anyway: the revenue administration. Registration, taxpayer numbers, withholding, the presumptive regime for micro-enterprise, and the beginning of the VAT build. Every country that has done this has taken five to eight years, and the welfare commitment of Chapter 14 is unfinanceable if it slips.

Phase 2 — years 3 to 7: the growth phase

The debt restructuring closes and the claims are settled in paper within the cap. Multilateral membership is pursued. VAT comes into force and the tax base begins to exist.

The energy build-out reaches scale — the bulk of the six gigawatts and the storage — weighted to the eastern provinces. Tourism reorients on the value-per-visitor metric, with the moratorium on state hotel construction in force and the heritage authorities established. The second track becomes visible: connectivity redundancy delivered, the residence regime open, service exports counted in the balance of payments. The medicines regulator reaches stringent-authority standard and the first product partnerships are signed.

The welfare convergence begins here rather than earlier, because this is the first phase in which there is revenue: public pay benchmarked and raised under Design 14.5, the universal cash benefits paid reliably for two full years before any ration item is withdrawn, and the pension reform legislated with its indexation formula.

Medium-scale privatisation proceeds under published auction. National assembly elections fall in this phase, at year five.

Phase 3 — years 7 to 15: the hard remainder

What was deliberately left: the large state sector, the utilities, the ports, the telecommunications incumbent, and the military conglomerate's core assets, sold only once the competition authority, the sector regulators, the courts, the press and the beneficial-ownership register are all functioning (Design 21.4.2, Design 21.5.5). Unbundling of the electricity network. Convergence of benefit levels toward the European standard on the published trajectory. And, if the conditions of Design 15.1.5 are met, the question of whether to reintroduce a national currency.

24.3 Who carries which phase

The political-economy question, which a design document is not entitled to skip.

Phase 0 is executable by the present government, and this is deliberate. Every measure in it is within the existing constitutional order, none requires surrendering the party's leading role, several would relieve pressures the government is currently absorbing, and the amnesty and the repeals would be enormously popular at almost no fiscal cost. A book that requires a rupture before its first measure is a book that does nothing until the rupture, and Cubans have been waiting for the rupture since 1991.

Phase 1 requires the party to accept an independent court, which is the hardest single concession in the book, because it is the concession from which the others follow. The coalition that could carry it is the one that already exists inside the Cuban state: the technocrats

who have been writing these reforms since 2011 and watching them be reversed; the enterprise managers who cannot operate without a currency that means something; and — this is the uncomfortable part — the armed forces, whose commercial interests are better served by a functioning economy with enforceable contracts than by an increasingly worthless monopoly over an increasingly empty one.

Phase 2 requires the bargain of Design 21.5.6 to have been struck explicitly: security, pensions, property and a legal route into ordinary political and business life, in exchange for the surrender of discretionary control. This is where a transition is either agreed or contested, and the book's position is that it should be negotiated openly and written down, because a bargain left implicit is renegotiated at every step.

Phase 3 requires a government formed after the elections of Phase 2. It is not the same government, and the design does not assume that it is friendly to what came before. That is the test of whether the commitment devices of Chapter 13 were real.

24.4 What it costs and where it comes from

Orders of magnitude over a decade, with assumptions stated. Every figure is marked unverified in the manuscript and listed in `verify.md`.

<i>Capital, cumulative over ten years</i>	
Photovoltaics and storage (Ch. 16)	\$6–8 bn
Grid transmission and distribution	\$2–3 bn
Water: leak reduction and treatment	\$1–2 bn
Cadastre, registry, revenue administration	\$0.3–0.5 bn
Strategic food reserve, initial	\$0.3–0.5 bn
<i>Recurrent, annual at maturity</i>	
Public pay settlement, health and education first	\$0.6–1.1 bn/yr
Benefit convergence, phased	rising
<i>Sources</i>	
Hotel construction redirected	\$1–1.5 bn/yr
Avoided fuel imports, from year six	\$1.2–1.5 bn/yr
Diaspora investment and remittance conversion	—
Concessional and bilateral finance	—
Privatisation proceeds (to the fund)	—
Tax base as it grows	—

One line in that table carries most of the argument, and it is best stated as a share rather than in dollars, because the peso conversion is the problem the note on sources described. In 2024 Cuba put 37.4 per cent of its total national investment into tourism-related construction and 2.7 per cent into agriculture, while running its hotels at occupancy around a quarter, importing most of its food and appealing for children's milk. Moving even half of the tourism share into generation, network repair, water and agriculture — which requires no foreign money, no legislation and nobody's permission — is of the same order as the entire energy programme in this book. The energy programme then saves, in perpetuity, more per year in foreign exchange than the hotels were ever going to earn.

The largest available source of finance for this plan is not foreign. It is Cuba's own investment budget, spent differently. That conclusion is the reason Chapter 22 matters more than any chapter about capital markets: the constraint is not the absence of money but the absence of anyone able to say no to the people currently allocating it.

24.5 What could go wrong

Six failure modes, in descending order of likelihood.

Capture. The incumbents run Phases 0 and 1, acquire the assets in Phase 2 before the institutions bite, and Cuba arrives in 2045 as a middle-income country with a captured state, a compliant press and an ownership structure fixed for a generation. This is the most likely failure and it is the one that is irreversible. Everything in Chapter 22 is aimed at it, and the single indicator that detects it early is Design 22.5.1.

The workforce leaves anyway. The reforms take four years to raise incomes and the exodus takes two to remove the people who would have benefited. Cuba ends the decade with better institutions and no one to use them. This is why net migration is the master indicator below and why Phase 0 is judged on whether the outflow slows rather than on growth.

Collapse before Phase 1 completes. A grid failure that is not recovered, a fuel supplier who stops, a hurricane in the wrong year, a fiscal crisis during the currency conversion. The design's answer is Phase 0's unglamorous insistence on keeping the existing fleet running and the food coming while the new things are built.

Sequencing failure. Elections before courts; privatisation before the registers; the ration withdrawn before the cash arrives; the currency converted before stabilisation. Each of these has a named clause against it and each will be argued for, by serious people, at the time.

The J-curve, politically. Reforms of this kind hurt before they help. Prices rise before supply responds; state employment falls before private employment absorbs it. The political system has to survive the trough, and a government that faces its first competitive election at the bottom of it may not. This is an argument for the pacing above — welfare convergence and elections both in Phase 2, after the first benefits are visible — and it is an honest weakness of any gradualist design.

An external swing. A sudden American normalisation would flood a country with no absorptive capacity, no registry, no planning system and no claims settlement, and the assets would be allocated

in eighteen months to whoever was positioned. A sharp tightening removes remittances and travel, which are the household economy. The design's premise is to assume neither, and the practical protection against the first is simply to have done Phase 1 before it happens.

24.6 How to tell whether this is working

A proposal that cannot be falsified is not a proposal. These are published annually, and each has a direction that constitutes failure.

Design 24.1. The indicators

- (1) **NET MIGRATION** The master indicator. If Cubans continue to leave at anything like the rate of 2021–25, nothing else in this book matters, because the design is labour-constrained and the labour is going.
- (2) **LOAD SHEDDING** Average daily hours of interruption, national and by province, with the eastern provinces reported separately. Everything is downstream of this.
- (3) **HEALTH AND EDUCATION OUTCOMES** Infant mortality, life expectancy, school attainment, and resident physicians, nurses and teachers per capita — *resident*, not counting missions abroad. These must not fall. If they fall, the design has failed on its own stated terms regardless of any growth figure.
- (4) **REAL WAGES AND THE PRICE OF THE BASKET** Not the official index — the cost of an actual consumption basket against actual median income.
- (5) **FOOD** Share of consumption domestically produced, and post-harvest losses.
- (6) **SERVICE EXPORTS EXCLUDING MEDICAL MISSIONS** The second track's existence test. If this is not visible by year five, Chapter 18 was wrong.
- (7) **CONCENTRATION** Per Design 22.5.1: the share of privatised assets acquired by the largest ten beneficial owners and by former officeholders. The early-warning indicator for capture.

- (8) OPENNESS OF CONTRACTING Share of public contracts by value awarded through open tender with more than two bidders.
- (9) TIME TO AUTHORISE Median days to obtain a business authorisation, and the share decided within the statutory period.
- (10) TAX REVENUE Revenue as a share of output, by instrument. The financing test for the welfare commitment.
- (11) REEF AND PROTECTED-AREA CONDITION Published annually, per Design 23.1.6.
- (12) DISAGGREGATION Every one of the above reported by province and, where individual, by race. Cuba declared the racial question solved in 1962, made it unmeasurable for thirty years, and entered its market opening with no idea where it stood. That is not repeated.

24.7 What this book does not claim

It does not claim to know what will happen. It is a design, not a forecast, and the difference is that a design states its assumptions in a form that lets someone else change them.

It does not claim that the measures here are the only ones that would work. Several of its recommendations — dollarisation most obviously, the ordering of courts before elections next — are genuinely contestable, and the book has tried to give the argument on the other side each time rather than to win by omission.

It does not claim that the embargo is unimportant, only that it is the one variable Cubans do not control, and that a design which waits on it is not a design.

It does not claim that the record of 1959–2026 is a simple story. Parts I and II were written at length precisely because it is not. A state that eliminated illiteracy in a year also ran labour camps; a state that held infant mortality through a thirty-five per cent contraction also handed thirty-year sentences to teenagers for walking in the street. Both libraries in the preface were telling the truth and neither was being honest, and any proposal that needs one of them to be false is a proposal that will fail on contact with the country.

And it does not claim that any of this is likely. The most probable future for Cuba is the continuation of the present: a slow contraction, an ageing remainder, the lights going out province by province, and the young leaving until there is nobody left to design for. That future requires no decisions at all, which is what makes it the default.

What the book claims is narrower and, I think, defensible. Cuba has assets that compose — an educated population, a medical and scientific capability with products and a reputation, an intact environment in a region that has spent its own, sunlight falling on a country that burns imported oil, and two million of its own people one flight away with capital and a reason to care. Those assets have been available for twenty years and have not converted, and the reason is not principally the embargo. It is that Cuba has never built the machinery that turns an asset into an investment: a currency that means something, a court that can be lost in fairly, a registry that says who owns what, a customs service that does not take a cut, a procurement system that publishes what it bought, and a rule that cannot be withdrawn the moment it starts to work.

That machinery is buildable. It is mostly not expensive. Almost none of it requires anyone's permission. And it is the difference between a country with sunshine and doctors and beaches that is poor, and one that is not.

Bibliography

Anders Åslund. *How Capitalism Was Built: The Transformation of Central and Eastern Europe, Russia, the Caucasus, and Central Asia*. Cambridge University Press, Cambridge, 2 edition, 2013.

Martin Carnoy, Amber K. Gove, and Jeffery H. Marshall. *Cuba's Academic Advantage: Why Students in Cuba Do Better in School*. Stanford University Press, Stanford, 2007.

Centers for Disease Control and Prevention. International notes: Epidemic neuropathy — Cuba, 1991–1994. *Morbidity and Mortality Weekly Report*, 43(10):183–192, 1994.

Alejandro de la Fuente. *A Nation for All: Race, Inequality, and Politics in Twentieth-Century Cuba*. University of North Carolina Press, Chapel Hill, 2001.

Susan Eva Eckstein. *Back from the Future: Cuba under Castro*. Routledge, New York, 2 edition, 2003.

Samuel Farber. *The Origins of the Cuban Revolution Reconsidered*. University of North Carolina Press, Chapel Hill, 2006.

Richard E. Feinberg. *Open for Business: Building the New Cuban Economy*. Brookings Institution Press, Washington, DC, 2016.

Julie M. Feinsilver. *Healing the Masses: Cuban Health Politics at Home and Abroad*. University of California Press, Berkeley, 1993.

- Foreign Claims Settlement Commission of the United States. Completed programs: Cuba — certified claims against the government of Cuba. US Department of Justice, 1972. Cuban Claims Program concluded 1972; 5,913 claims certified at a principal of \$1.9 billion.
- Manuel Franco, Usama Bilal, Pedro Ordúñez, Mikhail Benet, Alain Morejón, Benjamín Caballero, Joseph F. Kennelly, and Richard S. Cooper. Population-wide weight loss and regain in relation to diabetes burden and cardiovascular mortality in Cuba 1980–2010. *BMJ*, 346:f1515, 2013.
- Piero Gleijeses. *Conflicting Missions: Havana, Washington, and Africa, 1959–1976*. University of North Carolina Press, Chapel Hill, 2002.
- Piero Gleijeses. *Visions of Freedom: Havana, Washington, Pretoria, and the Struggle for Southern Africa, 1976–1991*. University of North Carolina Press, Chapel Hill, 2013.
- Joel S. Hellman. Winners take all: The politics of partial reform in postcommunist transitions. *World Politics*, 50(2):203–234, 1998.
- Inter-American Commission on Human Rights. Report no. 47/96, case 11.436, victims of the tugboat “13 de marzo” v. Cuba, 1996.
- Justicia 11J and Cubalex. Registro de personas privadas de libertad en el contexto de las protestas del 11 de julio de 2021. Ongoing documentation project, 2022. The only systematic count; produced by non-governmental monitors.
- John M. Kirk. *Healthcare without Borders: Understanding Cuban Medical Internationalism*. University Press of Florida, Gainesville, 2015.
- William M. LeoGrande and Peter Kornbluh. *Back Channel to Cuba: The Hidden History of Negotiations between Washington and Havana*. University of North Carolina Press, Chapel Hill, 2014.

- Carmelo Mesa-Lago. *Market, Socialist, and Mixed Economies: Comparative Policy and Performance — Chile, Cuba, and Costa Rica*. Johns Hopkins University Press, Baltimore, 2000.
- Carmelo Mesa-Lago and Jorge Pérez-López. *Cuba's Aborted Reform: Socioeconomic Effects, International Comparisons, and Transition Policies*. University Press of Florida, Gainesville, 2005.
- Carmelo Mesa-Lago and Jorge Pérez-López. *Cuba Under Raúl Castro: Assessing the Reforms*. Lynne Rienner, Boulder, 2013.
- Douglass C. North and Barry R. Weingast. Constitutions and commitment: The evolution of institutions governing public choice in seventeenth-century england. *The Journal of Economic History*, 49(4):803–832, 1989.
- Oficina Nacional de Estadística e Información. Anuario estadístico de cuba. Havana, annual, 2024a. Annual series; the edition relevant to each claim is named in the text.
- Oficina Nacional de Estadística e Información. Inversiones por actividad económica. Anuario Estadístico de Cuba, 2024b. Investment by sector; analysed by Pedro Monreal.
- Jr. Pérez, Louis A. *Cuba: Between Reform and Revolution*. Oxford University Press, New York, 5 edition, 2015.
- Marifeli Pérez-Stable. *The Cuban Revolution: Origins, Course, and Legacy*. Oxford University Press, New York, 3 edition, 2012.
- Archibald R. M. Ritter and Ted A. Henken. *Entrepreneurial Cuba: The Changing Policy Landscape*. First Forum Press, Boulder, 2015.
- Gérard Roland. *Transition and Economics: Politics, Markets, and Firms*. MIT Press, Cambridge, MA, 2000.
- Solargis and World Bank Group. Global solar atlas: Photovoltaic power potential. <https://globalsolaratlas.info>, 2020. Source for the specific-yield assumption in Chapter 16.

Julia E. Sweig. *Inside the Cuban Revolution: Fidel Castro and the Urban Underground*. Harvard University Press, Cambridge, MA, 2002.

Hugh Thomas. *Cuba: The Pursuit of Freedom*. Harper and Row, New York, 1971.